

Topological orders and edge excitations in fractional quantum Hall states

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Topological orders and edge excitations in fractional quantum Hall states

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Abstract

Fractional quantum Hall (FQH) liquids contain extremely rich internal structures which represent a whole new kind of ordering. We discuss the characterization and classification of the new orders (which are called topological orders). We also discuss the edge excitations in FQH liquids, which form the so-called chiral Luttinger liquids. The chiral Luttinger liquids at the edges also have very rich structures as a reflection of the rich topological orders in the bulk. Thus edge excitations provide us a practical way to measure topological orders in experiments.

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1. Introduction

The fractional quantum Hall (FQH) effect appears in two-dimensional (2D) electron systems in a strong magnetic field. Since its discovery in 1982 [1, 2], experiments on FQH systems have continued to reveal many new phenomena and surprises. These, together with the observed rich hierarchical structures [3], indicate that electron systems that demonstrate a FQH effect (those systems are called FQH liquids) contain extremely rich internal structures. In fact, FQH liquids represent a whole new state of matter. One needs to develop new concepts and new techniques to understand this new kind of state.

The first attempt to characterize the internal structures of FQH liquids was proposed in [4], where it was shown that a primary FQH liquid (with an inverse filling fraction ν^{-1} equal to an odd integer) contains an off-diagonal long range order in a non-local operator. Such an observation led to a description of FQH liquids in terms of Ginzburg–Landau–Chern–Simons effective theory [5]. Those developments have led to many interesting and important results and a deeper understanding of FQH liquids. However, in this article I shall try to describe the internal structures of FQH liquids from a more general point of view. It appears that some internal structures of FQH liquids (especially those in the so-called non-Abelian FQH liquids) cannot be described by the Ginzburg–Landau–Chern–Simons effective theory and the associated off-diagonal long range orders. Thus we need to develop more general concepts and formulations for the internal structures in FQH liquids. Let us start with a general discussion of correlated systems.

Gases are one of the simplest systems. The motion of a molecule in a gas hardly depends on the positions and motion of other molecules. Thus gases are weakly correlated systems. As we lower the temperature, the motions of molecules become more and more correlated. Below a certain temperature the motions of molecules are completely correlated and the molecules form a crystal which is a strongly correlated system. In a crystal, an individual atom can hardly move by itself. Excitations in a crystal always correspond to collective motions of many atoms (which are called phonons).

FQH liquids are another type of strongly correlated structure. Since electrons are much lighter than atoms, they have much stronger quantum motions. These quantum fluctuations prevent electrons in FQH systems from forming a crystal. Nevertheless, the motions of electrons in FQH liquids are highly correlated. For example, let us look at Laughlin's wavefunction for a filling fraction $\frac{1}{3}$ FQH liquid [2]:

$$\Psi_0 = \prod_{i < j} (z_i - z_j)^3 \exp \left(-\frac{1}{4} \sum_i |z_i|^2 \right), \quad (1.1)$$

where $z_i = x_i + iy_i$ is the coordinate i th electron. We see from Laughlin's wavefunction that

- (1) all electrons are in the first Landau level and have their own cyclotron motion and
- (2) the wavefunction has a third-order zero as any pair of electrons approach each other, and hence electrons try to stay away from each other as much as possible.

The Laughlin wavefunction represents a particular state in which all electrons have their own cyclotron motion in the first Landau level and *at the same time* avoid

collision with each other. This requires all the electrons in the Laughlin state to ‘dance’ collectively following a strict global dancing pattern. This collective dancing minimizes the ground-state energy and gives rise to a very stable correlated state. The strict collective dancing pattern makes FQH liquids highly organized systems which contain well defined internal orders. The highly correlated nature of FQH liquids also makes them incompressible liquids. A compression of FQH liquids create first-order zeros in the Laughlin state and breaks the global dancing pattern of electrons, hence costing finite energies.

We would like to remark that the above picture of collective dancing is quite vague and general. Collective motions of electrons exist in all correlated systems. To obtain a more concrete description of the internal structures in FQH liquids, one needs to find measurable quantum numbers that reflect the internal correlations in FQH liquids. As we shall see later, the correlations between electrons in FQH liquids are so special that we need a completely new set of quantum numbers to describe them.

Laughlin states represent only the simplest dancing pattern. Different FQH liquids contain different dancing patterns. Rich families of FQH liquids observed in experiments indicate that the possible dancing patterns are very rich. The internal orders (i.e. the dancing patterns) of FQH liquids are very different from the internal orders in other correlated systems, such as crystal, superfluids, etc. The internal orders in the latter systems can be described by order parameters associated with broken symmetries. As a result, the ordered states can be described by the Ginzburg–Landau effective theory. The internal order in HQH liquids is a new kind of ordering which cannot be described by order parameters associated with broken symmetries. Thus it would be very interesting to see what kind of effective theory describes the low-energy physics of FQH liquids.

Despite the absence of broken symmetries and associated order parameters, the internal orders in FQH liquids are universal in the sense that they are robust against arbitrary perturbations. These (universal) internal orders characterize a phase in phase diagram. This makes FQH liquids a new state of matter. A concept of ‘topological order’ was introduced to describe this new kind of ordering in FQH liquids [6, 7]. Later we shall give a more precise definition of the topological orders. We shall introduce measurable quantities to characterize different topological orders. The universality of the topological orders means that these characterization quantities are independent of various perturbations.

It is also useful to introduce the notion of topological fluids to describe quantum liquids with non-trivial topological orders. FQH liquids are not the only kind of topological fluid. Other examples of topological fluids include anyon superfluids [8], chiral spin states [6, 9], and short-ranged resonant valence bond (RVB) states for spin systems [10]. A general but brief discussion of topological order can be found in [11].

It is instructive to compare FQH liquids with crystals. FQH liquids are similar to crystals in the sense that they both contain rich internal patterns (or internal orders). The main difference is that the patterns in the crystals are static, which relates to the positions of atoms, while the patterns in FQH liquids are dynamic, which is associated with the ways that electrons dance around each other. However, many of the same questions for crystal orders can also be asked and should be addressed for topological orders. One important question is: how do we characterize and classify the topological orders? Another is: how do we experimentally measure the topological orders?

1.1. Characterization of topological orders

In the above, the concept of topological order (the dancing pattern) is introduced through the ground-state wavefunction. This is not quite correct because the ground-state wavefunction is not universal. To establish a new concept, such as topological order, one needs to find physical characterizations or measurements of topological orders or, in other words, one needs to find universal quantum numbers which are independent of details of interactions, effective mass, etc., but can take different values for different classes of FQH liquids. The existence of such quantum numbers implies the existence of topological orders.

One way to show the existence of the topological orders in FQH liquids is to study their ground-state degeneracies (in the thermodynamical limit). FQH liquids have a very special property. Their ground-state degeneracy depends on the topology of space [7, 12, 13]. For example, the $\nu = 1/q$ Laughlin state has q^g degenerate ground states on a Riemann surface of genus g [7]. The ground-state degeneracy in FQH liquids is *not* a consequence of symmetry of the Hamiltonian. The ground-state degeneracy is robust against arbitrary perturbations (even impurities that break all the symmetries in the Hamiltonian) [7]. The robustness of ground-state degeneracy indicates that the internal structures that give rise to ground-state degeneracy is universal, hence demonstrating the existence of universal internal orders, namely topological orders.

The ground-state degeneracy can be regarded as a new quantum number that (partially) characterizes the topological orders in FQH states. In particular, some hierarchical states with the same filling fraction can have different ground-state degeneracies, indicating different topological orders in those hierarchical states. Thus the ground-state degeneracies provide new information, in addition to the filling fraction, about the internal orders of FQH liquids. According to this picture we would like to suggest that the different hierarchical FQH states are characterized by the different topological orders instead of by their symmetry properties.

In compact space, the low-energy physics of FQH liquids is unique. There is only a finite number of low-energy excitations (i.e. the degenerate ground states); yet the low-energy dynamics are non-trivial since the ground state degeneracy depends on the topology of the space. Such special low-energy dynamics, which depend only on the topology of the space, are described by the so-called topological field theory which was under intensive study recently in the high-energy physics community [14, 15]. Topological theories are effective theories for FQH liquids just as the Ginzburg–Landau theory for superfluids (or other symmetry-broken phases).

The dependence of the ground-state degeneracy on the topology of the space indicates the existence of some kind of long-range order (the global dancing pattern mentioned above) in FQH liquids, despite the absence of long-range correlations for all local physical operators. In some sense we may say that FQH liquids contain hidden long-range orders.

1.2. Classification of topological orders

A long-standing problem has been how to label and classify the rich topological orders in FQH liquids. We are able to classify all crystal orders because we know the crystal orders are described by symmetry group, but our understanding of topological orders in FQH liquids is very poor, and the mathematical structure behind the topological orders is unclear.

Nevertheless, we have been able to find a simple and unified treatment for a class of FQH liquids: Abelian FQH liquids. It has been shown that the internal structures (the topological orders) in such fluids can be characterized by a symmetric integer matrix $\mathbf{K}$, a charge vector $\mathbf{t}$ and a spin vector $\mathbf{s}$ [16–21]. Various physical quantities can be determined from $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ that include ground-state degeneracy, quasiparticle quantum numbers, structure of the edge excitations, and exponents of electron and quasiparticle propagators along the edge. All quasiparticle excitations in this class of FQH liquids have Abelian statistics, which leads to the name Abelian FQH liquids. The low-energy effective theory of Abelian FQH liquids is a topological Chern–Simons theory with several $U(1)$ gauge fields $a_{I\mu}$ [16–20]

$$\mathcal{L} = \frac{1}{4\pi} K_{IJ} a_{I\mu} \partial_\nu a_{J\lambda} \varepsilon^{\mu\nu\lambda} - e A_\mu t_I \partial_\nu a_{I\lambda} \varepsilon^{\mu\nu\lambda}. \quad (1.2)$$

The above effective theory is the dual form of the Ginzburg–Landau–Chern–Simons effective theory discovered earlier [5]. Almost all FQH liquids observed in experiments were believed to be in this class.

We know that there are many different schemes to construct FQH states at filling fractions different from $1/m$ [3]. Different constructions in general lead to different wavefunctions, even when they have the same filling fraction. From the construction schemes themselves, it is usually difficult to see whether or not the FQH states described by those different wavefunctions belong to the same universality class. The above characterization (using $\mathbf{K}$, $\mathbf{t}$ and $\mathbf{s}$) of the topological orders has an advantage in that it is independent of specific construction schemes and provides a universal description of the universality classes of abelian FQH liquids. One can usually derive an effective theory for the constructed FQH states from the construction schemes [17–21]. If two different constructions lead to the same $\mathbf{K}$ matrix, charge vector $\mathbf{t}$ and spin vector $\mathbf{s}$ (up to a field redefinition of the gauge fields), then we can say that the two constructions generate the same FQH liquid. This approach allows us to determine [17, 18] that the two $\frac{2}{5}$ states obtained by hierarchical construction and Jain’s construction [3(f)] belong to the same universality class.

The physical meaning of the $\mathbf{K}$ matrix is quite transparent in a class of multilayer FQH states obtained by generalizing Laughlin’s construction for the $1/m$ state [22]. The wavefunction of those multilayer FQH states have the form

$$\prod_{a,b,i,j} (z_{ai} - z_{bj})^{k_{ab}/2} \exp(-\sum |z_{ai}|^2) \quad (1.3)$$

where z_{ai} is the coordinate of the i th electron in the a th layer, and k_{ab} are integers satisfying $k_{ab} = k_{ba}$ and k_{aa} odd. The $\mathbf{K}$ matrix that describes the multilayer FQH states is simply the matrix formed by the integer exponents in equation (1.3): $\mathbf{K} = (k_{ab})$. Thus $\mathbf{K}$ describes the pattern of zeros in the wavefunction which determines the way that electrons dance around each other. The zeros in the wavefunction can also be interpreted as a certain number of flux quanta being attached to each electron. For example the third-order zero in the Laughlin wavefunction in (1.1) implies that three flux quanta are attached to the electrons, since the phase of the wavefunction changes by $3 \times 2\pi$ as one electron moves around another. In this sense, $\mathbf{K}$ also describes the way that we attach flux quanta to electrons. It is obvious that the multilayer FQH states naturally admit the $\mathbf{K}$ -matrix description. However, it is less obvious that the hierarchical states and the many states obtained in Jain construction also admit the $\mathbf{K}$ -matrix description.

The above classification of FQH liquids is not complete. Not every HQH state is described by $\mathbf{K}$ matrices. In the last few years a new class of FQH states, namely non-Abelian FQH states, was proposed [23–25]. A non-Abelian FQH state contains quasiparticles with non-Abelian statistics. The observed filling fraction $\frac{5}{2}$ FQH state [26] is very likely to be one such state [27]. Recent studies in [23, 25, 28] reveal that the topological orders in some non-Abelian FQH states can be described by conformal field theories. However, we are still quite far from a complete classification of all possible topological orders in non-Abelian states.

1.3. Edge excitations—a practical way to measure topological orders in experiments

FQH liquids as incompressible liquids have a finite energy gap for all their bulk excitations. However, FQH liquids of finite size always contain one-dimensional (1D) gapless edge excitations; this is another unique property of FQH fluids. The structures of edge excitations are extremely rich, which reflects the rich bulk topological orders. Different bulk topological orders lead to different structures of edge excitations. Thus we can study and measure the bulk topological orders by studying structures of edge excitations.

The edge excitations of integer quantum Hall (IQH) liquids was first studied by Halperin [29]. He found that edge excitations are described by 1D Fermi liquids. In the last few years we have begun to understand the edge excitations of FQH liquids. We found that, owing to the non-trivial bulk topological order, electrons at the edges of (Abelian) FQH liquids form a new kind of correlated state: chiral Luttinger liquids [18, 30–33]. The electron propagator in chiral Luttinger liquids develops an anomalous exponent: $\langle c^\dagger(t, x)c(0) \rangle \propto (x - vt)^{-g}$, $g \neq 1$. (For Fermi liquids, $g = 1$.) The exponent g , in many cases, is a topological quantum number which does not depend on detailed properties of edges. Thus g is a new quantum number that can be used to characterize the topological orders in FQH liquids. Recently a Maryland–IBM group successfully measured the exponent g through the temperature dependence of tunnelling conductance between two edges [34], which was predicted to have the form $\sigma \propto T^{2g-2}$ [35]. This experiment demonstrates the existence of new chiral Luttinger liquids and opens the door to the experimental study of the rich internal and edge structures of FQH liquids.

The edge states of non-Abelian FQH liquids form more exotic 1D correlated systems which are not yet named. These edge states were found to be closely related to conformal field theories at 1 + 1 dimensions [25].

We know that crystal orders can be measured by X-ray diffraction experiments. In the following we would like to indicate that the topological orders in FQH liquids can be measured (in principle) through noise spectrum in an edge transport experiment. Let us first consider a 1D crystal driven through an impurity (figure 1(a)). Because of the crystal order, the voltage across the impurity has a narrow band noise at frequency $f = I/e$ if each unit cell has only one charged particle. More precisely the noise spectrum has a singularity: $S(f) \propto A\delta(f - (I/e))$. If each unit cell contains two charged particles (see figure 1(b)), we shall see an additional narrow band noise at $f = I/2e$: $S(f) \propto B\delta(f - (I/2e)) + A\delta(f - (I/e))$. In this example we see that the noise spectrum allows us to measure crystal orders in one dimension. A similar experiment can also be used to measure topological orders in FQH liquids. Let us consider a FQH sample with a narrow constriction (figure 2). The constriction induces a back scattering through quasiparticle tunnelling between the two edges. The back scattering causes a noise in the voltage across the constriction. In the weak

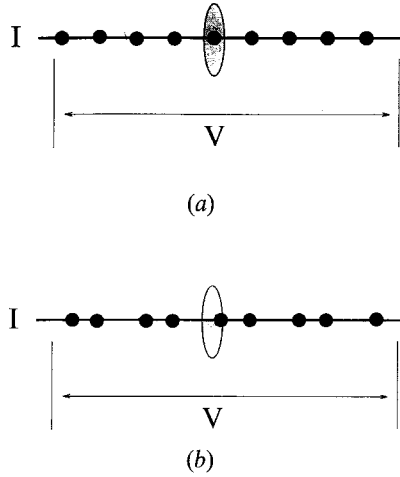


Figure 1. A 1D crystal passing an impurity will generate a narrow band noise in the voltage drop.

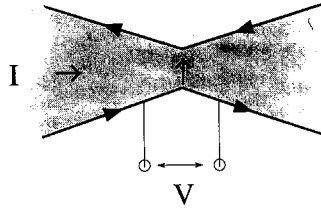


Figure 2. A FQH fluid passing through a constriction will generate band noises owing to the back scattering of the quasiparticles.

back-scattering limit, the noise spectrum contains singularities at certain frequencies which allows us to measure the topological orders in the FQH liquids.[†] To be more specific, the singularities in the noise spectrum have the form

$$S(f) \propto \sum_a C_a |f - f_a|^{\gamma_a}. \quad (1.4)$$

The frequencies and the exponents of the singularities (f_a, γ_a) are determined by the topological orders. For the Abelian state characterized by the matrix $\mathbf{K}$ and charge vector $\mathbf{t}$, the allowed values of the pair (f_a, γ_a) are given by

$$f_a = \frac{I}{ev} \mathbf{t}^T \mathbf{K}^{-1} \mathbf{l}, \quad \gamma_a = 2\mathbf{l}^T \mathbf{K}^{-1} \mathbf{l} - 1, \quad (1.5)$$

where $\mathbf{l}^T = (l_1, l_2, \dots)$ is an arbitrary integer vector, and $v = \mathbf{t}^T \mathbf{K}^{-1} \mathbf{t}$ is the filling fraction. The singularities in the noise spectrum are caused by quasiparticle tunnelling

[†] The discussion presented here applies only to the FQH states whose edge excitations all propagate in the same direction. This requires, for Abelian states, all the eigenvalues of $\mathbf{K}$ to have the same sign.

between the two edges. The frequency f_a of the singularity is determined by the electric charge of the tunnelling quasiparticle Q_q : $f_a = (I/e)(Q_q/ev)$. The exponent γ_a is determined by the statistics of the tunnelling quasiparticle θ_q : $\gamma = 2|\theta_q|/\pi - 1$. Thus the noise spectrum measures the charge and the statistics of the allowed quasiparticles.

1.4. Organization

In section 2 we shall construct the effective theory of FQH liquids, through which we will demonstrate that the Abelian FQH liquids are characterized by $\mathbf{K}$ matrices and charge vectors. In section 3 we shall derive the low-energy effective theory for the gapless edge excitations in FQH liquids from the bulk effective theory and demonstrate the close connection between the bulk topological orders and the structures of edge states. We shall also study some experimental consequence of the edge excitations. In particular we shall discuss the effects of long-range Coulomb interaction, smooth edge potential and impurities on the structures of FQH edges. Sections 4 and 5 contain some further study of topological orders, where we shall introduce spin vectors and discuss classification of the Abelian states. In section 6, we make an attempt to describe non-Abelian FQH states and their edge excitations using an operator algebraic approach.

I have tried to make each section more or less self-contained. Because of this, there are some overlaps between different sections.

2. Effective theory of fractional quantum Hall liquids

One way to understand the topological orders is to construct a low-energy effective theory for FQH liquids. The effective theory should capture the universal properties of FQH liquids and provide hints on how to characterize and label different topological orders in FQH liquids. In the following we shall review a way to construct effective theories that is closely related to the hierarchical construction proposed by Haldane and Halperin [3(a)]. The first effective field theory for the Laughlin states was written in the form of a Ginzburg–Landau theory with a Chern–Simons term [5]. In this section we shall adopt a different form of the effective theory, which contains only pure Chern–Simons terms. The pure Chern–Simons form is more compact and more convenient for studying hierarchical FQH states. The two forms of effective theory is related by duality transformations [3(e), 36].

2.1. Effective theory of the hierarchical fractional quantum Hall states

In this section we shall consider only single-layer spin-polarized quantum Hall (QH) systems. To construct the effective theory for the hierarchical states, we shall start with the effective theory (in the pure Chern–Simons form) of the Laughlin state. Then we shall use the hierarchical construction to obtain the effective theory for the hierarchical states.

First let us review the effective theory for Laughlin state [5, 7, 16,]. Consider a charged boson or fermion system in a magnetic field:

$$\mathcal{L} = -e\mathbf{A} \cdot \mathbf{J} + \text{kinetic energy}, \quad (2.1)$$

where

$$\begin{aligned} \mathbf{J} &= \sum_i \mathbf{v}_i \delta(\mathbf{r} - \mathbf{x}_i) \\ J^0 &= \sum_i \delta(\mathbf{r} - \mathbf{x}_i) \end{aligned} \quad (2.2)$$

are the current and the density of the particles and $(\mathbf{x}_i, \mathbf{v}_i)$ are the position and the velocity of the i th particle. We have also assumed that the charge of each particle is $-e$.

At a filling fraction $\nu = 1/m$, where m is an even integer for boson and an odd integer for fermion, the ground state of the above system is given by the Laughlin wavefunction [2]

$$\left(\prod_i (z_i - z_j)^m \right) \exp \left(-\frac{1}{4} \sum_i |z_i|^2 \right). \quad (2.3)$$

To construct the effective theory of such a state, we note that the (2.3) is an incompressible fluid and that the particle number current J_μ has the following response to a change in electromagnetic fields.[†]

$$-e\delta J_\nu = \sigma_{xy} \varepsilon^{\mu\nu\lambda} \partial_\nu \delta A_\lambda = \frac{\nu e^2}{2\pi} \varepsilon^{\mu\nu\lambda} \partial_\nu \delta A_\lambda \quad (2.4)$$

as a result of finite Hall conductance $\sigma_{xy} = \nu e^2/h$ (in this paper we always choose $\hbar = 1$). In a hydrodynamic approach to the incompressible Hall liquid, we can write the effective theory in terms of the current. We choose the Lagrangian in such a way that it produces the equation of motion (2.4). It is convenient to introduce a $U(1)$ gauge field a_μ to describe the conserved particle number current:

$$J_\mu = \frac{1}{2\pi} \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda}. \quad (2.5)$$

The current defined this way automatically satisfies the conservation law. Then the effective Lagrangian that produces (2.4) takes the following form:

$$\mathcal{L} = -m \frac{1}{4\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} - \frac{e}{2\pi} A_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda}. \quad (2.6)$$

Equation (2.6) describes only the linear response of the ground state to the external electromagnetic fields. To have a more complete description of the topological fluid such as a QH liquid, we need to introduce the boson or fermion excitations in our effective theory.

In the effective theory (2.6), inserting the source term

$$q a_0 \delta(\mathbf{x} - \mathbf{x}_0), \quad (2.7)$$

[†] Here we used the convention that the Greek letters μ, ν represent space-time indices 0, 1, 2 and the italic letters i, j represent spatial indices 1, 2.

which carries an a_μ charge q , will create an excitation of charge $Q = -qe/m$. This can be seen from the equation of motion $\delta\mathcal{L}/\delta a_0 = 0$:

$$j_0 = \frac{1}{2\pi} \varepsilon_{ij} \partial_i a_j = -\frac{e}{2\pi m} B + \frac{q}{m} \delta(\mathbf{x} - \mathbf{x}_0). \quad (2.8)$$

The first term indicates that the filling fraction $\nu \equiv 2\pi j_0/(-eB)$ is indeed $\nu = 1/m$, and the second term corresponds to the increase of the particle density associated with the excitation.

We also see that the excitation created by the source term (2.7) is associated with q/m unit of the a_μ flux. Thus, if we have two excitations carrying a_μ charge q_1 and q_2 , moving one excitation around the other will induce a phase $2\pi \times (\text{number of } a_\mu \text{ flux quanta}) \times a_\mu \text{ charge}$:

$$2\pi \frac{q_1}{m} q_2. \quad (2.9)$$

If $q_1 = q_2 \equiv q$, the two excitations will be identical. Interchanging them will induce half of the phase in equation (2.9):

$$\theta = \pi \frac{q^2}{m}. \quad (2.10)$$

Here θ is simply the statistical angle of the excitation that carries q units of the a_μ charge.

Our bosons or fermions carry a charge of $-e$. From the above discussion, we see that a charge $-e$ excitation can be created by inserting a source term of m units of the a_μ charge. Such an excitation has a statistical angle $\theta = \pi m$ (see equation (2.10)); thus it is a boson if m is even and a fermion if m is odd. Therefore we can identify the excitations of m units of the a_μ charge with the particles (the bosons or the fermions) that form the QH liquid. We would like to stress that the identification of the fundamental particles (the bosons or the fermions) in the effective theory is very important. It is this identification, together with the effective Lagrangian, that provides a complete description of the topological properties of the QH liquid. We shall see below that this identification allows us to determine the fractional charge and the fractional statistics of the quasiparticle excitations.

A quasihole excitation at a position described by a complex number $\xi = x_1 + ix_2$ is created by multiplying $\prod_i (\xi - z_i)$ by the ground-state wavefunction (2.3). Note that the phase of the wavefunction changes by 2π as a boson or a fermion going around the quasihole. Thus the quasihole also behaves like a vortex in the condensate of the bosons or the fermions.

Now let us try to create an excitation by inserting a source term of q units of the a_μ charge. Moving a boson or a fermion around such an excitation will induce a phase $2\pi q$ (see equation (2.9)). The single-valued property of the boson or fermion wavefunction requires such a phase to be multiples of 2π and hence q must be quantized as an integer for allowed excitations. From the charge of the excitations, we find that $q = -1$ corresponds to the fundamental quasihole excitation described above, while $q = 1$ corresponds to the fundamental quasiparticle excitation. The quasiparticle excitation carried an electric charge $-e/m$ and the quasihole e/m . Both have statistics $\theta = \pi/m$, as one can see from equation (2.10). We see that the effective theory reproduces the well known results for quasiparticles in Laughlin states [37].

The full effective theory with quasiparticle excitations is given by

$$\mathcal{L} = \left(-m \frac{1}{4\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} - \frac{e}{2\pi} A_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} \right) + l a_\mu j_\mu + \text{kinetic energy}, \quad (2.11)$$

where j_μ is the current of the quasiparticles which has the form of equation (2.2). For fundamental quasiparticles the integer l in equation (2.11) has $l = 1$, and for fundamental quasiholes $l = -1$. l takes other integer values for composite quasiparticles. Equation (2.11), together with the quantization condition on l , is a complete low-energy effective theory which captures the topological properties of the $1/m$ Laughlin state. It can be shown that the effective theory (2.11) is simply the dual form [7, 36] of the Ginzburg–Landau–Chern–Simons effective theory discovered earlier [5].

Now consider a $1/m$ FQH state formed by electrons, which corresponds to the fermion case discussed above. Let us increase the filling fraction by creating the fundamental quasiparticles, which are labelled $l = 1$. Equation (2.11) with $l = 1$ describes the $1/m$ state in the presence of these quasiparticles. Now, two equivalent pictures emerge.

- (1) In a mean-field theory approach, we may view the gauge field a_μ in equation (2.11) as a fixed background and do not allow a_μ to respond to the inserted source term j_μ . In this case the quasiparticle gas behaves like bosons in the ‘magnetic’ field $b = \partial_i a_j \varepsilon_{ij}$, as one can see from the second term in equation (2.11). These bosons do not carry any electric charge since the quasiparticle number current j_μ does not directly couple with the electromagnetic gauge potential A_μ . When the boson density satisfies

$$j_0 = \frac{1}{p_2} \frac{b}{2\pi}, \quad (2.12 a)$$

where p_2 is even, the bosons have a filling fraction $1/p_2$. The ground state of the bosons can again be described by a Laughlin state. The final electronic state that we obtained is just a second level hierarchical FQH state constructed by Haldane [3 (a)].

- (2) If we let a_μ respond to the insertion j_μ , then quasiparticles will be dressed by the a_μ flux. The dressed quasiparticles carry an electric charge of e/m and a statistics of $\theta = \pi/m$. When the quasiparticles have the density

$$j_0 = \frac{1}{(p_2 + \theta/\pi)} \frac{eB}{2\pi m} \quad (2.12 b)$$

where p_2 is even, the quasiparticle will have a filling fraction $1/(p_2 + \theta/\pi)$. In this case the quasiparticle system can form a Laughlin state described by the wavefunction

$$\prod_{i < j} (z_i - z_j)^{p_2 + \theta/\pi}.$$

The final electronic state obtained this way is again a second-level hierarchical FQH state. This construction was first proposed by Halperin [3 (b)].

The two constrictions in (1) and (2) lead to the same hierarchical state and are equivalent.

In the following we shall follow Haldane’s hierarchical construction to derive the

effective theory of hierarchical FQH states. Note that under the assumption (a), the boson Lagrangian (the second term of equation (2.11) with $l = 1$) is just equation (2.1) with an external electromagnetic field $-eA_\mu$ replaced by a_μ . Thus we can follow the same steps from equation (2.3) to equation (2.11) to construct the effective theory of the boson Laughlin state. Introducing a new $U(1)$ gauge field $\tilde{a}_\mu$ to describe the boson current, we find at the boson effective theory takes the form

$$\mathcal{L} = -\frac{p_2}{4\pi} \tilde{a}_\mu \partial_\nu \tilde{a}_\lambda \varepsilon^{\mu\nu\lambda} + \frac{1}{2\pi} a_\mu \partial_\nu \varepsilon^{\mu\nu\lambda}. \quad (2.13)$$

In equation (2.13) the new gauge field $\tilde{a}_\mu$ describes the density j^0 and the current j^i of the bosons and is given by

$$j^\mu = \frac{1}{2\pi} \partial_\nu \tilde{a}_\lambda \varepsilon^{\mu\nu\lambda}. \quad (2.14)$$

This reduces the coupling between a_μ and the boson current $a_\mu j_\mu$ to a Chern–Simons term between a_μ and $\tilde{a}_\mu$ (which becomes the second term in equation (2.13)). The total effective theory (including the original electron condensate) has the form

$$\mathcal{L} = \left(-\frac{p_1}{4\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} - \frac{e}{2\pi} A_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} \right) + \left(-\frac{p_2}{4\pi} \tilde{a}_\mu \partial_\nu \tilde{a}_\lambda \varepsilon^{\mu\nu\lambda} + \frac{1}{2\pi} a_\mu \partial_\nu \tilde{a}_\lambda \varepsilon^{\mu\nu\lambda} \right), \quad (2.15)$$

where $p_1 = m$ is an odd integer. Equation (2.15) is the effective theory of a second-level hierarchical FQH state.

The second level hierarchical FQH state contains two kinds of quasiparticle. One is a quasihole (or vortex) in the original electron condensate, and the other is a quasihole (or vortex) in the new boson condensate. The two kinds of quasihole are created by inserting the source terms $-j^\mu a_\mu$ and $-\tilde{j}^\mu \tilde{a}_\mu$ respectively, where $\tilde{j}^\mu$ and j^μ have a similar form to equation (2.2). The first kind of quasihole is created by multiplying $\prod_i (\xi - z_i)$ by the electron wavefunction, while the second kind is created by multiplying $\prod_i (\eta - \xi_i)$ by the boson Laughlin wavefunction (here ξ_i are the complex coordinates of the boson and η is the position of the quasihole). The effective theory for quasiholes in the second-level hierarchical states has the form

$$\tilde{j}^\mu \tilde{a}_\mu + j^\mu a_\mu + \text{kinetic energy}. \quad (2.16)$$

We can use the effective theory in equations (2.15) and (2.16) to calculate the quantum numbers of the quasiholes.

The total filling fraction can be determined from the equation of motion $\delta\mathcal{L}/\delta a_0 = \delta\mathcal{L}/\delta \tilde{a}_0 = 0$:

$$-eB = p_1 b - \tilde{b}, \quad b = p_2 \tilde{b}. \quad (2.17)$$

We find that

$$v = \frac{b}{-eB} = \frac{1}{p_1 - (1/p_2)}. \quad (2.18)$$

Equation (2.15) can be written in a more compact form by introducing $(a_{1\mu}, a_{2\mu}) = (a_\mu, \tilde{a}_\mu)$:

$$\mathcal{L} = \sum_{I,I'} \frac{1}{4\pi} K_{II'} a_{I\mu} \partial_\nu a_{I'\lambda} \varepsilon^{\mu\nu\lambda} - \frac{e}{2\pi} A_\mu \partial_\nu t_I a_{I\lambda} \varepsilon^{\mu\nu\lambda}, \quad (2.19)$$

where the matrix $\mathbf{K}$ has integer elements given by

$$\mathbf{K} = \begin{pmatrix} p_1 & -1 \\ -1 & p_2 \end{pmatrix}, \quad (2.20)$$

and $\mathbf{t}^T = (t_1, t_2) = (1, 0)$ will be called the charge vector. The filling fraction (2.18) can be rewritten as $\nu = \mathbf{t}^T \mathbf{K}^{-1} \mathbf{t}$.

A generic quasiparticle is labelled by two integers that consist of l_1 number of quasiparticles of the first kind and l_2 number of quasiparticles of the second kind. Such a quasiparticle carries l_1 units of the $a_{1\mu}$ charge and l_2 units of the $a_{2\mu}$ charge and is described by

$$(l_1 a_{1\mu} + l_2 a_{2\mu}) j^\mu. \quad (2.21)$$

After integrating out the gauge fields, we find that such a quasiparticle carried $\sum_J K_{JJ} l_J$ units of the $a_{1\mu}$ flux. Hence the statistics of such a quasiparticle are given by

$$\theta = \pi \mathbf{l}^T \mathbf{K}^{-1} \mathbf{l} = \frac{1}{p_2 p_1 - 1} (p_2 l_1^2 + p_1 l_2^2 + 2 l_1 l_2), \quad (2.22)$$

and the electric charge of the quasiparticle is

$$Q_q = -e \mathbf{t}^T \mathbf{K}^{-1} \mathbf{l} = -e \frac{p_2 l_1 + l_2}{p_2 p_1 - 1}. \quad (2.23)$$

The above construction can be easily generalized to the third-level hierarchical FQH states with the filling fraction

$$\nu = \frac{1}{p_1 - \frac{1}{(p_2 - (1/p_3))}} \quad (2.24)$$

by allowing the boson (described by current $\tilde{j}$) in equation (2.16) to condense into a $1/p_3$ Laughlin state. The effective theory of this third-level hierarchical state still has the form (2.19), but now I, I' run from 1 to 3. The new $\mathbf{K}$ matrix has the form

$$\mathbf{K} = \begin{pmatrix} p_1 & -1 & 0 \\ -1 & p_2 & -1 \\ 0 & -1 & p_3 \end{pmatrix} \quad (2.25)$$

and the charge vector $\mathbf{t}^T = (1, 0, 0)$. One can go further to obtain the effective theories of the n th-level hierarchical states. In the hierarchical construction, one always assumes that the quasiparticles from the last condensate ‘condense’ to obtain the next-level hierarchical state. Therefore the $\mathbf{K}$ matrix for a hierarchical FQH state has a tridiagonal form

$$K_{II'} = p_I \delta_{I,I'} - \delta_{I,I'-I} - \delta_{I,I'+1} \quad (2.26)$$

with the charge vector given by

$$t_I = \delta_{1I}. \quad (2.27)$$

In equation (2.26), p_1 is odd and $p_i|_{i>1}$ are even. The filling fraction of such a state

is given by [18]

$$v = \mathbf{t}^T \mathbf{K}^{-1} \mathbf{t} = \frac{1}{p_1 - \frac{1}{p_2 - \frac{1}{\cdots - (1/p_n)}}}. \quad (2.28)$$

We can also construct FQH states which are more general than those obtained from the standard hierarchical construction. The effective theory for these more general FQH states still has the form (2.19) but now I runs from unity to an integer n (n will be called the level of the FQH state). To obtain the form of the matrix $\mathbf{K}$, let us assume that at level $n-1$ the effective theory is given by equation (2.19) with $a_{I\mu}$, $I = 1, \dots, n-1$ and $\mathbf{K} = \mathbf{K}^{(n-1)}$. The quasiparticles carry integer charges of the $a_{I\mu}$ gauge fields. (This can always be achieved by properly normalizing the gauge fields.) Now consider an n th-level hierarchical state which is obtained by ‘condensation’ of quasiparticles with the $a_{I\mu}$ charge $l_I|_{I=1, \dots, n-1}$. The effective theory of this n th-level hierarchical state will be given by equation (2.19) with n gauge fields. The n th gauge field $a_{n\mu}$ comes from the new condensate. The matrix $\mathbf{K}$ is given by

$$\mathbf{K}^{(n)} = \begin{pmatrix} \mathbf{K}^{(n-1)} & -\mathbf{l} \\ -\mathbf{l}^T & p_n \end{pmatrix}, \quad (2.29)$$

with p_n even. The charge vector $\mathbf{t}$ is still given by equation (2.27). By iteration, we see that the generalized hierarchical states are always described by integer symmetric matrices, with K_{II} even except for K_{II} odd. The new condensate gives rise to a new kind of quasiparticle which again carries integer charge of the new gauge field $a_{n\mu}$. Hence, a generic quasiparticle always carries integral charges of the $a_{I\mu}$ field. Assume that those charges are l_I . Then the electric charge and the statistics of the quasiparticle are given by the following general formulae

$$\theta = \pi \mathbf{l}^T \mathbf{K}^{-1} \mathbf{l}, \quad Q_q = e \mathbf{t}^T \mathbf{K}^{-1} \mathbf{l}. \quad (2.30)$$

The filling fraction is given by

$$v = \mathbf{t}^T \mathbf{K}^{-1} \mathbf{t}. \quad (2.31)$$

From the above discussion we see that more general hierarchical states, in principle, can be obtained by assuming condensation of different types of quasiparticles. In the standard Haldane–Halperin hierarchical scheme, one assumes the quasiparticles from the last condensate condense to generate the next-level hierarchical states. This choice of condensing quasiparticles is valid if such quasiparticles have the smallest energy gap. However, in principle, one can not exclude the possibility that a different type of quasiparticles (rather than that from the last condensate) might have the smallest energy gap. In this case, we shall obtain a new hierarchical state from the condensation of different quasiparticles. Certainly, for the FQH systems used in real experiments, there are good reasons to believe that the quasiparticles from the last condensate do have the smallest energy gap and it is those quasiparticles that generate the next-level hierarchical states. It would be interesting to find out under which conditions other types of quasiparticle may have smaller energy gaps.

From the generic effective theory in equation (2.19) we obtain one of the important results in this paper; that the generalized hierarchical states can be labelled by an integer valued $\mathbf{K}$ matrix and a charge vector $\mathbf{t}$. Now we would like to ask a question: do different $(\mathbf{K}, \mathbf{t})$ describe different FQH states? Note that, through a

redefinition of the gauge fields $a_{I\mu}$, one can always diagonalize the $\mathbf{K}$ matrix into one with ± 1 as the diagonal elements. Thus it seems that all $\mathbf{K}$ matrices with the same signature describe the same FQH states, since they lead to the same effective theory after a proper redefinition of the gauge fields. Certainly this conclusion is incorrect. We would like to stress that the effective Lagrangian (2.19) alone does not provide a proper description of the topological orders in the hierarchical states. It is the effective Lagrangian (2.19) together with the quantization condition of the $a_{I\mu}$ charges that characterize the topological order. A $U(1)$ gauge theory equipped with a quantization condition on the allowed $U(1)$ charges is called a compact $U(1)$ theory. Our effective theory (2.19) is actually a compact $U(1)$ theory with all $U(1)$ charges quantized as integers. Thus the allowed $U(1)$ charges form an n -dimensional cubic lattice which will be called the charge lattice. Therefore, when considering the equivalence of two different $\mathbf{K}$ matrices, one can use only the field redefinitions that keep the charge quantization condition unchanged (i.e. keep the charge lattice unchanged). The transformations that map the charge lattice onto itself belong to the group $SL(n, Z)$ (a group of integer matrices with a unit determinate):

$$a_{I\mu} \rightarrow W_{IJ} a_{J\mu}, \quad \mathbf{W} \in SL(n, Z). \quad (2.32)$$

From the above discussion we see that the two FQH states described by $(\mathbf{K}_1, \mathbf{t}_1)$ and $(\mathbf{K}_2, \mathbf{t}_2)$ are equivalent (i.e. they belong to the same universality class) if there exists a $\mathbf{W} \in SL(n, Z)$ such that

$$\mathbf{t}_2 = \mathbf{W} \mathbf{t}_1, \quad \mathbf{K}_2 = \mathbf{W} \mathbf{K}_1 \mathbf{W}^T. \quad (2.33)$$

This is because, under equation (2.33), an effective theory described by $(\mathbf{K}_1, \mathbf{t}_1)$ simply changes into another effective theory described by $(\mathbf{K}_2, \mathbf{t}_2)$.

We would like to remark that in the above discussion we have ignored another topological quantum number: spin vector. Because of this, the equivalence condition (2.33) does not apply to clean systems. However, equation (2.33) does apply to disordered FQH systems because the angular momentum is not conserved in disordered systems and the spin vector is not well defined. A more detailed discussion of the spin vectors will be given in section 4.

In the following we list the $\mathbf{K}$ matrices, the charge vectors $\mathbf{t}$ and the spin vectors $\mathbf{s}$ for some common single-layer spin-polarized FQH states:

$$\begin{aligned} v = \frac{1}{m}, \quad \mathbf{t} = (1), \quad \mathbf{K} = (m), \quad \mathbf{s} = \left(\frac{m}{2}\right), \\ v = 1 - \frac{1}{m}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 1 & 1 \\ 1 & -(m-1) \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} \frac{1}{2} \\ \frac{(1-m)}{2} \end{pmatrix}, \\ v = \frac{2}{5}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}, \\ v = \frac{3}{7}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} \frac{1}{2} \\ 1 \\ 1 \end{pmatrix}. \end{aligned} \quad (2.34)$$

2.2. Effective theory of simple multilayer fractional quantum Hall states

The same approach used to construct the effective theory of the hierarchical states can also be used to construct the effective theory for the multilayer FQH states. The connection between the FQH wavefunction and the $\mathbf{K}$ matrix becomes very transparent for the multilayer FQH states. In this section we shall concentrate on double-layer FQH states. However, the generalization to the n -layer FQH states is straightforward.

We would like to construct an effective theory for the following simple double-layer FQH state:

$$\prod_{i < j} (z_{1i} - z_{1j})^l \prod_{i < j} (z_{2i} - z_{2j})^m \prod_{i,j} (z_{1i} - z_{2j})^n \exp \left[-\frac{1}{4} \left(\sum_i |z_{1i}|^2 + \sum_j |z_{2j}|^2 \right) \right], \quad (2.35)$$

where z_{Ii} is the complex coordinate of the i th electron in the I th layer. Here l and m are odd integers so that the wavefunction is consistent with the Fermi statistics of the electrons, while n can be any non-negative integer. The above wavefunction was first suggested by Halperin as a generalization of the Laughlin wavefunction [22]. It appears that these wavefunctions can explain some of the dominant FQH filling fractions observed in double-layer FQH systems.

We start with a single-layer FQH state in the first layer, namely

$$\prod_{i < j} (z_{1i} - z_{1j})^l \exp \left(-\frac{1}{4} \sum_i |z_{1i}|^2 \right), \quad (2.36)$$

which is a $1/l$ Laughlin state and is described by the effective theory

$$\mathcal{L} = -l \frac{1}{4\pi} a_{1\mu} \partial_\nu a_{1\lambda} \epsilon^{\mu\nu\lambda} - \frac{e}{2\pi} A_\mu \partial_\nu a_{1\lambda} \epsilon^{\mu\nu\lambda}, \quad (2.37)$$

where $a_{1\mu}$ is the gauge field that describes the electron density and current in the first layer.

Examining the wavefunction (2.35) we see that an electron in the second layer is bounded to a quasihole excitation in the first layer. Such a quasihole excitation is formed by n fundamental quasihole excitations and carries an $a_{1\mu}$ charge of $-n$. A gas of the quasiholes is described by the following effective theory:

$$\mathcal{L} = -na_{1\mu} j_\mu + \text{kinetic energy}, \quad (2.38)$$

where j_μ has the form (2.2). As we mentioned before, in the mean-field theory, if we ignore the response of the $a_{1\mu}$ field equation (2.38) simply described a gas of bosons in a magnetic field $-nb_1$ where $b_1 = \epsilon_{ij} \partial_i a_{1j}$. Now we would like to attach an electron (in the second layer) to each quasihole in equation (2.38). Such an operation has two effects; firstly the bound state of the quasiholes and the electrons can directly couple with the electromagnetic field A_μ since the electron carries the charge $-e$, and secondly the bound state behaves like a fermion. The effective theory for the bound states has the form

$$\mathcal{L} = (-eA_\mu - na_{1\mu})j_\mu + \text{kinetic energy}, \quad (2.39)$$

which now describes a gas of fermions (in the mean-field theory). These fermions see an effective magnetic field $-eB - nb_1$.

When the electrons (i.e. the fermions in equation (2.39)) in the second layer have

a density $(1/m)[(-eB - nb_1)/2\pi]$ (i.e. have an effective filling fraction $1/m$), they can form a $1/m$ Laughlin state, which corresponds to the $\prod_{i < j} (z_{2i} - z_{2j})^m$ part of the wavefunction. (Note here that $eB < 0$.) Introducing a new gauge field $j_\mu = (1/2\pi) \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda}$ to describe the fermion current j_μ in equation (2.39), the effective theory of the $1/m$ state in the second layer has the form

$$\mathcal{L} = -m \frac{1}{4\pi} a_{2\mu} \partial_\nu a_{2\lambda} \varepsilon^{\mu\nu\lambda}. \quad (2.40)$$

Putting equations (2.37), (2.39) and (2.40) together we obtain the total effective theory of the double layer state which has the form (2.19) with the $\mathbf{K}$ matrix and the charge vector $\mathbf{t}$ given by

$$\mathbf{K} = \begin{pmatrix} l & n \\ n & m \end{pmatrix}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (2.41)$$

We see that the $\mathbf{K}$ matrix is simply the exponents in the wavefunction. The filling fraction of the FQH state is still given by equation (2.31).

There are two kinds of (fundamental) quasihole excitation in the double-layer state. The first kind is created by multiplying $\prod_i (\xi - z_{1i})$ by the ground-state wavefunction and the second kind is created by $\prod_i (\xi - z_{2i})$. As the vortices in the two electron-condensates in the first and the second layers, a first kind of quasihole is created by the source term $-a_{1\mu} j^\mu$, and the second kind of quasihole by $-a_{2\mu} j^\mu$. Thus a generic quasiparticle in the double-layer state is a bound state of several quasiholes of the first and the second kinds and is described by equation (2.21). The quantum numbers of such quasiparticles are still given by equation (2.30).

In general, a multilayer FQH state of type (2.35) is described by a $\mathbf{K}$ matrix whose elements are integers and whose diagonal elements are odd integers. The charge vector has the form $\mathbf{t}^T = (1, 1, \dots, 1)$.

The double-layer FQH state (2.35) is usually labelled (l, m, n) . In the following we list the $\mathbf{K}$ matrices, the charge vectors $\mathbf{t}$ and the spin vectors $\mathbf{s}$ for some simple double-layer states:

$$\begin{aligned} v = \frac{1}{m}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} m & m \\ m & m \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} \frac{m}{2} \\ \frac{m}{2} \end{pmatrix} \\ v = \frac{2}{3}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} \frac{3}{2} \\ \frac{3}{2} \end{pmatrix}, \\ v = \frac{1}{2}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} \frac{3}{2} \\ \frac{3}{2} \end{pmatrix}, \\ v = \frac{2}{3}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 3 & 0 \\ 9 & 3 \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} \frac{3}{2} \\ \frac{3}{2} \end{pmatrix}, \\ v = \frac{2}{3}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{s} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}. \end{aligned} \quad (2.42)$$

From the above we see that the (332) double-layer state has the filling fraction

$\frac{2}{5}$, a filling fraction that also appears in single-layer hierarchical states. Now a question arises: do the double-layer $\frac{2}{5}$ state and the single-layer $\frac{2}{5}$ state belong to the same universality class? This question has experimental consequences. We can imagine the following experiment. We start with a (332) double-layer state in a system with very weak interlayer tunnelling. As we make the interlayer tunnelling stronger and stronger, while keeping the filling fraction fixed, the double-layer state will eventually become a single-layer $\frac{2}{5}$ state. The question is whether the transition between the double-layer (332) state and the single-layer $\frac{2}{5}$ is a smooth cross-over or a phase transition. If we ignore the spin vector, we see that the $\mathbf{K}$ matrices and the charge vectors of the two $\frac{2}{5}$ states are equivalent since they are related by a $SL(2, Z)$ transformation. Therefore, in the presence of disorders (in which case the spin vector is not well defined), the two $\frac{2}{5}$ states can change into each other smoothly. However, the cleaner samples may have a smaller activation gap for the dc conductance in the cross-over region. When we include the spin vector, the two $\frac{2}{5}$ states are not equivalent and, for pure systems, they are separated by a first-order phase transition.

The transition between the above two $\frac{2}{5}$ states should be similar to the transition in the following system. Let us consider an electron system with a strange kinetic energy such that the first Landau level has an energy E_1 and the second level has E_2 . Originally $E_1 < E_2$. Assume that by changing parameter we can make $E_1 > E_2$. This will cause a transition between the $\nu = 1$ state in the first Landau level and the $\nu = 1$ state in the second Landau level. For pure systems, such a transition must be first order since the two $\nu = 1$ states have different spin vectors. For disordered systems the transition can be a smooth cross-over. Certainly, it is always possible that some other states may appear between the two $\nu = 1$ (or $\nu = \frac{2}{5}$) states. The bottom line is that the two $\nu = 1$ states cannot be smoothly connected in a clean sample.

From equation (2.42) we also see that there are two different $\frac{2}{3}$ double-layer states. When the intralayer interaction is much stronger than the interlayer interaction (a situation in real samples) the ground-state wavefunction prefers to have higher-order zeros between electrons in the same layers. Thus, the (330) state should have a lower energy than the (112) state. The $\mathbf{K}$ matrix and the charge vector of the single-layer $\frac{2}{3}$ state are equivalent to those of the (112) state and are not equivalent to those of the (330) state. Thus, to change a double-layer $\frac{2}{3}$ state (i.e. the (330) state) into the single-layer $\frac{2}{3}$ state, one must go through a phase transition for both random and pure systems.

After replacing the layer index by the S_z spin index, all the results obtained in this section can be directly applied to describe spin unpolarized FQH states of spin- $\frac{1}{2}$ electron systems. In particular, one can show that the $\nu = \frac{2}{5}$ and the second $\nu = \frac{2}{3}$ states in equation (2.42) describe two spin singlet FQH states [38]. Owing to the different spin vectors, both states are inequivalent to their spin-polarized counterpart: the $\nu = \frac{2}{5}$ and the $\nu = \frac{2}{3}$ FQH states in equation (2.34). Thus for pure systems, the above spin singlet FQH states and the spin-polarized FQH states are separated by first-order transitions. However, for dirty systems, the first-order transition may be smeared into a smooth cross-over.

The double-layer (mmm) state is an interesting state since $\det(\mathbf{K}) = 0$. It turns out that the (mmm) state (in the absence of interlayer tunnelling) spontaneously breaks $U(1)$ symmetry and contains a superfluid mode [39]. More detailed discussions (including their experimental implications) can be found in [39, 40].

3. Edge excitations in Abelian fractional quantum Hall liquids

Owing to the repulsive interaction and strong correlation between the electrons, a FQH liquid is an incompressible state despite the fact that the first Landau level is only partially filled. All the bulk excitations in FQH states have finite energy gaps. FQH states and insulators are very similar in the sense that both states have finite energy gaps and short-ranged electron propagators. Because of this similarity it was puzzling that FQH systems apparently have very different transport properties from ordinary insulators. Halperin [29] first pointed out that the IQH states contain gapless edge excitations. Although the electronic states in the bulk are localized, the electronic states at the edge of the sample are extended (i.e. the electron propagator along the edge is long ranged) [41]. Therefore the non-trivial transport properties of the IQH states come from the gapless edge excitations [29, 42], for example a two-probe measurement of a QH sample can result in a finite resistance only when the source and the drain are connected by the edges. If the source and the drain are not connected by any edge, the two-probe measurement will yield an infinite resistance at zero temperature, a result very similar to the insulators. The edge transport picture has been supported by many experiments [43]. Halperin also studied the dynamical properties of the edge excitations of the IQH states and found that the edge excitations are described by a chiral 1D Fermi liquid theory.

Using the gauge argument in [29, 44, 45], one can easily show that FQH states also support gapless edge excitations. Thus it is natural to conjecture that the transport in FQH states is also governed by the edge excitations [46, 47]. However, since FQH states are intrinsically many-body states, the edge excitations in the FQH states cannot be constructed from a single-body theory. In other words, the edge excitations of FQH states should not be described by a Fermi liquid. In this case we need completely new approaches to understand the dynamical properties of the edge state of FQH liquids. Recent advances in fabrication of small devices make it possible to study in detail the dynamical properties of the edge states in FQH liquids. Thus it is important to develop a quantitative theory for FQH edge states to explain new experimental data.

There is another motivation to study the FQH states. We know that different FQH states were generally labelled by their filling fractions. However, it becomes clear that FQH states contain such extremely rich internal structures that the filling fraction alone is not sufficient to classify all the different universality classes of FQH states [6, 7]. One can easily construct different FQH states with the same filling fraction [3, 17–20]. From the last section we see that (generalized) hierarchical states and simple double-layer states can be labelled by $\mathbf{K}$ matrices and charge vectors $\mathbf{t}$. However, in the last section, $\mathbf{K}$ and $\mathbf{t}$ merely appear as parameters in some theoretical effective theory. An important question is whether one can find experiments that measure $\mathbf{K}$ and $\mathbf{t}$. Certainly a combination of $\mathbf{K}$ and $\mathbf{t}$ can be measured through the filling fraction, and the determinant of $\mathbf{K}$ can be measured through the ground-state degeneracy of the FQH liquid on a torus [7, 13]. Experimentally, however one can never put a QH state on a torus. Thus the results in [6, 7, 13] can be checked only in numerical calculations. In the following we shall see that edge excitations in FQH states provide an important (probably the only practical) probe to detect the topological orders in the bulk FQH states. Through tunnelling experiments between FQH edges, one can measure many different combinations of $\mathbf{K}$ and $\mathbf{t}$. Using the edge excitations we also can tell whether a FQH state is an Abelian FQH state or a non-Abelian FQH state. Thus the edge states provide us with a practical window

through which we can look into the internal structures in FQH states. The measurements of the edge states can provide new quantum numbers, in addition to the filling fractions, to characterize different quantum Hall states.

3.1. Hydrodynamical approach to the edge excitations: $1/m$ Laughlin state

The simplest (but not complete) way to understand the dynamics of edge excitations is to use the hydrodynamical approach. In this approach, one uses the fact that QH (IQH or FQH) states are incompressible irrotational liquids that contain no low-energy bulk excitations. Therefore the only low-lying excitations (below the bulk energy gap) are surface waves on a HQ droplet. These surface waves are identified as edge excitations of the QH state [31, 48, 49].

In the hydrodynamical approach we first study the classical theory of the surface wave on the QH droplet. Then we quantize the classical theory to obtain the quantum description of the edge excitations. It is amazing that the simple quantum description obtained from the classical theory provides a complete description of the edge excitations at low energies and allows us to calculate the electron and the quasiparticle propagators along the edges.

Consider a QH droplet with a filling fraction ν confined by a potential well. Owing to the non-zero conductance, the electric field of the potential well generates a persistent current flowing along the edge:

$$\mathbf{j} = \sigma_{xy} \hat{\mathbf{z}} \times \mathbf{E}, \quad \sigma_{xy} = \nu \frac{e^2}{h}. \quad (3.1)$$

This implies that the electrons near the edge drift with a velocity

$$v = \frac{E}{B} c, \quad (3.2)$$

where c is the velocity of the light. Thus the edge wave also propagates with the velocity v . Let us use the 1D density $\rho(x) = nh(x)$ to describe the edge wave, where $h(x)$ is the displacement of the edge, x is the coordinate along the edge, and $n = \nu/(2\pi l_B^2)$ is the 2D electron density in the bulk. We see that the propagation of the edge waves are described by the following wave equation:

$$\partial_t \rho - v \partial_x \rho = 0. \quad (3.3)$$

Note that the edge waves always propagate in one direction; there are no waves that propagate in the opposite direction.

The Hamiltonian (i.e. the energy) of the edge waves is given by

$$H = \int dx \frac{1}{2} e h \rho E = \int dx \pi \frac{v}{\nu} \rho^2. \quad (3.4)$$

In momentum space, equations (3.3) and (3.4) can be rewritten as

$$\begin{aligned} \dot{\rho}_k &= i v k \rho_k \\ H &= 2\pi \frac{v}{\nu} \sum_{k>0} \rho_k \rho_{-k}, \end{aligned} \quad (3.5)$$

where $\rho_k = \int dx (1/L^{1/2}) \exp(ikx) \rho(x)$, and L is the length of the edge. Comparing

equation (3.5) with the standard Hamiltonian equation

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}, \quad (3.6)$$

we find that, if we identify $\rho_k|_{k>0}$ as the ‘coordinates’, then the corresponding canonical ‘momenta’ can be identified as $p_k = i2\pi\rho_{-k}/vk$. We would like to stress that, because the edge waves are chiral, the displacement $h(x)$ contains both the ‘coordinates’ and the ‘momenta’.

Knowing the canonical coordinates and momenta, it is easy to quantize the classical theory. We simply view ρ_k and p_k as operators that satisfy $[p_k, \rho_{k'}] = i\delta_{kk'}$. Thus after quantization we have

$$[\rho_k, \rho_{k'}] = \frac{v}{2\pi} k\delta_{k+k'},$$

$$k, k' = \text{integer} \times \frac{2\pi}{L}, \quad (3.7)$$

$$[H, \rho_k] = v\rho_k.$$

The above algebra is called the $(U(1))$ Kac–Moody (K–M) algebra (for example [50]). A similar algebra has also appeared in the Tomonaga [51] model. Note that equation (3.7) simply describes a collection of decoupled harmonic oscillators (generated by (ρ_k, ρ_{-k})). Thus equation (3.7) is a 1D free-phonon theory (with only a single branch) and is exactly soluble. We shall show later that equation (3.7) provides a complete description of the low-lying edge excitations of the QH state.

To summarize, we find that the edge excitations in the QH states are described by a free-(chiral)-phonon theory at low energies. Not only do we show the existence of the gapless edge excitations, but also we obtain the density of states of the edge excitations. The specific heat (per unit length) of the edge excitations is found to be $(\pi/6)(T/v)$. The edge excitations considered here do not change the total charge of the system and hence are neutral. In the following, we shall discuss the charged excitations and calculate the electron propagator from the K–M algebra (3.7).

The low-lying charge excitations obviously correspond to adding (removing) electrons to (from) the edge. Those charged excitations carry integer charges and are created by electron operators $\Psi^\dagger$. The above theory of edge excitations is formulated in terms of a 1D density operator $\rho(x)$. So the central question is to write the electron operator in terms of the density operator. The electron operator on the edge creates a localized charge and should satisfy

$$[\rho(x), \Psi^\dagger(x')] = \delta(x - x')\Psi^\dagger(x'). \quad (3.8)$$

Since ρ satisfies the K–M algebra (3.7), one can show that the operators that satisfy (3.8) are given by [30]

$$\Psi \propto \exp\left(i \frac{1}{v} \phi\right) \quad (3.9)$$

where ϕ is given by $\rho = (1/2\pi) \partial_x \phi$.

Equation (3.8) implies only that the operator Ψ carries the charge e . In order to identify Ψ as an electron operator we need to show that Ψ is a fermionic operator.

Using the K–M algebra (3.7), we find that

$$\Psi(x)\Psi(x') = (-)^{1/\nu}\Psi(x')\Psi(x). \quad (3.10)$$

We see that the electron operator Ψ in equation (3.9) is fermionic only when $1/\nu = m$ is an odd integer, in which case the QH state is a Laughlin state [30, 52].

In the above discussion we have made an assumption that is not generally true. We have assumed that the incompressible QH liquid contains only *one* component of incompressible fluid which leads to one branch of edge excitations. The above result implies that, when $\nu \neq 1/m$, the edge theory with only one branch does not contain the electron operators and is not self-consistent. Therefore we conclude that the FQH states with $\nu \neq 1/m$ must contain more than one branch of edge excitation. (Here we have ignored the possibility of the pairing between the electrons.) Later (in section 6.1) we shall see that the one-branch assumption is true only for a simple Laughlin state with filling fraction $\nu = 1/m$. For hierarchical FQH states, there are several condensates corresponding to several components of incompressible fluid. Each component gives rise to a branch of the edge excitations. Thus a generic QH state may contain many branches of the edge excitations [32, 47] even when electrons are all in the first Landau level (see sections 3.2 and 3.3).

Now let us calculate electron propagator along the edge of the Laughlin states with $\nu = 1/m$. In this case the above simple edge theory is valid. Because ϕ is a free phonon field with a propagator

$$\langle \phi(x, t)\phi(0) \rangle = -\nu \ln(x - vt) + \text{constant}, \quad (3.11)$$

the electron propagator can be calculated as [30]

$$G(x, t) = \langle T(\Psi^\dagger(x, t)\Psi(0)) \rangle = \exp\left(\frac{1}{v^2} \langle \phi(x, t)\phi(0) \rangle\right) \propto \frac{1}{(x - vt)^m}. \quad (3.12)$$

The first thing that we see is that the electron propagator on the edge of a FQH state acquires a non-trivial exponent $m = 1/\nu$ that is not equal to one. This implies that the electrons on the edge of the FQH state are strongly correlated and cannot be described by Fermi-liquid theory. We shall call this type of an electron state a chiral Luttinger liquid.

The K–M algebra (3.7) and the electron operator (3.9) provided a complete description of both neutral and charged edge excitations at low energies. We would like to remark that the propagator (3.12) is correct only for large x and t . At a short distance the form of the propagator depends on the details of the electron interactions and the edge potentials. We would also like to emphasize that the exponent m of the edge propagator is determined by the bulk state. Such an exponent is a topological number that is independent of electron interactions, edge potential, etc. The quantization of the exponent is directly related to the fact that the exponent is linked to the statistics of the electrons (see equation (3.10)). Thus the exponent can be regarded as a quantum number that characterizes the topological orders in the bulk FQH states.

In the momentum space the electron propagator has the form

$$G(k, \omega) \propto \frac{(vk + \omega)^{m-1}}{vk - \omega}. \quad (3.13)$$

The anomalous exponent m can be measured in tunnelling experiments. The

tunnelling density of states of electron is given by

$$N(\omega) \propto |\omega|^{m-1}. \quad (3.14)$$

This implies that differential conductance has the form $dI/dV \propto V^{m-1}$ for a metal-insulator-FQH junction.

3.2. Hydrodynamical approach to the edge excitations: $\frac{2}{5}$ and $\frac{2}{3}$ states

In this section we shall use the hydrodynamical approach discussed above to study edge structures of second-level hierarchical states. We shall concentrate on the $\frac{2}{5}$ and $\frac{2}{3}$ states as examples. In particular we shall study the structures of the electron and the quasiparticle operators on the edges of the hierarchical states. We shall also see that the $\frac{2}{3}$ state contains two edge modes that propagate in the opposite directions, which is quite counter-intuitive.

First let us consider the $\nu = \frac{2}{5}$ FQH state. According to the hierarchical theory, the $\nu = \frac{2}{5}$ FQH state is generated by the condensation of quasiparticles on top of the $\nu = \frac{1}{3}$ FQH state. Thus the $\frac{2}{5}$ state contains two components of incompressible fluids. To be definite let us consider a special edge potential such that the FQH state consists of two droplets: one is the electron condensate with a filling fraction $\frac{1}{3}$ and radius r_1 , and the other is the quasiparticle condensate (on top of the $\frac{1}{3}$ state) with a filling fraction $\frac{1}{15}$ (note that $\frac{1}{3} + \frac{1}{15} = \frac{2}{5}$) and radius $r_2 < r_1$.

When $r_1 - r_2 > l_B$, the two edges are independent. Generalizing the hydrodynamical approach in section 3.1, we can show that there are two branches of the edge excitations whose low-energy dynamics are described by

$$[\rho_{Ik}, \rho_{Jk}] = \frac{v_I}{2\pi} k \delta_{IJ} \delta_{k+k'}, \quad (3.15)$$

$$H = 2\pi \sum_{I, k > 0} \frac{v_I}{v_I} \rho_{Ik} \rho_{I-k},$$

where $I = 1, 2$ labels the two branches, $(\nu_1, \nu_2) = (\frac{1}{3}, \frac{1}{15})$ are filling fractions of the electron condensate and the quasiparticle condensate, and v_I are the velocities of the edge excitations. ρ_I in equation (3.15) are the 1D electron densities given by $\rho_I = h_I v_I (1/2\pi l_B^2)$ where h_I are the amplitudes of the edge waves on the two droplets.

Because the electrons interact with each other, the edge velocities are determined by $v_I = cE_I^*/B$ where E_I^* are the effective electric fields that include contributions from both the edge potential and the electrons. In order for the Hamiltonian to be bounded from below, we require that $v_I v_I > 0$. We find that the stability of the $\nu = \frac{2}{5}$ FQH state requires both v_I to be positive.

Generalizing the discussion in section 3.1, the electron operators on the two edges are found to be

$$\Psi_I = \exp\left(i \frac{1}{v_I} \phi_I(x)\right), \quad I = 1, 2, \quad (3.16)$$

with $\partial_x \phi_I = (1/2\pi) \rho_I$. The electron propagators have the form

$$\langle T(\Psi_I(x, t) \Psi_I^\dagger(0)) \rangle = \exp(ik_I x) \frac{1}{(x - v_I t)^{-1/|v_I|}}, \quad I = 1, 2, \quad (3.17)$$

where $k_I = r_I/2l_B^2$.

According to the hierarchical picture, the $\nu = \frac{2}{3}$ FQH state is also formed by two condensates, an electron condensate with a filling fraction 1 and a hole condensate with a filling fraction $-\frac{1}{3}$. Thus the above discussion can also be applied to the $\nu = \frac{2}{3}$ FQH state by choosing $(v_1, v_2) = (1, -\frac{1}{3})$. Again there are two branches of the edge excitations but now with *opposite* velocities if the Hamiltonian is positive definite. This result, although surprising, is not difficult to understand. The stability of both the electron droplet and the hole droplet requires E_1^* and E_2^* to have opposite signs.

As we bring the two edges together ($r_1 - r_2 \sim l_B$) the interaction between the two branches of the edge excitations can no longer be ignored. In this case the Hamiltonian has the form

$$H = 2\pi \sum_{I, J, k > 0} V_{IJ} \rho_{Ik} \rho_{J-k}. \quad (3.18)$$

(The Hamiltonian may also contain terms that describe the electron hopping between edges, but those terms are irrelevant at low energies owing to the chiral property of the edge excitations. For example, one can show that those terms can never open an energy gap [45].)

The Hamiltonian (3.18) can still be diagonalized. For $v_1 v_2 > 0$, we may choose

$$\begin{aligned} \tilde{\rho}_{1k} &= \cos \theta \frac{1}{|v_1|^{1/2}} \rho_{1k} + \sin \theta \frac{1}{|v_2|^{1/2}} \rho_{2k}, \\ \tilde{\rho}_{2k} &= \cos \theta \frac{1}{|v_2|^{1/2}} \rho_{2k} - \sin \theta \frac{1}{|v_1|^{1/2}} \rho_{1k}, \\ \tan(2\theta) &= 2 \frac{|v_1 v_2|^{1/2} V_{12}}{|v_1| V_{11} - |v_2| V_{22}}. \end{aligned} \quad (3.19)$$

One can check that the $\tilde{\rho}$ values satisfy

$$[\tilde{\rho}_{Ik}, \tilde{\rho}_{Jk'}] = \frac{\text{sgn}(v_I)}{2\pi} k \delta_{IJ} \delta_{k+k'}, \quad (3.20)$$

$$H = 2\pi \sum_{I, k > 0} \text{sgn}(v_I) \tilde{v}_I \tilde{\rho}_{Ik} \tilde{\rho}_{I-k},$$

where the new velocities $\tilde{v}_I$ of the edge excitations are given by

$$\begin{aligned} \text{sgn}(v_1) \tilde{v}_1 &= \frac{\cos^2 \theta}{\cos 2\theta} |v_1| V_{11} - \frac{\sin^2 \theta}{\cos 2\theta} |v_2| V_{22}, \\ \text{sgn}(v_2) \tilde{v}_2 &= \frac{\cos^2 \theta}{\cos 2\theta} |v_2| V_{22} - \frac{\sin^2 \theta}{\cos 2\theta} |v_1| V_{11}. \end{aligned} \quad (3.21)$$

We see that there are still two branches of the edge excitations. However, in this case the edge excitations with a definite velocity are mixtures of those on the inner edge and the outer edge. One can also show that, as long as the Hamiltonian (3.18) is bounded from below, the velocities of the two branches $\tilde{v}_I$ are always positive. This result has been confirmed by numerical calculations [53].

After rewriting the electron operator Ψ_I in terms of $\tilde{\rho}_I$ by inverting equation

(3.19), we can calculate their propagators using equation (3.21):

$$\langle T(\Psi_I(x, t)\Psi_I^\dagger(0)) \rangle = \exp(ik_I x) \frac{1}{(x - \tilde{v}_1 t)^{\alpha_I}} \frac{1}{(x - \tilde{v}_2 t)^{\beta_I}}, \quad (3.22)$$

where

$$(\alpha_1, \alpha_2) = \left(\frac{1}{|v_1|} \cos^2 \theta, \frac{1}{|v_2|} \sin^2 \theta \right), \quad (\beta_1, \beta_2) = \left(\frac{1}{|v_1|} \sin^2 \theta, \frac{1}{|v_2|} \cos^2 \theta \right). \quad (3.23)$$

However, when the two edges are close to each other within the magnetic length, the Ψ_I are no longer the most general electron operators on the edge. The generic electron operator may contain charge transfers between the two edges. For the $\nu = \frac{2}{5}$ FQH state, the inner edge and outer edge are separated by the $\nu = \frac{1}{3}$ Laughlin state. Thus the elementary charge transfer operator is given by

$$\eta(x) = \exp \left[i \left(\phi_1 - \frac{v_1}{v_2} \sigma_2 \right) \right] = (\Psi_1 \Psi_2^\dagger)^{v_1}, \quad (3.24)$$

which transfers a $v_1 e = e/3$ charge from the outer edge to the inner edge. The generic electron operator then takes the form

$$\Psi(x) = \sum_{n=-\infty}^{+\infty} c_n \psi_n(x), \quad (3.25)$$

$$\psi_n(x) = \Psi_1(x) \eta^n(x).$$

To understand this result, we note that each operator ψ_n always creates a unit localized charge and is a fermionic operator regardless of the value of the integer n . Therefore each ψ_n is a candidate for the electron operator on the edge. For a generic interacting system the electron operator on the edge is expected to be a superposition of different ψ_n as represented in equation (3.25). Note that $\Psi_2 = \psi_{-1/v_1}$. The propagator of ψ_n can be calculated in a similar way as outlined above and is given by

$$\langle T(\psi_n(x, t)\psi_m^\dagger(0)) \rangle \propto \delta_{n,m} \exp \{ i[k_1 + nv_1(k_2 - k_1)]x \} \prod_I (x - \tilde{v}_I t)^{-\gamma_{In}}, \quad (3.26)$$

where γ_{In} are

$$\begin{aligned} \gamma_{1n} &= \left[\left(n + \frac{1}{|v_1|} \right) |v_1|^{1/2} \cos \theta - \frac{nv_1}{v_2} |v_2|^{1/2} \sin \theta \right]^2, \\ \gamma_{2n} &= \left[\left(n + \frac{1}{|v_1|} \right) |v_1|^{1/2} \sin \theta + \frac{nv_1}{v_2} |v_2|^{1/2} \cos \theta \right]^2. \end{aligned} \quad (3.27)$$

From equations (3.25) and (3.26) we see that the electron propagator has singularities at discrete momenta $k = k_1 + nv_1(k_2 - k_1)$. They are analogues to the $k_F, 3k_F, \dots$ singularities of the electron propagator in the interacting 1D electron systems.

For the $\nu = \frac{2}{3}$ FQH state, $v_1 v_2 < 0$. In this case we need to choose

$$\begin{aligned} \tilde{\rho}_{1k} &= \cosh \theta \frac{1}{|v_1|^{1/2}} \rho_{1k} + \sinh \theta \frac{1}{|v_2|^{1/2}} \rho_{2k}, \\ \tilde{\rho}_{2k} &= \cosh \theta \frac{1}{|v_2|^{1/2}} \rho_{2k} + \sinh \theta \frac{1}{|v_1|^{1/2}} \rho_{1k}, \\ \tanh(2\theta) &= 2 \frac{|v_1 v_2|^{1/2} V_{12}}{|v_1| V_{11} + |v_2| V_{22}}, \end{aligned} \quad (3.28)$$

to diagonalize the Hamiltonian. One can check that $\tilde{\rho}_I$ also satisfies the K–M algebra (3.20) but now

$$\begin{aligned}\operatorname{sgn}(v_1)\tilde{v}_1 &= \frac{\cosh^2 \theta}{\cosh(2\theta)}|v_1|V_{11} - \frac{\sinh^2 \theta}{\cosh(2\theta)}|v_2|V_{22}, \\ \operatorname{sgn}(v_2)\tilde{v} &= \frac{\cosh^2 \theta}{\cosh(2\theta)}|v_2|V_{22} - \frac{\sinh^2 \theta}{\cosh(2\theta)}|v_1|V_{11}.\end{aligned}\quad (3.29)$$

Again as long as the Hamiltonian H is positive definite, the velocities $\tilde{v}_I$ of the edge excitations always have opposite signs. The electron operator still has the form (3.25) with $\eta = (\Psi_1 \Psi_2^\dagger)^{v_1}$. The propagator of ψ_n is still given by equation (3.26) with

$$\begin{aligned}\gamma_{1n} &= \left[\left(n + \frac{1}{|v_1|} \right) |v_1|^{1/2} \cosh \theta + \frac{nv_1}{v_2} |v_2|^{1/2} \sinh \theta \right]^2, \\ \gamma_{2n} &= \left[\left(n + \frac{1}{|v_1|} \right) |v_1|^{1/2} \sinh \theta + \frac{nv_1}{v_2} |v_2|^{1/2} \cosh \theta \right]^2.\end{aligned}\quad (3.30)$$

From equations (3.27) and (3.30), we see that exponents in the electron propagator depend on the interedge interactions. However, we can show that the exponents satisfy a sum rule:

$$\sum_I \operatorname{sgn}(v_I) \gamma_{In} \equiv \lambda_n = \left(n + \frac{1}{|v_1|} \right)^2 v_1 + \frac{n^2 v_1^2}{v_2}. \quad (3.31)$$

λ_n always take odd-integer values and are independent of the details of the electron system. The quantization of λ_n again arises because the λ_n are directly linked to the statistics of the electrons:

$$\psi_n(x) \psi_n(x') = (-)^{\lambda_n} \psi_n(x') \psi_n(x). \quad (3.32)$$

We would like to point out that the values of λ_n are determined by the internal correlations (topological orders) of the bulk FQH state. λ_n can be changed only by changing the bulk topological orders through a 2D phase transition, despite the fact that they are properties of 1D edge excitations. Therefore λ_n are topological quantum numbers that can be used to characterize and to experimentally measure the 2D bulk topological orders.

The total exponents $g_e^{(n)}$ of the electron operators

$$g_e^{(n)} = \sum_I \gamma_{In} \quad (3.33)$$

are very important quantities. For the $\frac{2}{5}$ state, the minimum value of the exponents

$$g_e \equiv \min(g_e^{(n)}) = g_e^{(0)} = g_e^{(-1)} = 3 \quad (3.34)$$

controls the scaling properties of tunnelling of electrons between two edges. For example the tunnelling conductance scales at finite temperatures as [35]

$$\sigma \propto T^{2g_e-2}. \quad (3.35)$$

For the $\frac{2}{5}$ state the sum rule (3.31) implies that $g_e^{(n)}$ and g_e are topological quantum numbers that are independent of details of electron interactions and edge potential. However, for the $\frac{2}{3}$ state the sum rule cannot fix the value of g_e , and g_e is not universal. Later in section 3.5 we shall see that, in the presence of long-range Coulomb

interaction and/or in the presence of edge impurity, g_e for the $\frac{2}{3}$ state will take a universal value $g_e = 2$ [54].

3.3. Bulk effective theory and the edge states

In this section we shall directly derive the macroscopic theory of the edge excitations from the effective theory of the bulk FQH states [18, 20, 52]. In this approach we do not rely on a specific construction of the FQH states. The relation between the bulk topological orders and edge becomes quite clear.

We know that a hierarchical (or generalized hierarchical) FQH state contains many different condensates, that an electron condensate forms Laughlin states, and that additional quasiparticle condensates on top of that give rise to a hierarchical state. Each condensate corresponds to one component of the incompressible fluid. The idea is to generalize the hydrodynamical approach in section 3 to multi-component fluids and to obtain the low-energy effective theory of the edge excitations. To accomplish this, we would first like to write down the low-energy effective theory of the bulk FQH state. The effective theory contains information about the internal structures (the topological orders) in the bulk states, such as the number of the condensates and how different condensates interact with each other.

The different condensates in FQH states are not independent. The particles in one condensate behave like a flux tube to the particles in other condensates. To describe such a coupling, it is convenient to use $U(1)$ gauge fields to describe the density and the current of each condensate. In this case the couplings between different condensates are described by the Chern–Simons term of the gauge fields (see section 2) [18–20]. The most general Abelian FQH states of the electrons are classified by integer valued symmetry matrices $\mathbf{K}$ and described by the following effective theory:

$$\mathcal{L} = -\frac{1}{4\pi} K_{IJ} a_{I\mu} \partial_\nu a_{J\lambda} \varepsilon^{\mu\nu\lambda} - \frac{e}{2\pi} t_I A_\mu \partial_\nu a_{I\lambda} \varepsilon^{\mu\nu\lambda}. \quad (3.36)$$

In the following we shall choose the so-called symmetric basis with the charge vector $\mathbf{t}^T = (1, 1, \dots, 1)$. In this case the diagonal elements of $\mathbf{K}$ must be odd integers. We can always change the hierarchical basis with $\mathbf{t} = (1, 0, \dots, 0)$ to the symmetric basis with $\mathbf{t} = (1, 1, \dots, 1)$ by a field-redefinition of the $U(1)$ gauge fields. Let $\kappa = \dim(\mathbf{K})$. Then the FQH state described by equation (3.36) contains κ different condensates and there are κ kinds of different quasiparticle excitations.

The quasiparticle excitations can be viewed as vortices in different condensates. A generic quasiparticle is labelled by κ integers $l_I |_{I=1, \dots, \kappa}$, and can be generated by a source term

$$\mathcal{L} = l_I a_{I\mu} j^\mu. \quad (3.37)$$

Such a quasiparticle will be denoted ψ_I . j_μ in equation (3.37) has the form

$$\begin{aligned} \mathbf{j}(x) &= \mathbf{v} \delta(\mathbf{x} - \mathbf{x}_0), \\ j^0(x) &= \delta(\mathbf{x} - \mathbf{x}_0), \end{aligned} \quad (3.38)$$

which create a quasiparticle at $\mathbf{x}_0$ with velocity $\mathbf{v}$. The density and the current of the I th component of incompressible fluid (i.e. the I th condensate) are given by

$$J_{I\mu} = \frac{1}{2\pi} \varepsilon_{\mu\alpha\beta} \partial_\alpha a_{I\beta}. \quad (3.39)$$

As we create a quasiparticle ψ_1 , it will induce a change in the density of all the condensates, δJ_{I0} . From the equations of motion (3.36) and (3.37), we find that δJ_{I0} satisfies

$$\int d^2x \delta J_{I0} = l_J (K^{-1})_{JI} \int d^2x j_0 = l_J (K^{-1})_{JI}. \quad (3.40)$$

The charge and the statistics of the quasiparticle ψ_1 are given by

$$\theta_1 = \pi \mathbf{l}^T K^{-1} \mathbf{l}, \quad Q_1 = -e t_I \int d^2x \delta J_{I0} = -e \mathbf{l}^T K^{-1} \mathbf{t}. \quad (3.41)$$

A generic electron excitation can be viewed as a special kind of (generic) quasiparticle, $\psi_e = \psi_{\mathbf{l}_e}$, where the components of $\mathbf{l}_e$ are given by

$$l_{eI} = K_{IJ} L_J, \quad L_I = \text{integers}, \quad \sum_I L_I = 1. \quad (3.42)$$

We can show that these electron excitations satisfy the following properties.

- (a) They carry a unit charge (see equation (3.41)).
- (b) They have the fermionic statistics.
- (c) Moving an electron excitation $\psi_e = \psi_{\mathbf{l}_e}$ around any quasiparticle excitations, ψ_1 always induces a phase of multiple of 2π .
- (d) The excitations defined in equation (3.42) are all the excitations satisfying the above three conditions, (a)–(c).

We would like to point out that the effective theory (3.36) applies not only to the standard QH system in which all electrons are spin polarized and are in the first Landau level but also to the QH system in which electrons may occupy several Landau levels and/or occupy several layers and/or carry different spins. In this case the index I may label the condensates in different Landau levels, in different layers and/or with different spins.

To understand the relation between the effective theory and the edge states, let us first consider the simplest FQH state of the filling fraction $\nu = 1/q$ and try to re-derive the results in section 3.1 from the bulk effective theory. Such a FQH state is described by the $U(1)$ Chern–Simons theory with the action [7, 52].

$$S = \frac{q}{4\pi} \int a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} d^3x. \quad (3.43)$$

Suppose that our sample has a boundary. For simplicity we shall assume that the boundary is the x axis and the sample covers the lower half-plane.

There is one problem with the effective action (3.43) for FQH liquids with boundaries. It is not invariant under gauge transformations $a_\mu \rightarrow a_\mu + \partial_\mu f$ owing to the presence of the boundary: $\Delta S = (q/4\pi) \int_{y=0} dx dt f(\partial_0 a_1 - \partial_1 a_0)$. To solve this problem we shall restrict the gauge transformations to be zero on the boundary $f(x, y=0, t) = 0$. Owing to this restriction some degrees of freedom of a_μ on the boundary become dynamical.

We know that the effective theory (3.43) is derived only for a bulk FQH state without boundary. Here we shall take equation (3.43) with the restricted gauge transformation as the definition of the effective theory for a FQH state with boundary.

Such a definition is definitely self-consistent. In the following we shall show that such a definition reproduces the results obtained in section 3.1.

One way to study the dynamics of gauge theory is to choose the gauge condition $a_0 = 0$ and to regard the equation of motion for a_0 as a constraint. For the Chern–Simons theory such a constraint becomes $f_{ij} = 0$. Under this constraint, we may write a_i as $a_i = \partial_i \phi$. Inserting this into equation (3.43), one obtains [15] an effective 1D theory on the edge with an action:

$$S_{\text{edge}} = \frac{m}{4\pi} \int \partial_t \phi \partial_x \phi \, dx \, dt. \quad (3.44)$$

This approach, however, has a problem. It is easy to see that a Hamiltonian associated with the action (3.44) is zero and the boundary excitations described by equation (3.44) have zero velocity. Therefore this action cannot be used to describe any physical edge excitations in real FQH samples. The edge excitations in FQH states always have finite velocities.

The appearance of finite velocities of edge excitations is a boundary effect. The bulk effective theory defined by equation (3.36) does not contain information about the velocities of the edge excitations. To determine the dynamics of the edge excitations from the effective theory, we must find a way to input the information about the edge velocity. The edge velocities must be treated as the external parameters that are not contained in the bulk effective theory. The problem is how to put these parameters in the theory.

Let us now note that the condition $a_0 = 0$ is not the only choice for the gauge-fixing condition. A more general gauge-fixing condition has the form

$$a_t = a_0 + v a_x = 0. \quad (3.45)$$

Here a_x is the component of the vector potential parallel to the boundary of the sample and v is a parameter that has the dimensions of velocity.

It is convenient to choose new coordinates that satisfy

$$\begin{aligned} \tilde{x} &= x - vt, \\ \tilde{t} &= t, \quad \tilde{y} = y. \end{aligned} \quad (3.46)$$

In these coordinates the components of the gauge field are given by

$$\begin{aligned} \tilde{a}_t &= a_t + v a_x, \\ \tilde{a}_{\tilde{x}} &= a_x, \\ \tilde{a}_{\tilde{y}} &= a_y. \end{aligned} \quad (3.47)$$

The gauge-fixing condition becomes that discussed before in the new coordinates. It is easy to see that the form of the Chern–Simons action is preserved under the transformations (3.46) and (3.47):

$$S = \frac{q}{4\pi} \int d^3x \, a_\mu \partial_\nu a_\lambda \epsilon^{\mu\nu\lambda} = \frac{q}{4\pi} \int d^3x \, \tilde{a}_\mu \partial_\nu \tilde{a}_\lambda \epsilon^{\tilde{\mu}\tilde{\nu}\tilde{\lambda}}. \quad (3.48)$$

Repeating the previous derivation, we find that the edge action is given by

$$S = \frac{q}{4\pi} \int d\tilde{t} \, d\tilde{x} \, \partial_{\tilde{t}} \phi \partial_{\tilde{x}} \phi. \quad (3.49)$$

In terms of the original physical coordinates the above action acquires the form

$$S = \frac{q}{4\pi} \int dt dx (\partial_t + v \partial_x) \phi \partial_x \phi, \quad (3.50)$$

which is a chiral boson theory (for example [55]). It is easy to see that the edge excitations described by equation (3.50) have a non-zero velocity. The quantization of chiral boson theory has been discussed in detail in [55]. The canonical momentum $\pi(x)$ is equal to $\pi = \delta L / \delta \dot{\phi} = (q/4\pi) \partial_x \phi$. The ‘coordinate’ ϕ and ‘momentum’ π obey the commutation relations:

$$\begin{aligned} [\pi(x), \phi(y)] &= \frac{1}{2} \delta(x - y), \\ [\phi(x), \phi(y)] &= \frac{\pi}{q} \text{sgn}(x - y). \end{aligned} \quad (3.51)$$

The Hamiltonian (3.50) of the theory is given by

$$H = -\frac{qv}{4\pi} \int dx \partial_x \phi \partial_x \phi. \quad (3.52)$$

The Hilbert space contains only left-moving degrees of freedom (or right-moving degrees of freedom if $v < 0$). The theory (3.51) and (3.52) describes free left-moving (or right-moving) phonons (i.e. the edge density waves). One can easily show that (3.51) and (3.52) are equivalent to the K–M algebra (3.7) by identifying $\rho = (1/2\pi) \partial_x \phi$.

In the following we would like to show that $\rho = (1/2\pi) \partial_x \phi$ can actually be interpreted as the 1D electron density on the edge. First we note that the coupling between the electrons and the external electromagnetic potential is given by $\int A_\mu J_\mu d^3x = \int (1/2\pi) A \partial_\nu a d^3x$ (see equation (3.39)). However, for a finite sample, such coupling should be written as

$$\int A_\mu J_\mu d^3x \sim \int \frac{1}{2\pi} a_\mu \partial_\nu A_\lambda \varepsilon^{\mu\nu\lambda} d^3x \quad (3.53)$$

which is invariant under the gauge transformation of the electromagnetic field A_μ and under the restricted gauge transformation of the gauge field a_μ . Assuming that A_μ is independent of y and $\tilde{y}$ and $A_y = A_{\tilde{y}} = 0$, we see that, from $\tilde{a}_{\tilde{t}} = \partial_{\tilde{t}} \phi$,

$$\begin{aligned} \int \frac{1}{2\pi} a_{\tilde{t}} \partial_{\tilde{t}} A_{\tilde{x}} \varepsilon^{\tilde{t}\tilde{x}\tilde{z}} d^3\tilde{x} &= - \int d\tilde{x} d\tilde{y} d\tilde{t} \frac{1}{2\pi} \partial_{\tilde{y}} \phi (\partial_{\tilde{x}} A_{\tilde{t}} - \partial_{\tilde{t}} A_{\tilde{x}}) \\ &= - \int d\tilde{x} d\tilde{t} \frac{1}{2\pi} \phi (\partial_{\tilde{x}} A_{\tilde{t}} - \partial_{\tilde{t}} A_{\tilde{x}}) |_{\tilde{y}=0} \\ &= \int dx dt \frac{1}{2\pi} (A_t + v A_x) \partial_x \phi |_{\tilde{y}=0}, \end{aligned} \quad (3.54)$$

where we have used the equation of motion $(\partial_t + v \partial_x) \phi = \partial_t \phi = 0$ and the transformations (3.46) and (3.47). Equation (3.54) clearly indicates that the 1D edge electron density is given by $(1/2\pi) \partial_x \phi = \rho$. This identification, together with the algebra (3.51) and (3.52), completes our proof that our treatment of the effective theory (3.43) and (3.45) reproduces the edge theory obtained in section 3.1.

The velocity v of the edge excitations enters our theory through the gauge-fixing

condition. Note that under the restricted gauge transformations the gauge-fixing conditions (3.45) with different v cannot be transformed into each other. They are physically inequivalent. This agrees with our assumption that v in the gauge-fixing condition is physical and actually determines the velocity of the edge excitations.

The Hamiltonian (3.52) is bounded from below only when $vq < 0$. The consistency of our theory requires v and q to have opposite signs. Therefore the sign of the velocity (the chirality) of the edge excitations is determined by the sign of the coefficient in front of the Chern–Simons terms.

The above results can be easily generalized to the generic FQH states described by equation (3.36) because the matrix $\mathbf{K}$ can be diagonalized. The resulting effective edge theory has the form

$$S_{\text{edge}} = \frac{1}{4\pi} \int dt dx (K_{IJ} \partial_t \phi_I \partial_x \phi_J - V_{IJ} \partial_x \phi_I \partial_x \phi_J). \quad (3.55)$$

The Hamiltonian is given by

$$H_{\text{edge}} = \frac{1}{4\pi} \int dt dx V_{IJ} \partial_x \phi_I \partial_x \phi_J. \quad (3.56)$$

Therefore V must be a positive definite matrix. Using this result, one can show that a positive eigenvalue of $\mathbf{K}$ corresponds to a left-moving branch and a negative eigenvalue corresponding to a right-moving branch.

The effective theory of the $\nu = \frac{2}{5}$ FQH state is given by

$$\mathbf{K} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}. \quad (3.57)$$

Since $\mathbf{K}$ has two positive eigenvalues, the edge excitations of the $\nu = \frac{2}{5}$ FQH state have two branches moving in the same direction. The $\nu = 1 - 1/n$ FQH state is described by the effective theory with

$$\mathbf{K} = \begin{pmatrix} 1 & 0 \\ 0 & -n \end{pmatrix}. \quad (3.58)$$

The two eigenvalues of $\mathbf{K}$ now have opposite signs; hence, the two branches of the edge excitations move in opposite directions.

3.4. Charged excitations and electron propagator on the edges of generic fractional quantum Hall states

In the last two sections we studied the dynamics of edge excitations in FQH liquids at low energies. We found that the low-lying edge excitations are described by a free-phonon theory that is exactly soluble. In this section we shall concentrate on the generic charge excitations. In particular we shall calculate the propagators of the electrons and the quasiparticles for the most general (Abelian) FQH states [20] (see section 5). The key point again is to write the electron or the quasiparticle operators in terms of the phonon operator ρ_I . Once we do so, the propagators can be easily calculated because the phonons are free (at low energies and long wavelengths).

We know that for the FQH state described by equation (3.36), the edge states

are described by the action (3.55). The Hilbert space of the edge excitations forms a representation of K–M algebra:

$$[\rho_{Ik}, \rho_{Jk'}] = (K^{-1})_{IJ} \frac{1}{2\pi} k \delta_{k+k'},$$

$$k, k' = \text{integer} \times \frac{2\pi}{L},$$
(3.59)

where $\rho_I = (1/2\pi)\partial_x \phi_I$ is the edge density of the I th condensate in the FQH state, $I, J = 1, \dots, \kappa$, and κ is the dimension of $\mathbf{K}$. The electron density on the edge is given by

$$\rho_e = -e \sum_I \rho_I. \quad (3.60)$$

The dynamics of the edge excitations are described by the Hamiltonian

$$H = 2\pi \sum_{IJ} V_{IJ} \rho_{I,k} \rho_{J,-k}, \quad (3.61)$$

where V_{IJ} is a positive definite matrix.

Let us first try to write down the quasiparticle operator Ψ_I on the edge which creates a quasiparticle labelled l_I . We know that inserting the quasiparticle on the edge will cause a change $\delta\rho_I$ in the edge density of the I th condensate (see equation (3.40)) that satisfies

$$\int dx \delta\rho_I = l_I (K^{-1})_{II}. \quad (3.62)$$

Because Ψ_I is a local operator that causes only a local change in the density, we have

$$[\rho_I(x), \Psi_I(x')] = l_I (K^{-1})_{II} \delta(x - x') \Psi_I(x'). \quad (3.63)$$

Using the K–M algebra (3.59), one can show that the quasiparticle operators that satisfy equation (3.63) are given by

$$\Psi_I \propto \exp(i\phi_I l_I). \quad (3.64)$$

The charge of the quasiparticle Ψ_I is determined from the commutator $[\rho_e, \Psi_I]$ and is found to be

$$Q_I = -e \sum_{IJ} l_I (K^{-1})_{IJ}. \quad (3.65)$$

From equations (3.42) and (3.64), we see that the electron operator can be written as

$$\Psi_{e,L} \propto \prod_I \Psi_I^{l_I} \propto \exp\left(i \sum_I l_I \phi_I\right),$$

$$l_I = \sum_J K_{IJ} L_J, \quad \sum_I L_I = 1. \quad (3.66)$$

The above operators carry a unit charge as one can see from equation (3.65). The

commutation of $\Psi_{e,L}$ can be calculated as

$$\begin{aligned}\Psi_{e,L}(x)\Psi_{e,L}(x') &= (-)^{\lambda}\Psi_{e,L}(x')\Psi_{e,L}(x), \\ \lambda &= \sum_{IJ} L_I K_{IJ} L_J,\end{aligned}\tag{3.67}$$

Because the diagonal elements of $\mathbf{K}$ are odd integers, we can show that $(-)^{\lambda} = -1$. The electron operators defined in (3.66) are indeed fermion operators.

Since all the operators $\Psi_{e,L}$ for different choices of L_I carry a unit charge and are fermionic, each $\Psi_{e,L}$ can be a candidate for the electron operators. In general the true electron operator is a superposition of $\Psi_{e,L}$ values:

$$\Psi_e = \sum_L C_L \Psi_{e,L}.\tag{3.68}$$

In this paper, when we say there are many different electron operators on the edge, we really mean that the true physical electron operator is a superposition of those operators.

Using the K–M algebra (3.59) and the Hamiltonian (3.61), we can calculate the propagators of a generic quasiparticle operator:

$$\Psi_1 \propto \exp\left(i \sum_I l_I \phi_I\right)\tag{3.69}$$

(which includes the electron operators for suitable choices of $\mathbf{l}$). First we note that, after a suitable redefinition of ρ_I ,

$$\tilde{\rho}_I = \sum_J U_{IJ} \rho_J,\tag{3.70}$$

$\mathbf{K}$ and $\mathbf{V}$ can be simultaneously diagonalized; that is, in terms of $\tilde{\rho}_I$, equations (3.59) and (3.61) become

$$[\tilde{\rho}_{Ik}, \tilde{\rho}_{Jk'}] = \sigma_I \delta_{IJ} \frac{1}{2\pi} k \delta_{k+k'},\tag{3.71}$$

$$H = 2\pi \sum_I |v_I| \tilde{\rho}_{I,k} \tilde{\rho}_{I,-k},$$

where $\sigma_I = \pm 1$ is the sign of the eigenvalues of $\mathbf{K}$. The velocity of the edge excitations created by $\tilde{\rho}_I$ is given by $v_I = \sigma_I |v_I|$.

To prove the above result we first redefine ρ_I to transform $\mathbf{V}$ into the identity matrix: $\mathbf{V} \rightarrow \mathbf{U}_1 \mathbf{V} \mathbf{U}_1^T = 1$. This is possible because $\mathbf{V}$ is a positive definite symmetric matrix. Now $\mathbf{K}$ becomes a new symmetric matrix $\mathbf{K}_1 = \mathbf{U}_1 \mathbf{K} \mathbf{U}_1^T$ whose eigenvalues have the same sign as the eigenvalues of $\mathbf{K}$ (although the absolute values may differ). Then we make an orthogonal transformation to diagonalize $\mathbf{K}_1$: $(K_1)_{IJ} \rightarrow \sigma_I |v_I| \delta_{IJ}$. After a trivial rescaling of the densities, we obtain equation (3.71). In terms of $\tilde{\rho}_I$ the operator Ψ_1 has the form

$$\begin{aligned}\Psi_1 &\propto \exp\left(i \sum_I \tilde{l}_I \tilde{\phi}_I\right), \\ \tilde{l}_J &= \sum_I l_I U_{IJ}^{-1}.\end{aligned}\tag{3.72}$$

From equations (3.71) and (3.72) we see that the propagator of Ψ_I has the following general form:

$$\langle \Psi_I^\dagger(x, t) \Psi_I(0) \rangle \propto \exp(i l_I k_I x) \prod_I (x - v_I t + i \sigma_I \delta)^{-\gamma_I}, \quad \gamma_I = \tilde{l}_I^2, \quad (3.73)$$

where $v_I = \sigma_I |v_I|$ is the velocities of the edge excitations. γ_I in equation (3.73) satisfy the sum rule

$$\sum_I \sigma_I \gamma_I \equiv \lambda_1 = \sum_I l_I K_{IJ}^{-1} l_J. \quad (3.74)$$

In order to prove the above sum rule, we have used the relation

$$(UKU^T)_{IJ} = \sigma_I \delta_{IJ}. \quad (3.75)$$

From equations (3.67) and (3.74) we see that the sum rule is directly related to the statistics of Ψ_I and that λ_1 is a topological quantum number. If Ψ_I represents the electron operator, λ_1 will be an odd integer. From equation (3.73) we also see that the operator Ψ_I creates an excitation with momentum near $\sum_I l_I k_I$.

As an application of the formalism developed above, let us discuss the hierarchical states with the filling fractions $\nu = p/(pq + 1)$ (q even) in more detail. Those states include $\nu = \frac{2}{5}, \frac{3}{7}, \frac{2}{9}, \dots$ FQH states. The hierarchical states with $\nu = p/(pq + 1)$ are described by the p by p matrices $\mathbf{K} = 1 + q\mathbf{C}$, where $\mathbf{C}$ is the pseudo-identity matrix: $C_{IJ} = 1$, $I, J = 1, \dots, p$. $\mathbf{K}^{-1} = 1 - [q/(pq + 1)]\mathbf{C}$. Because all the edge excitations move in the same direction, we have

$$\langle \Psi_I^\dagger(x = 0, t) \Psi_I(0) \rangle \propto t^{-\lambda_1}, \quad (3.76)$$

where λ_1 is given by equation (3.74). The fundamental quasiparticle is given by $\mathbf{I}^\Gamma = (1, 0, \dots, 0)$ and carries the charge $1/(pq + 1)$. The exponent in its propagator is $\lambda_1 = 1 - q/(pq + 1)$. The quasiparticle with the smallest exponent is given by $\mathbf{I}^\Gamma = (1, \dots, 1)$ and carries the charge $p/(pq + 1)$. The exponent is $\lambda_1 = p/(pq + 1)$, that is less than $1 - q/(pq + 1)$ (note we have $q \geq 2$ and $p \geq 1$). Such a quasiparticle (with the charge $p/(pq + 1)$) dominates the tunnelling between two edges of the *same* FQH fluid at low energies.

The electron operators are given by $\Psi_{e,L} = \psi_I$ with $\mathbf{I}$ satisfying $\sum_I l_I = pq + 1$. The exponent in the propagator is given by $\lambda_1 = \sum_I l_I^2 - q(pq + 1)$. The electron operator with a minimum exponent in its propagator is given by $\mathbf{I}^\Gamma = (q, \dots, q, q + 1)$. The value of the minimum exponent is $\lambda_1 = q + 1$. Such an electron operator dominates the tunnelling between edges of two *different* FQH fluids at low energies.

3.5. Some simple phenomenological consequences of chiral Luttinger liquids at fractional quantum Hall edges

In the last four sections we have shown that the electrons at the edges of a FQH liquid form a chiral Luttinger liquid. As one of the characteristic properties of chiral Luttinger liquids, the electron and the quasiparticle propagators have anomalous exponents:

$$\langle \Psi_e^\dagger(t, x = 0) \Psi_e(0) \rangle \sim t^{-g_e}, \quad \langle \Psi_q^\dagger(t, x = 0) \Psi_q(0) \rangle \sim t^{-g_q}. \quad (3.77)$$

Those anomalous exponents can be measured by tunnelling experiments between the FQH edges.

Two situations need to be considered separately: case I, in which two edges of

FQH liquids are separated by an insulator; case II, in which two edges are separated by a FQH liquid. In case I, only electrons can tunnel between the edges, and in case II the quasiparticles supported by the FQH liquid that separates the two edges can tunnel. The tunnelling operator A is given by $A \propto \Psi_{e1} \Psi_{e1}^\dagger$ for case I, where Ψ_{e1} and Ψ_{e2} are the electron operators on the two edges. The tunnelling operator A has the form $A \propto \Psi_{q1} \Psi_{q2}^\dagger$ for case II, where Ψ_{q1} and Ψ_{q2} are the quasiparticle operators on the two edges. The physical properties of the tunnelling can be calculated from the correlation of the tunnelling operator, which in turn can be expressed as a product of the electron or quasiparticle propagators on the two edges.

Let $g = g_e$ for case I and $g = g_q$ for case II. Then, at zero temperature, one can show that [35] the anomalous exponents of the electron and quasiparticle operators lead to nonlinear tunnelling I - V curve

$$I \propto V^{2g-1}. \quad (3.78)$$

The noise spectrum of the tunnelling current also contains a singularity at the frequency $f = QV/h$:

$$s(f) \propto \left| f - \frac{QV}{h} \right|^{2g-1} \quad (3.79)$$

where V is the voltage difference between the two edges and Q is the electric charge of the tunnelling electron or the tunnelling quasiparticle. At finite temperature T , the zero bias conductance also has a power law dependence:

$$\sigma \propto T^{2g-2}. \quad (3.80)$$

We see that the anomalous exponents can be easily measured by the tunnelling experiments. The noise spectrum further reveals the charges of the tunnelling particles.

The exponents g_e and g_q are calculated in section 3.4. To summarize the results in a simple way, it is convenient to divide the FQH edges into three classes. In class A, all edge excitations move in one direction, the direction of the normal drift velocity along the edge. Such a direction will be called the normal direction. In class B, one branch of edge excitations propagates in the normal direction, and all other branches propagate in the opposite direction. In class C, there are two or more branches propagating in the normal direction and one or more branches propagating in the opposite direction.

For the class A edges, the exponents g_e and g_q are directly related to the statistics of the electrons and quasiparticles. In this case, g_e and g_q are universal and independent of details of electron interactions and edge potentials. Table 1 lists some FQH states which support the class A edge states, as well as the corresponding values

Table 1. Some FQH states which support the class A edge states, together with the g_e , g_q and charge values.

FQH state	g_e	g_q	Charge Q_q/e
$\nu = 1/m$	m	$1/m$	$1/m$
$\nu = \frac{2}{5}$	3	$\frac{2}{5}$	$\frac{2}{5}$
$\nu = n/(2n+1)$	3	$n/(2n+1)$	$n/(2n+1)$
$(331)_{\nu=1/2}$	3	$\frac{3}{8}$	$\frac{1}{4}$
$(332)_{\nu=2/5}$	3	$\frac{2}{5}$	$\frac{2}{5}$

of the exponents g_e and g_q and the electric charge of the associated particles (note only the minimum values of g_e and g_q are listed).

We also see from section 3.4 that, owing to the presence of the modes propagating in both directions, g_e and g_q are not universal for the class B and class C edges. Their values depend on the electron interactions and the edge potentials. However, the electrons in the real FQH sample interact through a quite special interaction: the long-range Coulomb interaction. In this case the edge Hamiltonian has a special form

$$H = \sum_{i,j} \left[V_{ij} \rho_i \rho_j + V_c \left(\sum_i t_i \rho_i \right)^2 \right], \quad (3.81)$$

with $V_c \gg V_{ij}$ (in fact, $V_c \sim -\ln k$ at long wavelengths). In equation (3.81), t_i is the charge vector, and $\sum_i t_i \rho_i$ is the total charge density on the edge. In the limit $V_c \gg V_{ij}$ the edge has a special structure. One edge mode propagates at a large velocity $v \sim V_c$ in the normal direction. All other modes propagate at smaller velocity $v \sim V_{ij}$. What is more important is that only the fast mode carries charge and all the other slow modes are neutral. The separation between the charge and the neutral modes in the presence of the long-range Coulomb interaction make it very difficult to detect more than one edge branch in the edge magnetoplasma experiments [56]. In general, the edge magnetoplasma experiments can see only a single charge mode. The neutral modes are hard to detect because of the weak coupling. Thus the edge magnetoplasma experiment in [57] does not contradict our picture that the $\frac{2}{3}$ FQH state contains edge modes that propagate in both directions.

In addition to separating the charge and the neutral modes, the long-range Coulomb interaction also makes the exponents g_e and g_q universal for the class B edges. The exponent g_1 in the propagator of a quasiparticle labelled by an integer vector $\mathbf{l}$ can be calculated as follows. The calculation applies only to the class B edges in the limit $V_c \gg V_{ij}$. We first separate $\mathbf{l}$ into a charge part and a neutral part:

$$\mathbf{l} = \mathbf{l}_c + \mathbf{l}_n, \quad (3.82)$$

where the neutral part $\mathbf{l}_n$ satisfies

$$\mathbf{t}^T \mathbf{K}^{-1} \mathbf{l}_n = 0. \quad (3.83)$$

Therefore we have

$$\mathbf{l}_n = \mathbf{l} - \mathbf{t} \frac{\mathbf{t}^T \mathbf{K}^{-1} \mathbf{l}}{\mathbf{t}^T \mathbf{K}^{-1} \mathbf{t}}, \quad \mathbf{l}_c = \mathbf{t} \frac{\mathbf{t}^T \mathbf{K}^{-1} \mathbf{l}}{\mathbf{t}^T \mathbf{K}^{-1} \mathbf{t}}. \quad (3.84)$$

One can show that the operator $\sum_i l_{n,i} \rho_i$ generates the slow neutral modes and $\sum_i l_{c,i} \rho_i$ generates the fast charge mode. Following the calculation outlined in section 3.4, one can show that

$$\begin{aligned} g_1 &= |\mathbf{l}_c^T \cdot \mathbf{K}^{-1} \mathbf{l}_c| + |\mathbf{l}_n^T \cdot \mathbf{K}^{-1} \mathbf{l}_n| \\ &= |\mathbf{l}^T \cdot \mathbf{M} \mathbf{l}|, \end{aligned} \quad (3.85)$$

where

$$\mathbf{M} = \mathbf{K}^{-1} - \frac{2}{\mathbf{t}^T \mathbf{K}^{-1} \mathbf{t}} \mathbf{K}^{-1} \mathbf{t} \mathbf{t}^T \mathbf{K}^{-1}. \quad (3.86)$$

The simple result (3.85) arises from two facts: firstly the charge and the neutral modes separate, and secondly all the neutral modes propagate in one direction. From equation (3.85) we can calculate the minimum values of g_e and g_q . Table 2 lists some

Table 2. Some FQH states which support the class B edge states, together with the g_e , g_q and charge values.

FQH state	g_e	g_q	Charge Q_q/e
$\nu = \frac{2}{3}$	2	$\frac{2}{3}$	$\frac{1}{3}, \frac{2}{3}$
$\nu = 1 - 1/m$	$(m+1)/(m-1)$	$(1/m)(m+1)/(m-1)$	$1/m$
$\nu = \frac{3}{5}$	$\frac{7}{3}$	$\frac{3}{5}$	$\frac{3}{5}$
$\nu = (n+1)/(2n+1)$	$3 - 2/(n+1)$	$(n+1)/(2n+1)$	$(n+1)/(2n+1)$
$\nu = \frac{2}{7}$	4	$\frac{2}{7}$	$\frac{1}{7}, \frac{2}{7}$

FQH states which support the class B edge states, and the corresponding values of the exponents g_e and g_q (again only the minimum values of g_e and g_q are listed).

In section 3.3 we argued that the edge excitations of the a FQH characterized by a matrix $\mathbf{K}$ are described by $U(1)$ K -M algebra characterized by the same matrix (see equation (3.59)). This result is correct only for sharp edges where the electron density drops to zero in a range of the order of the magnetic length near the edge. However, the edge potentials in real QH samples are very smooth, and the electron density gradually drops to zero in a range of a few thousands Ångströms in order to minimize the electrostatic energy [58]. Thus the edge structures for smooth and sharp edge potentials are quite different. It was shown in [59] that, as the edge potential changes from a sharp one to a smooth one, the QH edge will undergo a so-called edge reconstruction. One or more pairs of edge branches are generated after the edge reconstruction. Each pair contains two branches that propagate in opposite directions and behaves like the usual non-chiral Luttinger liquid in one dimension. After the edge reconstruction, even a $\nu = 1$ IQH state can have several edge branches. The edge excitations on a smooth edge are generally described by the following matrix [59]:

$$\mathbf{K}_{\text{edge}} = \mathbf{K}_{\text{bulk}} \oplus \begin{pmatrix} \mathbf{K}' & 0 \\ 0 & -\mathbf{K}' \end{pmatrix},$$

where $\mathbf{K}_{\text{bulk}}$ is the matrix characterizing the bulk FQH state and $\mathbf{K}'$ is another interior matrix that depends on the edge potential. For example, the edge excitations on a smooth $\frac{2}{3}$ edge could be described by

$$\mathbf{K}_{\text{edge}} = \begin{pmatrix} -3 & & & \\ & 1 & & \\ & & -1 & \\ & & & 1 \end{pmatrix}.$$

In an ideal FQH sample where electrons and/or quasiparticles are not allowed to be scattered between different edge branches, a bar of the FQH liquid has a quantized two-terminal conductance only when the FQH liquid has a class A edge state. However, in the presence of the long-range Coulomb interaction, any FQH liquids will have a (approximately) quantized two-terminal conductance $\sigma = \nu e^2/h$, owing to the separation of the charge and the neutral modes in the presence of the long-range interaction.

In the real sample, there are many mechanisms that allow electrons and

quasiparticles to scatter between different edge branches. These mechanisms include inelastic scattering of phonons and elastic scattering of impurities. These scatterings equilibrate different edge branches. Using the sum rule obtained in [45, 47], one can show that an equilibrated edge always gives a quantized two-terminal conductance. Thus, in addition to the long range Coulomb interaction, there are many mechanisms that cause the two-terminal conductance to be quantized at $\sigma = \nu e^2/h$, even for the class B and C edges.

In the following we shall briefly discuss some effects of impurities on the properties of FQH edge states. We know that the electrons in the non-chiral Luttinger liquids are localized in the presence of impurities. Because of this, the additional pairs of the edge branches generated by the smooth edge will be localized by the impurities. Thus it is possible that the dirty smooth edges may have very similar physical properties to the clean sharp edges at sufficiently low energies. It is also obvious that the impurities cannot localize the electrons in the class A edge owing to the lack of back scattering.

The situation becomes very interesting for the edge state of the $\nu = \frac{2}{3}$ FQH liquid, which contains two branches propagating in opposite directions. In this case the impurities do cause back scattering. However, the back scattering will not cause localization of the electrons on the edge. Mobile edge excitations are required to cancel the gauge anomaly from the bulk Chern–Simons term, so that the effective action for a finite QH system will be invariant under the gauge transformation of the electromagnetic gauge potential [45]. A detailed dynamical theory for the disordered $\frac{2}{3}$ edge has been proposed in [54], where it was shown that even at the strong-disordered fixed point the electrons on the edge remain delocalized. It was further argued that the strongly disordered fixed point has a very special dynamical property, namely the charge mode and the neutral mode separate even in the absence of long range interaction. The separation of the charge and the neutral modes leads to the universal g_e and g_q whose values coincide with those calculated above for the long-range interaction. In [54] some experimental consequences of the disordered edges were also studied in detail. However, a different point of view was raised in [60].

4. Shift and spin vectors: new topological quantum numbers for fractional quantum Hall liquids

In section 2 we studied the effective theory of (Abelian) FQH liquids. We found that the effective theory depends on the $\mathbf{K}$ matrix and the charge vector $\mathbf{t}$. This result seems to suggest that the (Abelian) FQH liquids are completely characterized by $\mathbf{K}$ and $\mathbf{t}$. However, in this section we shall see that there is a new kind of topological quantum numbers for FQH liquids. This new type of quantum number does not directly appear in the effective theory on the plane. However, it can still be measured by experiments in planar systems, because of a new set of selection rules induced by these quantum numbers.

4.1. Quantum Hall states on a sphere and the shift

To understand the appearance of new quantum numbers, spin vectors, let us first consider the $\nu = 1$ IQH state on a sphere. Assume that the magnetic field is uniform on the sphere and let N_ϕ be the total number of the flux quanta passing through the sphere. The Hamiltonian has an $su(2)$ symmetry associated with the rotation of the sphere. Thus the degenerate eigenstates form irreducible representations of the $su(2)$

group. The ground states form a representation with $su(2)$ spin $S^{su(2)} = N_\phi/2$ and thus have $2S^{su(2)} + 1 = (N_\phi + 1)$ -fold degeneracy [12]. These $N_\phi + 1$ states form the first Landau level on the sphere. The next energy level has $N_\phi + 3$ degeneracy and forms the $S^{su(2)} = (N_\phi/2) + 1$ representation of $su(2)$, which corresponds to the second Landau level. In general the states in the n th Landau level form the $S^{su(2)} = (N_\phi/2) + n - 1$ representation of the $su(2)$ rotation of the sphere.

The electron wavefunctions on the sphere can be expressed in a simple form through the following spinor coordinates [12]:

$$(u, v) = \left(\cos \theta \exp\left(\frac{i\phi}{2}\right), \sin \theta \exp\left(\frac{-i\phi}{2}\right) \right). \quad (4.1)$$

The $N_\phi + 1$ states in the first Landau level are described by the wavefunction $u^m v^n$, $m = N_\phi - n = 0, \dots, N_\phi$, which form the $S^{su(2)} = m + n$ representation with $S_z^{su(2)} = m$.

We see that the $\nu = 1$ IQH state with a filled first Landau level contains $N_e = N_\phi + 1$ number of electrons. Thus N_e/N_ϕ is not exactly equal to one. We shall call $\mathcal{S}$ in the relation

$$N_\phi = \nu^{-1} N_e - \mathcal{S} \quad (4.2)$$

the shift. The shift of a QH state depends on the topology of the space. The $\nu = 1$ IQH state has a shift $\mathcal{S} = 1$ on a sphere, and a shift $\mathcal{S} = 0$ on a torus [13].

We can construct another $\nu = 1$ IQH state with filled *second* Landau level. Such a state contains $N_e = N_\phi + 3$ electrons and has a shift $\mathcal{S} = 3$ on the sphere. On the torus the shift still vanishes.

Both the above two IQH states are described by the same effective theory with $\mathbf{K} = 1$ and $\mathbf{t} = 1$ on the plane. This indicates that the topological quantum number $\mathcal{S}$ cannot be determined from $\mathbf{K}$ and $\mathbf{t}$. Thus the $\mathbf{K}$ matrix and the charge vector do not provide a complete description of the QH liquids. We need to find some additional quantum numbers which will at least enable us to determine the shift [21, 61, 62].

To gain some understanding of the origin of the new quantum numbers, let us consider in more detail the above two $\nu = 1$ IQH states. The two IQH states differ only in the different cyclotron motions of the electrons. We know that, as the Landau level index increases, the orbital angular momentum carried by the cyclotron motion also increases. In the following we shall call such the orbital angular momentum the orbital spin. Note here that the orbital spin is associated with the $U(1)$ rotation of the plane and should be distinguished from the ordinary spin quantum number of the electron and the $su(2)$ quantum $S^{su(2)}$ discussed above. In this section we shall consider only spinless electrons (which correspond to the spin polarized electrons in experiments).

As an object with no-zero orbital spin moves along a loop C on a curved 2D space, a non-zero Berry phase will be induced. The Berry phase is given by

$$\phi = S_{\text{ob}} \oint_C dx^i \omega_i = S_{\text{ob}} \int_{S_C} d^2x R, \quad (4.3)$$

where S_C is the area enclosed by the loop C . Here R is the curvature and ω_i the connection whose curl gives rise to the curvature. The relation between ω_i and $R = \varepsilon_{ij} \partial_i \omega_j$ is the same as relation between the gauge potential a_i and the 'magnetic field' $b = \varepsilon_{ij} \partial_i a_j$. When a 2D space is embedded in a three-dimensional space, the

integral $\oint_C dx^i \omega_i = \int_{S_C} d^2x R$ has a geometrical meaning. The norm vector of the 2D surface spans a solid angle Ω as it moves along the loop C . The above integral is simply determined by this solid angle

$$\oint_C dx^i \omega_i = \int_{S_C} d^2x R = \Omega. \quad (4.4)$$

Thus the total curvature of a sphere is $\int_{\text{sphere}} d^2x R = 4\pi$.

From the above discussion we see that, because electrons carry a non-zero orbital spin, the total flux seen by the electron is the sum of the magnetic flux and the Berry phase induced through the coupling of the orbital spin to the curvature of the space. It is this Berry phase that causes the shift. Thus electrons with different orbital spins will cause different shifts. According to this picture we also obtain the well known fact that the shift always vanishes on torus [13], since the curvature on torus is zero.

According to the semiclassical picture of the cyclotron motion, the electrons in the n th Landau level have an orbital spin $S_{\text{ob}} = n + \text{constant}$. Thus the orbital spins of the electrons in the first and the second Landau levels differ by one, and the induced Berry phases differ by two flux quanta on the sphere. Therefore the shifts of the $\nu = 1$ state in the first Landau level and the second Landau level differ by two.

To determine the absolute value of the orbital spin for each Landau level, let us move an electron in the n th Landau level around a loop C on the sphere. There are two ways to calculate the phase Φ induced by such a motion. The phase Φ can be obtained as the sum of the phase induced by the magnetic field and the Berry phase induced by the orbital spin

$$\Phi = \oint_C (dx^i - eA_i + S_{\text{ob}}\omega_i) = 2\pi(N_\phi + 2S_{\text{ob}}) \frac{\Omega}{4\pi}, \quad (4.5)$$

where Ω is the solid angle spanned by the loop C on the sphere. The phase Φ can also be calculated from the fact that the electrons in the n th Landau level form an $S^{su(2)} = N_\phi/2 + n - 1$ representation of the $su(2)$. An electron in the n th Landau level and at position (θ, ϕ) on the sphere is an eigenstate of $\mathbf{n} \cdot \mathbf{S}^{su(2)}$ with an eigenvalue $N_\phi/2 + n - 1$, where $\mathbf{n}$ is the unit vector in the direction (θ, ϕ) and $\mathbf{S}^{su(2)}$ are the three generators of the $su(2)$ rotation of the sphere. As $\mathbf{n}$ traces out the loop C , a Berry phase will be induced owing to the non-zero $su(2)$. The induced phase is

$$\Phi = S^{su(2)}\Omega = [N_\phi + 2(n - 1)] \frac{\Omega}{2}. \quad (4.6)$$

Comparing equation (4.5) with equation (4.6) we find the orbital spin for n th Landau level to be

$$S_{\text{ob}} = n - 1. \quad (4.7)$$

The orbital spin for first Landau level vanishes. Equations (4.5)–(4.7) reveal a direct relation between the following quantities:

- (a) the orbital spin of electrons;
- (b) the number of effective flux quanta, $N_\phi^* = N_\phi + 2S_{\text{ob}}$, seen by the electrons;
- (c) the $S^{su(2)}$ quantum number of the electrons, $S^{su(2)} = N_\phi^*/2$;
- (d) the number of states in the corresponding Landau level, $N = 2S^{su(2)} + 1 = N_\phi^* + 1$;
- (e) the value of the shift, $\mathcal{S} = N - N_\phi = 2S_{\text{ob}} + 1$, for the $\nu = 1$ state.

4.2. Spin vectors in quantum Hall liquids

In the last section we saw that the shift is closely related to the coupling between the orbital spin and the curvature of the space. In this section we shall attempt to include such coupling in the effective theory so that we shall be able to calculate the shift from the effective theory. In the process of building such an effective theory we find that it is necessary to introduce a new topological quantum number, the spin vector, in our description of QH liquids.

First let us consider a system of bosons or fermions on a sphere with a uniform magnetic field. We also assume that the bosons or fermions carry an orbital spin S_{ob} . Those particles are described by

$$\mathcal{L} = -et_1 A_\mu J^\mu + S_{\text{ob}} \omega_i J^i + \text{kinetic energy}. \quad (4.8)$$

Here, in addition to the orbital spin, we have also included two other complications. The first is that we allow the magnetic field B to have either positive or negative values. The second is that we include t_1 in equation (4.8). $t_1 = 1$ if the particles carry a charge of $-e$, and $t = -1$ if the particles carry a charge e .

Let $N_\phi = -eB/\Phi_0$ be the number of the flux quanta. If $N_\phi t_1 > 0$, then a Laughlin state of the boson or the fermion with a filling fraction $1/m$ is described by the wavefunction

$$\Phi = \prod_{i < j} (v_i u_j - v_j u_i)^m, \quad (4.9)$$

where (u, v) is the spinor coordinate (see equation (4.1)). If $t_1 N_\phi < 0$, the $1/m$ Laughlin state will have the form

$$\Phi = \prod_{i < j} (v_i^* u_i^* - v_j^* u_j^*)^m. \quad (4.10)$$

Here we always assume that $m > 0$. For a given i , Φ is a homogeneous polynomial of (u_i, v_i) (or (u_i^*, v_i^*)) of degree $m(N_p - 1)$ where N_p is the number of the bosons or fermions. Thus the single-particle states form an $S^{su(2)} = m(N_p - 1)/2$ representation of the $su(2)$ rotation of the sphere. Knowing that all particles are in the first Landau level we conclude that the total number of the effective flux quanta seen by the particles is given by $N_\phi^* = 2S^{su(2)} = m(N_p - 1) = |t_1 N_\phi + 2S_{\text{ob}}|$. This implies that $|N_\phi| = mN_p - [m + 2S_{\text{ob}} \text{sgn}(t_1 N_\phi)]$. The shift is defined through $|N_\phi| = mN_p - \mathcal{S}$ when N_ϕ can be negative. So the shift of our $1/m$ state is given by

$$\mathcal{S} = 2S_{\text{ob}} \text{sgn}(t_1 N_\phi) + m. \quad (4.11)$$

Equation (4.11) is valid for both positive and negative t_1 and N_ϕ . We also would like mention that the number of N_p of the particles is assumed to be positive in the above discussion.

Now the question is how to reproduce the shift (4.11) from the effective theory. Here we propose the Chern–Simons effective theory of the $1/m$ state on the sphere to have the form

$$\mathcal{L} = \left(-m \text{sgn}(t_1 N_\phi) \frac{1}{4\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} - \frac{et_1}{2\pi} A_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} + s \omega_i \partial_\nu a_\lambda \varepsilon^{i\nu\lambda} \right), \quad (4.12)$$

where the third term describes the coupling of the curvature. The new quantum number s will be called the spin vector (here it has only one component). From the equation of motion $\partial \mathcal{L} / \partial a_0 = 0$, the total number of the particles can be shown to be

$$\begin{aligned}
N_p &= \int d^2x \frac{1}{2\pi} \varepsilon_{ij} \partial_i a_j \\
&= \int d^2x \frac{1}{2m\pi \operatorname{sgn}(t_1 N_\phi)} \varepsilon_{ij} \partial_i (-t_1 e A_j + 2\pi s \omega_j) \\
&= \frac{1}{m \operatorname{sgn}(t_1 N_\phi)} (t_1 N_\phi + 2s).
\end{aligned} \tag{4.13}$$

We see that the inclusion of $\operatorname{sgn}(t_1 N_\phi)$ in equation (4.12) ensures that $N_p > 0$. Equation (4.13) implies that $N_p = (1/m)[|N_\phi| + 2 \operatorname{sgn}(t_1 N_\phi)s]$. Thus the shift is simply given by $\mathcal{S} = 2 \operatorname{sgn}(t_1 N_\phi)s$ and we find the spin vector for the above $1/m$ state to be (by comparing with equation (4.11))

$$s = S_{\text{ob}} + \frac{m}{2} \operatorname{sgn}(t_1 N_\phi). \tag{4.14}$$

It is interesting that to see that the spin vector has two contributions. The first contribution, from the orbital spin S_{ob} , is easy to understand. One way to understand the second contribution is the following. We know that the $1/m$ state can be viewed as a boson condensation of composite bosons. A composite boson is a bound state of the boson or the fermion with m units of flux quanta. The binding of the flux changes not only the statistics of the particle but also the orbital spin of the boson or the fermion to a new value, and s in equation (4.14) can be regarded as the orbital spin of the composite bosons.

We would like to mention that, if the finite orbital spin S_{ob} is due to the bosons or fermions occupying the n th Landau level, then

$$S_{\text{ob}} = (n - 1) \operatorname{sgn}(t_1 N_\phi), \tag{4.15}$$

which generalizes equation (4.7) to include the possibility that t_1 and N_ϕ may be negative. Thus equation (4.14) can be rewritten as

$$s = \left(n - 1 + \frac{m}{2} \right) \operatorname{sgn}(t_1 N_\phi). \tag{4.16}$$

Equation (4.16) only applies to the Lagrangian

$$\mathcal{L} = \left(-K_{11} \frac{1}{4\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} - \frac{et_1}{2\pi} A_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} + s\omega_i \partial_\nu a_\lambda \varepsilon^{i\nu\lambda} \right), \tag{4.17}$$

with K_{11} and $t_1 N_\phi$ to have the same sign. In this case $\varepsilon_{ij} \partial_i a_j / 2\pi$ is positive and can be regarded as the density of the particles. When K_{11} and $t_1 N_\phi$ have opposite signs, we should view $-\varepsilon_{ij} \partial_i a_j / 2\pi$ as the density of the particles, or we can make a transformation $a_\mu \rightarrow -a_\mu$ and $t_1 \rightarrow -t_1$ in equation (4.17) to make K_{11} and $t_1 N_\phi$ have the same sign. Thus equation (4.16) can be generalized to

$$s = \left(n - 1 + \frac{|K_{11}|}{2} \right) \operatorname{sgn}(t_1 N_\phi), \tag{4.18}$$

which applies to the $1/m$ state described by equation (4.17) where K_{11} can have any signs. Here we have assumed that the condensed particles occupy the n th Landau level.

Applying the above results to the electron systems we see that the $1/m$ state in the first Landau level has a spin vector $s = m/2$ while a $1/m$ state in the n th Landau level has a spin vector $s = n - 1 + (m/2)$ (here we have assumed that $t_1 > 0$ and $N_\phi > 0$).

Now, knowing the effective theory of the first-level hierarchical states, we can include the quasiparticle excitations (or electron excitations in the next layer) and construct the effective theory of the second-level hierarchical states (or double-layer FQH states).

Consider a κ th-level hierarchical state (or a κ -layer FQH state) of an electron system on a sphere. Motivated by the effective theory of the Laughlin state (4.17), let us assume the effective theory for the FQH state to have the form

$$\mathcal{L} = -\frac{1}{4\pi} K_{II'} a_{I\mu} \partial_\nu a_{I'\lambda} \varepsilon^{\mu\nu\lambda} - \frac{e}{2\pi} t_I A_\mu \partial_\nu a_{I\lambda} \varepsilon^{\mu\nu\lambda} + s_I \omega_i \partial_\nu a_{I\lambda} \varepsilon^{i\nu\lambda}, \quad (4.19)$$

where the curvature couples with all the condensates through the spin vector $\mathbf{s}^T = (s_1, s_2, \dots)$. From the equation of motion $\partial \mathcal{L} / \partial a_0 = 0$, we obtain the number of electrodes

$$N_e = \mathbf{t}^T \mathbf{K}^{-1} \mathbf{t} N_\phi + 2 \mathbf{t}^T \mathbf{K}^{-1} \mathbf{s}. \quad (4.20)$$

For an electron system $\mathbf{t}^T \mathbf{K}^{-1} \mathbf{t} N_\phi = \nu N_\phi$ is always positive, and N_e is positive as expected. However, for a hole system (where the particles carry a charge of e), $\mathbf{t}^T \mathbf{K}^{-1} \mathbf{t} N_\phi = \nu N_\phi$ is always negative. To include this possibility, equation (4.20) should be generalized to

$$N_e = |\mathbf{t}^T \mathbf{K}^{-1} \mathbf{t} N_\phi + 2 \mathbf{t}^T \mathbf{K}^{-1} \mathbf{s}|. \quad (4.21)$$

Now the positive N_e can be regarded as the number of electrons or holes. We can see that the shift, for electron or hole systems, is given by

$$\mathcal{S} = \frac{2 \mathbf{t}^T \mathbf{K}^{-1} \mathbf{s}}{\mathbf{t}^T \mathbf{K}^{-1} \mathbf{t}} \text{sgn}(N_\phi) \quad (4.22)$$

on the sphere. We see that the spin vector determines the shift of the QH state.

Now the question is how to determine the value of the spin vector. The idea is that the FQH state described by equation (4.19) contains κ components, and each component is described by a Laughlin state. Thus we may use the result (4.18) to calculate the spin vector $\mathbf{s}$.

Let us first concentrate on the first component described by $a_{1\mu}$. Equation (4.19) can be rewritten as

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4\pi} K_{11} a_{1\mu} \partial_\nu a_{1\lambda} \varepsilon^{\mu\nu\lambda} + \left(-\sum_{I>1} \frac{1}{2\pi} K_{I1} a_{I\mu} - \frac{e}{2\pi} t_1 A_\mu \right) \partial_\nu a_{1\lambda} \varepsilon^{\mu\nu\lambda} \\ & + s_1 \omega_i \partial_\nu a_{1\lambda} \varepsilon^{i\nu\lambda} + \dots \end{aligned} \quad (4.23)$$

where $\dots$ represents terms that do not contain $a_{1\mu}$. Equations (4.23) and (4.17) have the same form except $-et_1 A_\mu$ in equation (4.17) is replaced by $-et_1 A_\mu - \sum_{I>1} K_{I1} a_{I\mu}$. Thus from equation (4.18) we see that

$$s_1 = \frac{|K_{11}|}{2} \text{sgn} \left(t_1 N_\phi - \sum_{I>1} K_{I1} \frac{b_I}{2\pi} \right) \quad (4.24)$$

if we assume the condensate occupies the first Landau level. Since $b_I/2\pi = (K^{-1})_{IJ}t_J N_\phi$, equation (4.24) can be rewritten as

$$s_1 = \frac{|K_{11}|}{2} \operatorname{sgn} \left[\left(t_1 - \sum_{I=2, J=1}^{\kappa} K_{I1} (K^{-1})_{IJ} t_J \right) N_\phi \right] = \frac{K_{11}}{2} \operatorname{sgn} \left(\sum_J (K^{-1})_{IJ} t_J N_\phi \right). \quad (4.25)$$

Generalizing the above to a general condensate, we get

$$s_I = \frac{K_{II}}{2} \operatorname{sgn} \left(\sum_J (K^{-1})_{IJ} t_J N_\phi \right). \quad (4.26)$$

Note that $t_I \sum_J (K^{-1})_{IJ} t_J N_\phi$ is the number of charges (in the unit of electron charge $-e$) in the I th condensate.

To obtain equation (4.26) we have assumed that each condensate forms a Laughlin state in the first Landau level, which is one of the assumptions in the hierarchical construction. To construct more general states we may assume (although it may not be energetically favourable) that the I th condensate forms a Laughlin state in the $n_{I\text{th}}$ Landau level. In this case we should replace equation (4.26) by

$$s_I = \frac{K_{II}}{2} \operatorname{sgn} \left(\sum_J (K^{-1})_{IJ} t_J N_\phi \right) + (n_I - 1) \operatorname{sgn} \left(K_{II} \sum_J (K^{-1})_{IJ} t_J N_\phi \right), \quad (4.27)$$

where n_I is an integer.

An important consequence of equation (4.27) is the quantization of the spin vector. We see that twice the spin vector, $2\mathbf{s}$, is always an integer vector.

We would like to point out that equations (4.26) and (4.27) apply to both the hierarchical FQH states and the multilayer FQH state. In general we expect the stable states to have $n_I = 1$. For the multilayer states with $t_I = 1$ and K_{IJ} all positive, $(K^{-1})_{IJ} t_J N_\phi$ is the number of electrons in the I th layer which is positive. Therefore

$$s_I = \frac{K_{II}}{2}. \quad (4.28)$$

In particular for the (lmn) state,

$$\mathbf{s} = \begin{pmatrix} l/2 \\ m/2 \end{pmatrix}. \quad (4.29)$$

For the hierarchical states with $\mathbf{t}^T = (1, 0, 0, \dots)$, equation (4.26) reduces to

$$s_I = \frac{K_{II}}{2} \operatorname{sgn} [(K^{-1})_{I1} N_\phi], \quad (4.30)$$

which allows us to calculate the spin vectors for the hierarchical states listed in equation (2.34).

Combining equations (4.28) and (4.30) with equation (4.22), we can calculate the shift for the hierarchical or the multilayer states. Note that the shift $\mathcal{S}$ is independent of $\operatorname{sgn}(N_\phi)$ due to a cancellation, which is expected.

In the following we shall calculate the orbital spin quantum number of quasiparticle excitations [21, 61, 62] from the new effective theory (4.19) that contains the spin vector. Let us create a generic quasiparticle labelled l_i in the FQH state (4.19) on the sphere. The charge and the statistics of the quasiparticle are given by equation (2.30). In the presence of the quasiparticle the relation between the number of

electrons and the number of the flux quanta is no longer given by equation (4.21). There is an additional shift due to the quasiparticle. Again from the equation of motion $\partial(\mathcal{L} + \mathcal{L}_q)/\partial a_0 = 0$, we find that

$$N_e = \mathbf{t}^T \mathbf{K}^{-1} \mathbf{t} N_\phi + 2\mathbf{t}^T \mathbf{K}^{-1} \mathbf{s} + \mathbf{t}^T \mathbf{K}^{-1} \mathbf{l}. \quad (4.31)$$

To calculate the orbital spin of the quasiparticle, we move quasiparticle around a loop C which spans a solid angle Ω . There are two ways to calculate the induced phase Φ . First the phase Φ can be obtained as a sum of the phase induced by a magnetic field and the phase induced by the orbital spin:

$$\Phi = \frac{Q_q}{e} N_\phi \frac{\Omega}{2} + 2S_{\text{ob}}^q \frac{\Omega}{2}, \quad (4.32)$$

where $Q_q = e\mathbf{t}^T \mathbf{K}^{-1} \mathbf{l}$ is the charge and S_{ob}^q is the orbital spin of the quasiparticle. In fact, equation (4.32) can be viewed as the definition of the orbital spin S_{ob}^q of the quasiparticle. Note that N_ϕ in equation (4.32) is the number of the flux quanta in the presence of the quasiparticle and is given by equation (4.31). The same phase can also be calculated from the coupling

$$l_I a_{I\mu} j_\mu. \quad (4.33)$$

First let us assume that the densities of each condensate (i.e. the field strength of $a_{I\mu}$) is uniform on the sphere even in the presence of quasiparticle. This can be achieved by including the term

$$\mathcal{L}_K = g(f_{\mu\nu}^I)^2, \quad f_{\mu\nu}^I = \partial_\mu a_{I\nu} - \partial_\nu a_{I\mu} \quad (4.34)$$

in the effective theory (4.19) and assuming that g is very very large. In this limit, the phase induced by the coupling (4.33) can be easily calculated. From the equation of motion we find that

$$\int d^2x b_I = \int d^2x \varepsilon_{ij} \partial_i a_{Ij} = 2\pi K_{IJ}^{-1} (t_J N_\phi + 2s_J + l_J), \quad (4.35)$$

where we integrate over the whole sphere. Since b_I are constants on the sphere, we have

$$\Phi = l_I \frac{\Omega}{4\pi} \int d^2x b_I = l_I K_{IJ}^{-1} (t_J N_\phi + 2s_J + l_J) \frac{\Omega}{2}. \quad (4.36)$$

Although Φ in equation (4.36) is calculated in the large- g limit, in the following we would like to show that Φ is topological and is independent of g . Note that the solid angle Ω spanned by the loop C is defined only up to a multiple of 4π . The consistency of the theory requires that the ambiguity of Ω should only cause an ambiguity of multiples of 2π in the phase Φ . Thus the coefficient in front of $\Omega/2$ must be quantized as an integer for any value of g :

$$\mathbf{l}^T \mathbf{K}^{-1} \mathbf{t} N_\phi + 2\mathbf{l}^T \mathbf{K}^{-1} \mathbf{s} + \mathbf{l}^T \mathbf{K}^{-1} \mathbf{l} = \text{integer}. \quad (4.37)$$

This implies that the coefficient in front of $\Omega/2$ is independent of g and that equation (4.36) is valid for all values of g . Comparing equations (4.32) and (4.36), we find that

$$S_{\text{ob}} = \frac{1}{2} \mathbf{l}^T \mathbf{K}^{-1} \mathbf{l} + \mathbf{l}^T \mathbf{K}^{-1} \mathbf{s}. \quad (4.38)$$

It is interesting to see that the orbital spin of the quasiparticle consists of two contributions. The first term $\frac{1}{2} \mathbf{l}^T \mathbf{K}^{-1} \mathbf{l} = \theta/2\pi$ is directly related to the statistic θ of

the quasiparticle and satisfies the spin-statistics theorem. The second term is due to the spin vector of the condensates, which violate the spin-statistics theorem [21, 61, 62].

We would like to remark that for a given $\mathbf{K}$ matrix, equation (4.37) sometimes cannot be satisfied for any integer choice of N_ϕ , if $\mathbf{l}$ takes certain values. This implies that the quasiparticle labelled by these $\mathbf{l}$ cannot be created individually on a sphere [13 (a), (b)]. For example, one cannot create a single charge- $\frac{1}{4}$ quasiparticle on top of the $\nu = \frac{1}{2}$ (331) FQH state on a sphere, although one can create two of them by inserting a unit flux. This selection rule, however, does not invalidate the above calculation of S_{ob} , because mathematically we can always choose a non-integer value of N_ϕ .

From the construction discussed above we can easily obtain the spin vectors for the FQH states listed in equations (2.34) and (2.42). Using equation (4.28) we can calculate the orbital spin quantum numbers of various excitations. In particular, one can show that some neutral plasma modes carry non-trivial orbital spin quantum numbers. The conservation of the angular momentum implies that to these plasma modes through, for example, Raman scattering, the photon must transfer a certain amount of angular momentum to the plasma mode. By performing angle-resolved circularly polarized Raman scattering we can measure the amount of transferred angular momentum, which in turn measures the orbital spin of the plasma mode. Such measurement allows us to obtain information about the spin vector of the FQH state [61].

4.3. Wavefunctions and $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ of fractional quantum Hall liquids

In this section we shall study the relation between $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ (the $\mathbf{K}$ matrix, charge vector $\mathbf{t}$ and the spin vector $\mathbf{s}$) and the wavefunctions of FQH liquids. The results obtained here allow us easily to calculate $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ for various FQH liquids obtained in various construction schemes.

Let $\Psi_{\mathbf{K}_1 \mathbf{t}_1 \mathbf{s}_1}(\{z_i\})$ and $\Psi_{\mathbf{K}_2 \mathbf{t}_2 \mathbf{s}_2}(\{w_i\})$ be two wavefunctions of two FQH liquids labelled $(\mathbf{K}_1, \mathbf{t}_1, \mathbf{s}_1)$ and $(\mathbf{K}_2, \mathbf{t}_2, \mathbf{s}_2)$. Putting them together we obtain a new FQH liquid labelled $(\mathbf{K}_3, \mathbf{t}_3, \mathbf{s}_3)$:

$$\Psi_{\mathbf{K}_1 \mathbf{t}_1 \mathbf{s}_1}(\{z_i\}) \Psi_{\mathbf{K}_2 \mathbf{t}_2 \mathbf{s}_2}(\{w_i\}) = \Psi_{\mathbf{K}_3 \mathbf{t}_3 \mathbf{s}_3}(\{z_i\}, \{w_i\}). \quad (4.39)$$

Obviously

$$\begin{aligned} \mathbf{K}_3 &= \begin{pmatrix} \mathbf{K}_1 & 0 \\ 0 & \mathbf{K}_2 \end{pmatrix} = \mathbf{K}_1 \oplus \mathbf{K}_2, \\ \mathbf{t}_3 &= \begin{pmatrix} \mathbf{t}_1 \\ \mathbf{t}_2 \end{pmatrix} = \mathbf{t}_1 \oplus \mathbf{t}_2, \\ \mathbf{s}_3 &= \begin{pmatrix} \mathbf{s}_1 \\ \mathbf{s}_2 \end{pmatrix} = \mathbf{s}_1 \oplus \mathbf{s}_2. \end{aligned} \quad (4.40)$$

We also know that $\Psi_{\mathbf{K} \mathbf{t} \mathbf{s}}^t(\{z_i\})$ is a wavefunction for the holes (with charge e) which will be called the charge conjugate state of the $\Psi_{\mathbf{K} \mathbf{t} \mathbf{s}}$ state. The Hall conductance of the charge conjugate state has an opposite sign to that of the original state. Note that the transformation

$$\mathbf{t} \rightarrow -\mathbf{t}, \quad \mathbf{s} \rightarrow -\mathbf{s}, \quad \mathbf{K} \rightarrow -\mathbf{K}, \quad a_{I\mu} \rightarrow a_{I\mu}, \quad (4.41)$$

in the effective theory (4.19) leaves the equation motion unchanged. However, the

transformation changes the sign of the Hall conductance and the sign of electric current $J_e^\mu = -(e/2\pi)t_I \partial_\nu a_{I\lambda} \varepsilon^{\mu\nu\lambda}$. Thus,

$$\Psi_{\mathbf{K}\mathbf{t}\mathbf{s}}^\dagger = \Psi_{-\mathbf{K}, -\mathbf{t}, -\mathbf{s}} \quad (4.42)$$

The charge conjugation changes the signs of $\mathbf{t}$, $\mathbf{s}$ and $\mathbf{K}$.

Given a spin-polarized FQH state in a single layer at a filling fraction $\nu_{\mathbf{K}\mathbf{t}}$ and labelled $(\mathbf{K}, \mathbf{t}, \mathbf{s})$, there is a particle-hole conjugate state at the filling fraction $\nu = 1 - \nu_{\mathbf{K}\mathbf{t}}$. This conjugate state is obtained simply by putting the $\nu = 1$ IQH state and the charge conjugate of the $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ state together. Thus the label of the $1 - \nu_{\mathbf{K}\mathbf{t}}$ state is given by

$$\mathbf{K}_{1-\nu_{\mathbf{K}\mathbf{t}}} = 1_{1 \times 1} \oplus (-\mathbf{K}), \quad \mathbf{t}_{1-\nu_{\mathbf{K}\mathbf{t}}} = 1 \oplus (-\mathbf{t}), \quad \mathbf{s}_{1-\nu_{\mathbf{K}\mathbf{t}}} = \frac{1}{2} \oplus (-\mathbf{s}). \quad (4.43)$$

Similarly, given a spin-polarized FQH state in double layers at a filling fraction $\nu_{\mathbf{K}\mathbf{t}}$ and labelled $(\mathbf{K}, \mathbf{t}, \mathbf{s})$, there is a particle-hole conjugate state at the filling fraction $\nu = 2 - \nu_{\mathbf{K}\mathbf{t}}$. Such a conjugate state is labelled

$$\mathbf{K}_{2-\nu_{\mathbf{K}\mathbf{t}}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \oplus (-\mathbf{K}), \quad \mathbf{t}_{2-\nu_{\mathbf{K}\mathbf{t}}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \oplus (-\mathbf{t}), \quad \mathbf{s}_{2-\nu_{\mathbf{K}\mathbf{t}}} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \oplus (-\mathbf{s}). \quad (4.44)$$

From the two FQH states $\Psi_{\mathbf{K}_1\mathbf{t}_1\mathbf{s}_1}(\{z_i\})$ and $\Psi_{\mathbf{K}_2\mathbf{t}_2\mathbf{s}_2}(\{w_i\})$, we can construct a new FQH state by multiplying the two wavefunctions together and setting $z_i = w_i$:

$$\Psi_{\mathbf{K}_1\mathbf{t}_1\mathbf{s}_1}(\{z_i\}) \Psi_{\mathbf{K}_2\mathbf{t}_2\mathbf{s}_2}(\{w_i\})|_{z_i=w_i} = \Psi_{\mathbf{K}_3\mathbf{t}_3\mathbf{s}_3}(\{z_i\}). \quad (4.45)$$

To calculate $(\mathbf{K}_3, \mathbf{t}_3, \mathbf{s}_3)$ it is convenient to use the $SL(n, Z)$ transformation

$$\mathbf{K} \rightarrow \mathbf{W}\mathbf{K}\mathbf{W}^T, \quad \mathbf{t} \rightarrow \mathbf{W}\mathbf{t}, \quad \mathbf{t} \rightarrow \mathbf{W}\mathbf{t}, \quad \mathbf{s} \rightarrow \mathbf{W}\mathbf{s} \quad (4.46)$$

to transform $\mathbf{t}_1$ and $\mathbf{t}_2$ into $\mathbf{t}_{1,2}^T = (1, 0, 0, \dots)$. This is possible if the elements in $\mathbf{t}_{1,2}$ do not contain common integer factors (i.e. the largest common divisor is one). In the new basis $\mathbf{K}_{1,2}$ have the form

$$\mathbf{K}_{1,2} = \begin{pmatrix} m_{1,2} & \mathbf{k}_{1,2}^T \\ \mathbf{k}_{1,2} & \tilde{\mathbf{K}}_{1,2} \end{pmatrix}, \quad (4.47)$$

where $m_{1,2}$ are odd integers, $\mathbf{k}_{1,2}$ are integer vectors in $\dim \mathbf{K}_{1,2} - 1$ dimensions, and $\tilde{\mathbf{K}}_{1,2}$ are $(\dim \mathbf{K}_{1,2} - 1) \times (\dim \mathbf{K}_{1,2} - 1)$ integer matrices with even diagonal elements (see section 6). Accordingly the spin vectors have the form

$$\mathbf{s}_{1,2} = \begin{pmatrix} s_{1,2}^1 \\ \tilde{\mathbf{s}}_{1,2} \end{pmatrix}, \quad (4.48)$$

where $s_{1,2}^1$ are half odd integers and $\tilde{\mathbf{s}}_{1,2}$ are integer vectors in $\dim \mathbf{K}_{1,2} - 1$ dimensions.

Using the $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ in the new basis, one can show that $(\mathbf{K}_3, \mathbf{t}_3, \mathbf{s}_3)$ are

given by

$$\begin{aligned} \mathbf{K}_3 &= \begin{pmatrix} m_1 + m_2 & \mathbf{k}_1^T & \mathbf{k}_2^T \\ \mathbf{k}_1 & \mathbf{K}_1 & 0 \\ \mathbf{k}_2 & 0 & \tilde{\mathbf{K}}_2 \end{pmatrix}, \\ \mathbf{t}_3^T &= (1, 0, 0, \dots), \\ \mathbf{s}_3 &= \begin{pmatrix} s_1^1 + s_2^1 \\ \tilde{s}_1 \\ \tilde{s}_2 \end{pmatrix}. \end{aligned} \quad (4.49)$$

The results in equation (4.49) are obtained from the following consideration. Before imposing the constraint $z_i = w_i$, the effective theory just describes the two independent FQH states and has the form

$$\mathcal{L} = -\frac{1}{4\pi} K_{1IJ} a_{I\mu}^1 \partial_\nu a_{J\lambda}^1 \varepsilon^{\mu\nu\lambda} + s_{1I} \omega_i \partial_\nu a_{I\lambda}^1 \varepsilon^{i\nu\lambda} - \frac{1}{4\pi} K_{2IJ} a_{I\mu}^2 \partial_\nu a_{J\lambda}^2 \varepsilon^{\mu\nu\lambda} + s_{2I} \omega_i \partial_\nu a_{I\lambda}^2 \varepsilon^{i\nu\lambda}. \quad (4.50)$$

After imposing the constraint $z_i = w_i$, the charge currents in the two FQH states $j_{1,2}^\mu = (1/2\pi) \varepsilon^{\mu\nu\lambda} \partial_\nu a_{1\lambda}^1$ are always equal to each other $j_1^\mu = j_2^\mu$ and the relative fluctuations $j_1^\mu - j_2^\mu$ are not allowed. Thus the constraint $z_i = w_i$ can be realized in the effective theory by setting $a_{1\mu}^1 = a_{1\mu}^2$. This changes the effective theory (4.50) into one with $(\mathbf{K}, \mathbf{s})$ given by equation (4.49). The charge vector can be obtained by realizing that A_μ only couples with the charge current $j_1^\mu = j_2^\mu$.

We would like to remark that equation (4.49) is not always valid. Let χ_n be the wavefunction of the $\nu = n$ integer QH state with first n Landau levels filled. Then the FQH state described by $\chi_1(\{z_i\})(\chi_2(\{z_i\}))^2$ is in fact a $\nu = \frac{1}{2}$ non-Abelian state rather than an Abelian state as implied by equation (4.49). More detailed discussions can be found in [63].

However, equation (4.49) does apply to the following wavefunction $(\chi_1)^p \Psi_{\mathbf{Kts}}$. Note that $(\chi_1)^p$ is just the wavefunction of the $1/p$ Laughlin state. From equation (4.49), one can show that multiplying $(\chi_1)^p$ to a FQH wavefunction $\Psi_{\mathbf{Kts}}$ changes $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ to

$$\mathbf{K} \rightarrow \mathbf{K} + p\mathbf{t}\mathbf{t}^T, \quad \mathbf{t} \rightarrow \mathbf{t}, \quad \mathbf{s} \rightarrow \mathbf{s} + \frac{p}{2}\mathbf{t}. \quad (4.51)$$

A sequence of FQH states, $\nu = \frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \dots, n/(2n+1), \dots$, under Jain's construction, are described by the wavefunctions $\chi_1^2 \chi_n$. The $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ for the IQH state χ_n are given by

$$\mathbf{K} = 1_{n \times n}, \quad \mathbf{t}^T = (1, \dots, 1), \quad \mathbf{s}^T = \left(\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots, \frac{2n-1}{2} \right). \quad (4.52)$$

Thus the $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ for the $\nu = n/(2n+1)$ state are

$$\mathbf{K} = 1_{n \times n} + 2\mathbf{C}, \quad \mathbf{t}^T = (1, \dots, 1), \quad \mathbf{s}^T = \left(\frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \dots, \frac{2n+1}{2} \right), \quad (4.53)$$

where $\mathbf{C}$ is a matrix with all its elements equal to 1. The hierarchical construction

can also generate a sequence of FQH states with filling fraction $\nu = n/(2n + 1)$. The $\mathbf{K}$ matrix obtained from the hierarchical construction is a $n \times n$ tridiagonal matrix with off diagonal elements equal to -1 and diagonal elements equal to 2 , except $K_{11} = 3$. The charge and the spin vectors are given by $\mathbf{t}^T = (1, 0, \dots)$ and $\mathbf{s}^T = (\frac{3}{2}, 1, 1, \dots)$. Although the two sets of $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ obtained from Jain's construction and the hierarchical construction are very different, one can show that they are equivalent under a $SL(n, Z)$ transformation. In fact the matrix $W_{IJ} = \delta_{IJ} - \delta_{I-1, J}$ transforms the $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ in equation (4.53) into those obtained from the hierarchical construction through equation (4.46). Therefore the $\nu = n/(2n + 1)$ states obtained from the Jain's construction and the hierarchical construction belong to the same universality class.

Another interesting example is that the following three sets of $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ describe the same $\nu = \frac{2}{3}$ FQH state:

$$\begin{aligned} \mathbf{K} &= \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, & \mathbf{t}^T &= (1, 1), & \mathbf{s}^T &= (\frac{1}{2}, -\frac{1}{2}), \\ \mathbf{K} &= \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, & \mathbf{t}^T &= (1, 0), & \mathbf{s}^T &= (\frac{1}{2}, -1), \\ \mathbf{K} &= \begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}, & \mathbf{t}^T &= (1, -1), & \mathbf{s}^T &= (\frac{1}{2}, -\frac{3}{2}). \end{aligned} \quad (4.54)$$

In fact,

$$\mathbf{W} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{W} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

map and the first set of $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ into the second and the third sets respectively. The first set of $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ is obtained from the Jain's construction with a wavefunction $\chi_2^*(\chi_1)^2$, the second set from the hierarchical construction, and the third set is obtained by simply putting a $\nu = 1$ QH state of electrons and a $\nu = -\frac{1}{3}$ QH states of holes together.

The above three sets of $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ describe the same spin-polarized single-layer $\frac{2}{3}$ state. However, a double layer or a spin singlet $\frac{2}{3}$ state is described by

$$\mathbf{K} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{t}^T = (1, 1), \quad \mathbf{s}^T = (\frac{1}{2}, \frac{1}{2}), \quad (4.55)$$

which is inequivalent to the $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ of the spin-polarized single-layer $\frac{2}{3}$ state. The two kinds of $\frac{2}{3}$ state are separated by a first-order transition (or other FQH states) in clean systems.

5. Classification of Abelian Hall states

In sections 2 and 4, we have derived the effective theories and the form of $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ for some FQH liquids from some specific construction schemes. In this section we shall derive the form of the $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ from some general principles that an electron system must satisfy. This allows us to obtain a classification of most general Abelian FQH states.

For a class of generalized hierarchical states, the effective theory is given by

equation (4.19) with $(\mathbf{K}, \mathbf{t}, \mathbf{s})$, satisfying

$$\begin{aligned}
 K_{11}^h &= \text{odd integer}, \\
 K_{II}^h |_{I>1} &= \text{even integer}, \\
 K_{IJ}^h &= \text{integer}, \quad \text{for } I \neq J, \\
 t_I^h &= \delta_{1I}, \\
 s_I^h &= \pm \frac{K_{II}^h}{2}.
 \end{aligned} \tag{5.1}$$

We shall call this basis the hierarchical basis, as indicated by the superscript h . We also constructed the effective theory for the simple multilayer FQH states with $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ having the form

$$\begin{aligned}
 K_{II}^s &= \text{odd integer}, \\
 K_{IJ}^s &= \text{integer}, \quad \text{for } I \neq J, \\
 t_I^s &= 1, \\
 s_I^s &= \pm \frac{K_{II}^s}{2}.
 \end{aligned} \tag{5.2}$$

Such a basis will be called the symmetric basis, which is indicated by the superscript s .

First we would like to mention that using the transformation (4.46) and the matrix $W_{IJ} = \delta_{IJ} - \delta_{1-I, J}$ in the group $SL(\dim(\mathbf{K}), \mathbb{Z})$, we can transform $(\mathbf{K}^s, \mathbf{t}^s)$ in the symmetric basis satisfying equation (5.2) into a corresponding representation in the hierarchical basis satisfying equation (5.1). $\mathbf{W}^{-1}$ is also an element of $SL(\dim(\mathbf{K}), \mathbb{Z})$ and will transform in the opposite direction. Thus the matrices satisfying equations (5.1) and (5.2) have a one-to-one correspondence. However, in general the spin vectors in the symmetric basis do not transform into the corresponding spin vector in the hierarchical basis. To obtain the spin vectors in equations (5.1) and (5.2) we have assumed that the quasiparticles and the electrons always condense into Laughlin states in the ‘first’ Landau level. If we allow them to form Laughlin states in higher Landau levels, more general spin vectors will be generated:

$$\begin{aligned}
 s_I^h &= \pm \frac{K_{II}^h}{2} + \text{integer}, \\
 s_I^s &= \pm \frac{K_{II}^s}{2} + \text{integer}.
 \end{aligned} \tag{5.3}$$

In this case $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ in the two basis will have a one-to-one correspondence under the transformation $\mathbf{W}$.

Even though the $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ given above are quite general, it is not clear whether these $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ cover all possible Abelian FQH states or not. In the following we shall derive the effective theory of the most general Abelian FQH states from some general principles. This discussion will help us understand why the $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ have the form described in equations (5.1) and (5.2) and what the physical reasons are behind it. We shall show that the generalized hierarchical states and the multilayer states represent the most general Abelian FQH states (if we generalize the spin vectors to those in equation (5.3)).

Our first working assumption is that the effective theory of Abelian FQH states is described by a Lagrangian of the form (4.19). Certainly this assumption alone is not enough. For arbitrary choices of $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ the Lagrangian (4.19) may not describe an electron system. So the problem that we are facing is not how to derive the effective theory of an electron system, but the reverse, that is how to determine whether an effective theory is consistent with the underlying electron system or not. This leads us to our second working assumption. In order for the effective theory to describe an electron system, the effective theory must contain κ independent electron operators, where κ denotes the rank of $\mathbf{K}$. (Here we ignore the possibility of electron pairing). We shall discuss why we need κ electron operators later.

To implement our second assumption, it is convenient to work in the electron basis:

$$\mathcal{L} = -\frac{1}{4\pi} \tilde{a}_{I\mu} \tilde{K}_{IJ} \varepsilon^{\mu\nu\lambda} \partial_\nu \tilde{a}_{J\lambda} - \frac{e}{2\pi} \tilde{t}_I A_\mu \varepsilon^{\mu\nu\lambda} \partial_\nu \tilde{a}_{I\lambda} + s_I \omega_i \tilde{e}^{i\nu\lambda} \partial_\nu \tilde{a}_{I\lambda} + \tilde{a}_{I\mu} \tilde{j}_I^\mu. \quad (5.4)$$

We have used a tilde on the symbols to remind ourselves that we are in the electron basis. The electron basis is defined by the requirement that a source term that carries a unit $\tilde{a}_{I\mu}$ charge generates an electron. Thus $\tilde{j}_I$ in equation (5.4) represents the current of the I th electron excitations above the ground state.

In order for the electrons to have Fermi statistics, we must have

$$(\tilde{K}^{-1})_{IJ} \equiv K_{IJ} = \begin{cases} \text{odd integer} & \text{for } I = J, \\ \text{integer} & \text{for } I \neq J. \end{cases} \quad (5.5)$$

Furthermore, in order for the electrons to carry unit charge, $\tilde{t}$ must satisfy

$$\sum_J (\tilde{K}^{-1})_{IJ} \tilde{t}_J = 1 \quad \text{for all } I. \quad (5.6)$$

This equation implies that

$$\tilde{t}_J = \sum_L \tilde{K}_{JL}. \quad (5.7)$$

To find the condition imposed on the spin vector $\tilde{s}_I$, let us put the FQH state on a sphere. We know that the ground state on the sphere form the singlet representation of the $su(2)$ rotation of the sphere. Any neutral excitations above the ground state must carry integer $su(2)$ spins. As we add an electron to the system, the electron excitation must form an $S^{su(2)} = (N_\phi/2) + \text{integer representation of } su(2)$. From the relation $S^{su(2)} = N_\phi/2 + S_{\text{ob}}$ we find the orbital spin S_{ob} of an electron excitation must be an integer. Therefore (see equation (4.38))

$$S_{\text{ob}}^I = \sum_J (\tilde{K}^{-1})_{IJ} \tilde{s}_J + \frac{1}{2} (\tilde{K}^{-1})_{II} = \text{integer}. \quad (5.8)$$

Given the effective theory in the electron basis, we now need to determine the allowed quasiparticle excitations. A generic quasiparticle is created by a source term

$$\mathcal{L} = c_I a_{I\mu} j^\mu, \quad (5.9)$$

whose a_I charge is equal to c_I . Since the electron wavefunction must be single valued (even in the presence of the quasiparticle), the phase induced by moving an electron

around the quasiparticle must be a multiple of 2π . This requires c_I to satisfy

$$\sum_{J=1}^{\kappa} (\tilde{K}^{-1})_{IJ} c_J = \text{integer for all } I. \quad (5.10)$$

(Note that moving a quasiparticle described by c_I around that described by c'_I induces a phase $2\pi c_I (\tilde{K}^{-1})_{IJ} c'_J$.) The set of all c_J satisfying equation (6.7) form a κ -dimensional lattice, with the basis vectors of the lattice $\mathbf{c}^{(L)}$: $L = 1, 2, \dots, \kappa$. The components of $\mathbf{c}^{(L)}$ are given by

$$c_J^{(L)} = \tilde{K}_{JL}. \quad (5.11)$$

The $a_{I\mu}$ charge $\mathbf{c}$ of any allowed excitations, including the electron excitations, will be linear combinations of $\mathbf{c}^{(L)}$ with integer coefficients:

$$\mathbf{c} = \sum_L l_L \mathbf{c}^{(L)}. \quad (5.12)$$

Such a quasiparticle (labelled ψ_I) is created by the following source term:

$$\mathcal{L} = j^\mu \tilde{a}_{I\mu} (\tilde{K}^{-1})_{IJ} l_J. \quad (5.13)$$

Now let us redefine the gauge fields and change to the so-called quasiparticle basis, in which the fundamental quasiparticles described by $\mathbf{c}^{(L)}$ will carry the unit charge of the new gauge fields. From equation (5.11) we see that the substitution $a_I = \sum_J \tilde{K}_{IJ} \tilde{a}_J$ in equations (5.4) and (5.13) will lead to

$$\mathcal{L} = -\frac{1}{4\pi} a_{I\mu} K_{IJ} \epsilon^{\mu\nu\lambda} \partial_\nu a_{J\lambda} - \frac{e}{2\pi} t_I A_\mu \epsilon^{\mu\nu\lambda} \partial_\nu a_{I\lambda} + s_I \omega_i \epsilon^{i\nu\lambda} \partial_\nu a_{I\lambda} + l_I a_{I\mu} j^\mu. \quad (5.14)$$

The $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ in equation (5.14) are related to $(\tilde{\mathbf{K}}, \tilde{\mathbf{t}}, \tilde{\mathbf{s}})$ in equation (5.4) through

$$\mathbf{K} = \tilde{\mathbf{K}}^{-1}, \quad \mathbf{t} = \mathbf{K}\tilde{\mathbf{t}}, \quad \mathbf{s} = \mathbf{K}\tilde{\mathbf{s}}. \quad (5.15)$$

Equation (5.14) is exactly the same as the effective theory (4.19) that we wrote down in section 4, with $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ satisfying equations (5.2) and (5.3) as one can see from equations (5.6) and (5.8).

We have reached an important result: the most general Abelian FQH states of unpaired electrons are described by the effective theory (5.14) with $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ satisfying equation (5.2). In other words, the topological orders in the Abelian FQH states are labelled by integer valued symmetric matrices with odd diagonal elements and a spin vector $\mathbf{s}$, up to an equivalency condition $\mathbf{K} \sim \mathbf{W}\mathbf{K}\mathbf{W}^T$ and $\mathbf{s} \sim \mathbf{W}\mathbf{s}$ with $\mathbf{W}$ an element of $SL(\kappa, Z)$ which leaves the vectors $\mathbf{t} = (1, 1, \dots, 1)$ invariant. The determinant of $\mathbf{W}$ must be equal to 1 in order to preserve the charge quantization condition.

The above discussions also reveal that the condition K_{II}^s odd is a consequence of the Fermi statistics of the underlying electrons. FQH liquids formed by bosons will have K_{II}^s even in the symmetric basis.

We would next like to say a few words about why we need $\kappa = \dim(\mathbf{K})$ independent electron operators. If we had less than κ electron operators, the condition (5.10) would become $\sum_{J=1}^{\kappa'} (\tilde{K}^{-1})_{IJ} c_J = \text{integer}$ only for $I = 1, \dots, \kappa'$, where $\kappa' < \kappa$. This is not enough to fix c_J on a lattice. In this case, excitations with arbitrarily small charge would be allowed. Such excitations may be continuously connected to the ground state and would be, we believe, gapless. This argument suggests that, in order for the effective theory to have finite energy gap, we require the presence of κ different

electron operators. (Of course, we also require $\det \mathbf{K} \neq 0$ in order for the gauge fluctuations to have finite gaps.)

We would like to remark that here we have only shown that a generic Abelian state must be described by $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ that satisfies equation (5.1) or (5.2). However, $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ here merely appear as some parameters in the effective theory. Only certain combinations are physically measurable. Thus different $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ may describe the same topological orders. At least the $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ related through the $SL(n, Z)$ transformation (4.46) describe the same topological order. However, it may be possible that $(\mathbf{K}, \mathbf{t}, \mathbf{s})$ with even different dimensions might describe the same topological order. More details can be found in [6]. An attempt to describe topological orders using only measurable quantities can be found in [6]. Clearly more work needs to be done to obtain a more complete and more satisfactory picture of Abelian topological orders.

6. Algebraic approach and non-Abelian fractional quantum Hall liquids

So far in this article we have discussed only a small class of FQH liquids: Abelian FQH liquids. It was first pointed out by Moore and Read [23] that another type of FQH liquid with quasiparticles of non-Abelian statistics is also possible. Non-Abelian FQH liquids cannot be described by Abelian Chern–Simons effective theory; thus the approaches previously used to study the edge states and quasiparticle quantum numbers can no longer be used for non-Abelian states. We need to develop new methods to calculate the physical properties of non-Abelian states.

At the moment, we know two ways to construct non-Abelian FQH states. The first method uses the parton construction and the second method uses the conformal field theory (CFT). The parton construction allows us to construct FQH states whose effective theory is a non-Abelian Chern–Simons theory [24, 28, 33]. For example $\chi_1(\chi_2)^2$ is a $\nu = \frac{1}{2}$ non-Abelian state described by a $U(1) \times SU(2)$ Chern–Simons theory. The CFT approach allows us to construct more general non-Abelian states [23, 25, 28, 64]. The effective theories of many non-Abelian states obtained from CFT are not known at this time.

In this section we shall concentrate on the edge properties of non-Abelian FQH states. We shall develop an algebraic method which allows us to calculate the properties of the edge states for the FQH states constructed using CFT. Our approach is motivated by an observation by Moore and Read [23] that the Laughlin wavefunction can be written as a correlation function between the charge operators (i.e. the vertex operators) in the $U(1)$ K–M algebra. The $U(1)$ K–M algebra is a CFT which is also called the Gaussian model. (For readers not familiar with CFT, we suggest [65, 66].) The observation of Moore and Read links the Laughlin wavefunction to a CFT. We shall call this CFT the bulk CFT. From our discussion in section 3 we see that the edge excitations of the Laughlin state are also described by a CFT: the $U(1)$ K–M algebra. Such a CFT will be called the edge CFT of the FQH liquid. It is very interesting to see that, for a Laughlin state, the bulk CFT is identical with the edge CFT. It turns out that such a relation can be extended to more general Abelian and non-Abelian states. In this section we shall explain how the bulk CFT and the edge CFT are connected and how to use such a relation to calculate physical properties of non-Abelian edge states.

However, we would like to stress that, to relate the bulk CFT that generates the bulk wavefunction to the edge CFT that describes the edge spectrum, we need to introduce a concept of minimal CFT for the bulk CFT. This is because, once a FQH

wavefunction can be written as a correlation function in a certain CFT, then the wavefunction can also be expressed as a correlation in many other CFTs which contain the original CFT. The minimal (bulk) CFT for a FQH state reproduces the wavefunction, and by definition it is contained by any other CFT that reproduces the wavefunction. As we shall see later, it is this minimal bulk CFT that is identical with the edge CFT that describes the edge excitations of the corresponding FQH state. Thus it is very important to identify the minimal bulk CFT.

6.1. Ideal Hamiltonians, zero-energy states and edge excitations

In this section we shall describe a microscopic theory for the edge excitations in the Laughlin states. Let us consider an electron gas in the first Landau level. We choose the interaction between electrons to be $V(\mathbf{r}) \propto -\delta^2\delta(\mathbf{r})$. Note that firstly $H_V = \sum_{ij} V(\mathbf{r}_i - \mathbf{r}_j)$ is positive definite (i.e. $\langle\psi|H_V|\psi\rangle \geq 0$ for any $|\psi\rangle$) and secondly H_V has zero expectation value for the $\nu = \frac{1}{3}$ Laughlin wavefunction,

$$\Phi_3(\{z_i\}) = \prod_{i < j} (z_i - z_j)^3 \prod_k \exp(-\frac{1}{4}|z_k|^2). \quad (6.1)$$

Thus Φ_3 is an exact ground state of our Hamiltonian with zero energy. This Hamiltonian H_V will be called the ideal Hamiltonian of the $\frac{1}{3}$ Laughlin state. However, the Laughlin state (6.1) is not the only state with zero energy. One can easily check that the following types of state all have zero energy [48]:

$$\Phi(\{z_i\}) = P(\{z_i\})\Phi_3(\{z_i\}), \quad (6.2)$$

where $P(z_i)$ is a symmetric polynomial of z_i so that $\Phi(\{z_i\})$ remains an antisymmetric function. In fact, the reverse is also true; all the zero-energy states are of the form (6.2). This is because, in order for a fermion state to have zero energy, Φ must vanish at least as fast as $(z_i - z_j)^3$ when any two electrons i and j are brought together (the possibility of $(z_i - z_j)^2$ is excluded by the fermion statistics). The Laughlin wavefunction is zero only when $z_i = z_j$; therefore $P = \Phi/\Phi_3$ is a finite function. Since Φ and Φ_3 are both antisymmetric functions in the first Landau level, P is a finite symmetric holomorphic function that can only be a symmetric polynomial.

Among all the states in equation (6.2), the Laughlin state describes a circular droplet with smallest radius. All other states, having higher angular momenta, are deformations and/or inflations of the droplet of the Laughlin state. Thus the states generated by P , being low-energy excitations above the ground state Φ_3 , correspond to the edge excitations of the Laughlin state.

Now let us study the structures of the zero-energy space (i.e. the space of symmetric polynomials). We know that the space of symmetric polynomials is generated by the polynomials $s_n = \sum_i z_i^n$ (through multiplication and addition). Let $M_0 = 3[N(N-1)/2]$ be the total angular momentum of the Laughlin state (6.1). Then the state Φ will have an angular momentum $M = \Delta M + M_0$ where ΔM is the order of the symmetric polynomial P . Since we have only one order-zero and order-one symmetric polynomial $s_0 = 1$ and $s_1 = \sum_i z_i$, the zero-energy states for $\Delta M = 0, 1$ are non-degenerate. However, when $\Delta M = 2$, we have two zero-energy states generated by $P = s_2$ and $P = s_1^2$. For general ΔM the degeneracy of the zero energy states is given by

$$\begin{array}{cccccccc} \Delta M = & 0 & 1 & 2 & 3 & 4 & 5 & 6, \\ \text{degeneracy of} & 1 & 1 & 2 & 3 & 5 & 7 & 11. \end{array} \quad (6.3)$$

Here we would like to point out that the degeneracy in equation (6.3) is exactly what we expected from the macroscopic theory. We know for a circular droplet, the angular momentum ΔM can be regarded as the momentum along the edge $k = 2\pi \Delta M/L$ where L is the perimeter of the QH droplet. According to the macroscopic theory the (neutral) edge excitations are generated by the density operators ρ_k . One can easily check that the edge states generated by the density operators have same degeneracies as those in equation (6.3) for every ΔM , for example the two states at $\Delta M = 2$ are generated by $\rho_{\kappa_0}^2$ and $\rho_{2\kappa_0}$ where $\kappa_0 = 2\pi/L$. Therefore the space generated by the K–M algebra (3.7) and the space of the symmetric polynomials are identical.

There seems to be one problem in the above identification. The states generated by symmetric polynomials P all have zero energy, while the edge excitations in the macroscopic theory have finite energies: $E \propto k \propto \Delta M$. This is because we did not include the confining potential in our ideal Hamiltonian. A confining potential for the FQH droplet will lift the degeneracy and give edge excitations an energy $E \propto \Delta M$.

Now let us ask a physical question. Do the symmetric polynomials generate all the low-energy states? If this is true, the space of the low-energy edge excitations will be identical with the space of symmetric polynomial (or the zero-energy space) [48]. Unfortunately, up to now we do not have analytical proof of the above statement. This is because, although states orthogonal to the states generated by the symmetric polynomials have non-zero energies, it is not clear that those energies remain finite in the thermodynamical limit. It might be possible that the energy gap approaches zero in the thermodynamical limit. To resolve this problem, for now we have to rely on numerical calculations. In figure 3 we present the energy spectrum of a system of six electrons in the first 22 orbits for the Hamiltonian introduced at the beginning of this section. The degeneracies of the zero-energy states at $M = 45, \dots, 51$ (or $\Delta M = 0, \dots, 6$) are found to be 1, 1, 2, 3, 5, 7, 11, which agrees with equation (6.3). More importantly, we can clearly see a finite energy gap that separates all other states from the zero-energy states. Thus the numerical results imply that all the low-lying edge excitations of the Laughlin state are generated by the symmetric

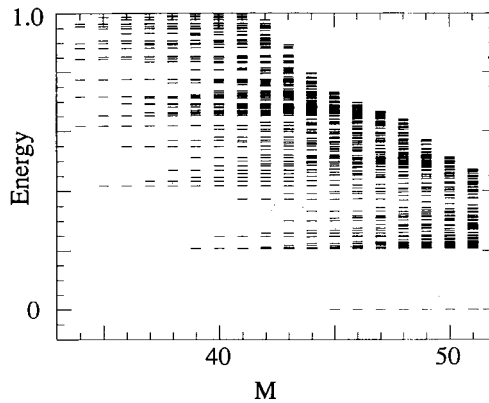


Figure 3. The energy spectrum of a system of six electrons in the first 22 orbits for the two-body Hamiltonian H_V . The degeneracies of the zero-energy states at $M = 45, \dots, 51$ are found to be 1, 1, 2, 3, 5, 7, 11.

polynomials or the K–M algebra (3.7). Using the above assumption of the energy gap, we see that the edge excitations in an FQH liquid can be studied through the zero-energy states of the ideal Hamiltonian for that FQH liquid.

6.2. Edge excitations and current algebra

In this section we are going to study edge excitations of the Laughlin state using a current algebra approach. The algebraic approach developed here can be easily generalized to study the edge excitations of non-Abelian states.

We note that the $\nu = 1/m$ Laughlin state

$$\prod_{i < j} (z_i - z_j)^m \exp \left(-\frac{1}{4} \sum_i |z_i|^2 \right) \equiv \Phi_m(\{z_i\}) \exp \left(-\frac{1}{4} \sum_i |z_i|^2 \right) \quad (6.4)$$

is a zero-energy state of Hamiltonian with the following two-body interaction [67]:

$$V(z_1 - z_2) = - \sum_{n=1}^{(m-1)/2} C_n \partial_{z_1}^{2n-1} \delta(z_1 - z_2) \partial_{z_1}^{2n-1}, \quad (6.5)$$

where $C_n > 0$. (The Haldane pseudopotentials [12] for the above interaction are given by $V_{2l+1} > 0$ for $2l+1 < m$ and $V_{2l+1} = 0$ for $2l+1 \geq m$.) Thus equation (6.5) defines an ideal Hamiltonian for the $1/m$ Laughlin state. Other zero-energy states of the ideal Hamiltonian form the edge excitation of the $1/m$ state. From equation (6.5) we see that any antisymmetric wavefunction of form

$$\tilde{\Phi}_m(\{z_i\}) \exp \left(-\frac{1}{4} \sum_i |z_i|^2 \right) \quad (6.6)$$

has zero energy, if and only if $\tilde{\Phi}_m$ does not contain zeros of order less than m , that is

$$\tilde{\Phi}_m(z_i) = O[(z_1 - z_2)^m] \quad (6.7)$$

as $z_1 \rightarrow z_2$.

Now the question is how to construct the Hilbert space of these zero-energy states. One approach is to use the symmetric polynomial discussed in the last section. Here we shall use a more complicated approach, current algebra in CFT.

Let us concentrate on the holomorphic part of the wavefunction (Φ_m in equations (6.4) and (6.6)). To construct the edge states we need to construct holomorphic functions that satisfy equation (6.7). We notice that the Laughlin–Jastrow function Φ_m can be written as a correlation of the vertex operators (or primary fields) ψ_e , in the Gaussian model as follows [23, 68]:

$$\Phi_m = \lim_{z_\infty \rightarrow \infty} (z_\infty)^{2h_N} \left\langle \Psi_N(z_\infty) \prod_{i=1}^N \psi_e(z_i) \right\rangle, \quad (6.8)$$

where

$$\psi_e(z) = \exp [im^{1/2}\phi(z)], \quad \Psi_N(z) = \exp [iNm^{1/2}\phi(z)]. \quad (6.9)$$

Here N is the number of electrons, and $h_N = mN^2/2$ is the conformal dimension of Ψ_N . The scalar field ϕ in the Gaussian model is normalized so that $\langle \phi(z)\phi(0) \rangle = \ln z$. (Note that the ϕ field here is just the ϕ field that we introduced in the $U(1)$ K–M algebra (3.9) in section 3.1.) The factor $(z_\infty)^{2h_N}$ is included in equation (6.8) so that

the limit $z_\infty \rightarrow \infty$ gives rise to a finite function.[‡] Let $j(z) \equiv \partial\phi$ be the $U(1)$ current in the Gaussian model. Since j is a local operator, we find that the correlation

$$\tilde{\Phi}_m \propto \left\langle \oint dz \alpha(z) j(z) \Psi_N(z_\infty) \prod_{i=1}^N \psi_e(z_i) \right\rangle, \quad (6.10)$$

with an appropriately chosen holomorphic function $\alpha(z)$, satisfies equation (6.7) because $\psi_e(z)$ has the following operator product expansion (OPE) as $z \rightarrow 0$:

$$\begin{aligned} \psi_e(z) \psi_e(0) &\propto z^m \exp[i2m^{1/2}\phi(z)] + O(z^{m+1}), \\ j(z) \psi_e(0) &\propto \frac{1}{z} \psi_e(0) + O(1). \end{aligned} \quad (6.11)$$

The OPE between ψ_e guarantees the correlation to have m th orders zero as $z_i \rightarrow z_j$. The integration contour of z in equation (6.10) encloses all the $\psi_e(z_i)$ operators but not Ψ_N . If $\alpha(z)$ has no poles except at z_∞ , then equation (6.10) is finite for finite z_i , since $z_\infty \rightarrow \infty$. Therefore equation (6.10) is a wavefunction with zero energy, and hence a wavefunction for the edge excitations. Introducing

$$j(z) = \sum_{n=-\infty}^{\infty} \frac{j_n}{(z - z_\infty)^{n+1}} \quad (6.12)$$

and choosing

$$\alpha(z) = (z - z_\infty)^{-n} \quad (n \geq 0), \quad (6.13)$$

we find that, by shrinking the contour around z_∞ and letting $z_\infty \rightarrow \infty$, equation (6.10) becomes

$$\Phi_m^{(n)} = \lim_{z_\infty \rightarrow \infty} (z_\infty)^{2h_N + 2n} \left\langle \Psi_N^{(n)}(z_\infty) \prod_{i=1}^N \psi_e(z_i) \right\rangle, \quad (6.14)$$

where

$$\Psi_N^{(n)}(z_\infty) = j_n \Psi_N(z_\infty). \quad (6.15)$$

Again the factor $z_\infty^{2h_N + 2n}$ is included in (6.15) to ensure the existence of a finite limit.[‡]

The above discussion can be generalized to the case with several insertions of the current operator. The general statement is the following. The ground-state wavefunction can be written as a correlation of $\psi_e(z_i)$ with the primary field Ψ_N inserted at infinity. To generate edge excitations above the ground state, we simply replace the primary field Ψ_N by its current descendants [69], which are generally of the form (with $n_1, n_2, \dots \geq 0$)

$$\Psi_N^{(n_1, n_2, \dots)} = (j_{-n_1} j_{-n_2} \dots) \Psi_N. \quad (6.16)$$

Let us call $l = \sum_i n_i$ the level of the descendant field $\Psi_N^{(n_1, n_2, \dots)}$. The wavefunction associated with it is given by

$$\Phi_m^{(n_1, n_2, \dots)} = \lim_{z_\infty \rightarrow \infty} (z_\infty)^{2h_N + 2l} \left\langle \Psi_N^{(n_1, n_2, \dots)}(z_\infty) \prod_{i=1}^N \psi_e(z_i) \right\rangle. \quad (6.17)$$

This wavefunction has zero energy and can be identified as an edge excitation.

We would like to show that all the edge states generated by the level- l descendant fields have a total angular momentum $L = M_0 + l$, where $M_0 = mN(N-1)/2$ is the

[‡] Consider an operator Ψ of dimension h . The correlation function $\langle \Psi(z) \dots \rangle$ (here $\dots$ representing other operators) as $z \rightarrow \infty$ is proportional to $\langle \Psi(z) \Psi^\dagger(0) \rangle (1 + O(z^{-1}))$, where $\Psi^\dagger$ is the conjugate of Ψ . Thus $\langle \Psi(z) \dots \rangle \propto z^{-2h}$ as $z \rightarrow \infty$.

total angular momentum of the ground state. We first note that the angular momentum of the ground state can be expressed in terms of conformal dimensions h_e and h_N of ψ_e and Ψ_N . Under a conformal transformation $z \rightarrow w = f(z)$, the correlation of primary fields satisfies

$$\left\langle \Psi_N(z_\infty) \prod_{i=1}^N \psi_e(z_i) \right\rangle = [f'(z_\infty)]^{h_N} \prod_i [f'(z_i)]^{h_e} \left\langle \Psi_N(w_\infty) \prod_{i=1}^N \psi_e(w_i) \right\rangle. \quad (6.18)$$

Choosing $f(z) = \lambda z$, we have

$$\left\langle \Psi_N(\lambda z_\infty) \prod_{i=1}^N \psi_e(\lambda z_i) \right\rangle = \lambda^{-h_N - Nh_e} \left\langle \Psi_N(z_\infty) \prod_{i=1}^N \psi_e(z_i) \right\rangle, \quad (6.19)$$

which implies that

$$\Phi_m(\lambda z_i) = \lambda^{h_N - Nh_e} \Phi_m(z_i). \quad (6.20)$$

For $\lambda = \exp(i\theta)$, this transformation is simply a rotation by angle θ in the complex plane. Thus the angular momentum of the ground state is

$$M_0 = h_N - Nh_e. \quad (6.21)$$

With $h_N = mN^2/2$ and $h_e = m/2$, we see that $M_0 = mN(N-1)/2$, as expected for the Laughlin $1/m$ state. More generally, since the dimension of a current descendant $\Psi_N^{(n_1, n_2, \dots)}(z)$ is the sum $h_N + l$ [69], where l is its level, we find that under $z \rightarrow w = \lambda z$:

$$\Psi_N^{(n_1, n_2, \dots)}(z) = \lambda^{h_N + l} \Psi_N^{(n_1, n_2, \dots)}(w). \quad (6.22)$$

Thus, from equation (6.17) and (6.22) we can see that the edge-excited state described by $\Phi_m^{(n_1, n_2, \dots)}$ carries an angular momentum of $L = h_N + l - Nh_e = M_0 + l$. As a general rule, valid for any FQH states that can be generated by CFT, the angular momentum of an edge excitation is equal to that of the ground state plus the level of the descendant level of the associated insertion at infinity.

Note that the descendant fields $\Phi_m^{(n_1, n_2, \dots)}$ with different $(n_1, n_2, \dots)$ may not be linearly independent. Using the standard approach in CFT, one can show that, from the OPE $j(z)j(0) = 1/z^2$, j_n satisfies the $U(1)$ K-M algebra:

$$[j_n, j_m] = 2\pi n \delta_{m+n}. \quad (6.23)$$

From the $U(1)$ K-M algebra (6.23), it is easy to show that the number D_l of linearly independent descendant fields (6.16) at level l is given by the partition number of l . The values of D_l for small l are given in equation (6.3). The values of D_l for generic l can be expressed in terms of the character of the $U(1)$ K-M algebra for the primary field Ψ_N [70]:

$$\text{ch}_N(\xi) \equiv \xi^{h_N} \sum_l D_l \xi^l = \xi^{h_N} \left(1 / \prod_{n>0} (1 - \xi^n) \right). \quad (6.24)$$

If there is a one-to-one correspondence between the edge states and the descendant fields of Ψ_N , then we can use the above formula to obtain the number D_L of edge excitations at any given angular momentum L :

$$\text{Ch}_N(\xi) \equiv \sum_L D_L \xi^L = \text{ch}_N(\xi) \xi^{-Nh_e} = \left(1 / \prod_{n>0} (1 - \xi^n) \right) \xi^{M_0}. \quad (6.25)$$

The one-to-one correspondence between the edge states and the descendant fields of Ψ_N means that linearly independent descendant fields always generate linearly independent wave functions of z_i through equation (6.17). However, we do not expect this correspondence to be true for finite N and arbitrarily large level $l = \sum_k n_k$. On the other hand, it is conceivable that this one-to-one correspondence holds for any finite level l when N is very large or in the limit $N \rightarrow \infty$. Although we do not know, at present, how to prove this statement within the CFT approach for a generic Abelian FQH state, its validity for the $1/m$ states can be seen in the following way. It is known [48, 49] that the edge states can be generated by multiplying the ground-state wavefunction by symmetric polynomials of electron coordinates z_i . Mathematically, the number of linearly independent symmetric polynomials of degree l is precisely given by the partition number of the integer l , which is just the number of linearly independent descendant fields.

In the above we have generated the edge excitations of the Laughlin state by inserting the current operator. One may try to generate more edge states by inserting some other operators, such as the energy–momentum tensor $\mathbf{T}$, because the insertions of the energy–momentum tensor also maintain the structure of zeros (6.7) of the wavefunctions. However, one can show that the edge states generated by the energy–momentum tensor is contained in those generated by the current since $\mathbf{T} \propto \mathbf{j}^2$. For the Laughlin state, j is the ‘fundamental operator’ that generates all the edge excitations.

From the discussions in this section, we see a close relationship between the (minimum) bulk CFT that generates the bulk wavefunction and the edge CFT that generates the spectrum of the edge excitations, at least for the $1/m$ Laughlin states. In the next section, we shall show that this relationship can be generalized to non-Abelian FQH systems.

6.3. Non-Abelian fractional quantum Hall liquids and their edge excitations

In this section we are going to study the physical properties of some simple non-Abelian FQH liquids [23, 25, 63, 64] using the algebraic approach. In general, the holomorphic part of the many-body wavefunctions for a non-Abelian FQH state consists of two factors. One of them is of usual Laughlin–Jastrow form $\prod_{i < j} (z_i - z_j)^r$ (r is a fraction or integer), which we call the $U(1)$ part. The connection of this part to the $U(1)$ K–M algebra (or the Gaussian model) is well understood. Here we concentrate on the other part, which is not of the Laughlin–Jastrow form. We call it the non-Abelian part, because it is this part that is presumably responsible to the appearance of non-Abelian statistics for quasiparticles.

Let us first consider a simple non-Abelian FQH state, namely the p -wave-paired FQH state for spinless electrons discussed in [23]. The total wavefunction is given by the product of a Pfaffian wavefunction Φ_{Pf} and Laughlin wavefunction Φ_m

$$\begin{aligned}\Phi_p &= \Phi_{\text{Pf}} \Phi_m, \\ \Phi_{\text{Pf}} &= \mathcal{A} \left(\frac{1}{(z_1 - z_2)} \frac{1}{(z_3 - z_4)} \cdots \right), \\ \Phi_m &= \left(\prod_{i < j} (z_i - z_j)^m \right) \exp \left(-\frac{1}{4} \sum_i |z_i|^2 \right),\end{aligned}\tag{6.26}$$

where $\mathcal{A}$ is the antisymmetrization operator and m is an even integer. The filling fraction of the p -wave-paired state is $\nu = 1/m$.

Let us first analyse the structure of zeros of this wavefunction assuming, for simplicity, $m = 2$. Let $z_1 = z_3 + \delta_1$ and $z_2 = z_3 + \delta_2$, we find that Φ_p has the following expansion (we expand δ_2 first):

$$\Phi_p = \sum_{k \text{ odd}} (\delta_2)^k A_k(z_1, z_3, \dots) = \sum_{k \text{ odd}} (\delta_2)^k \sum_l (\delta_1)^l A_{kl}(z_3, z_4, \dots). \quad (6.27)$$

One can directly check that the coefficients

$$A_{12} = 0. \quad (6.28)$$

Therefore Φ is the exact ground state of Hamiltonian H with the following three-body potential:

$$V = -V_1 \partial_{z_3} \delta(z_2 - z_3) \partial_{z_2} \partial_{z_1}^2 \delta(z_1 - z_3) \partial_{z_1}^2. \quad (6.29)$$

This is because the Hamiltonian H is positive definite if $V_1 > 0$, and Φ is a zero-energy state of H . It was checked numerically that Φ is the non-degenerate ground state of H (on a sphere) with a finite energy gap. This implies that the constraint (6.28) uniquely fixes the ground-state wavefunction [63, 71].

On a plane, the ground state of H is identified as the minimum angular momentum state with zero energy. Numerical results indicate that the ground state on the plane is non-degenerate [63, 71]. Other zero-energy states all have higher angular momenta and are identified as the edge excitations above the ground state.

It was shown in [23] that the so-called non-Abelian part Φ_{Pf} can be written as a correlation of the dimension- $\frac{1}{2}$ primary fields ψ in the Ising model. ψ is just the fermion field in a free Majorana fermion theory. Therefore the total ground-state wavefunction (the holomorphic part) can be written as

$$\Phi_p = \lim_{z_\infty \rightarrow \infty} (z_\infty)^{2h_N} \left\langle \Psi_N(z_\infty) \prod_{i=1}^N \psi_\epsilon(z_i) \right\rangle, \quad (6.30)$$

where

$$\psi_\epsilon(z) = \psi(z) \exp [im^{1/2} \phi(z)], \quad (6.31)$$

$$\Psi_N(z_\infty) = \exp [-iNm^{1/2} \phi(z_\infty)]. \quad (6.32)$$

Here $\phi(z)$ is the chiral scalar field in the Gaussian model whose correlation produces the $U(1)$ part Φ_m . In equation (6.30), $h_N = mN^2/2$ is the dimensions of Ψ_N and N the total number of the electrons.

Let $z_1 = z_3 + \delta_1$ and $z_2 = z_3 + \delta_2$ and assume $m = 2$, we find that the OPE of the operators ψ_ϵ has the following form:

$$\begin{aligned} \psi_\epsilon(z_1) \psi_\epsilon(z_2) \psi_\epsilon(z_3) &= \psi_\epsilon(z_1) \delta_2 \exp [2im^{1/2} \phi(z_3)] + \dots \\ &= \delta_2 (\delta_1)^4 \psi(z_3) \exp [3im^{1/2} \phi(z_3)] + \dots \end{aligned} \quad (6.33)$$

We find that the lower-order zeros (such as the term containing $\delta_1^2 \delta_2^2$) are absent as described by equation (6.28). Thus the correlations between ψ_ϵ automatically generate wavefunctions that satisfy (6.28) and become the zero-energy states of the three-body Hamiltonian H .

From the above, we see that the wavefunction of a p-wave-paired FQH state can be written as a correlation in the $U(1) \times \text{Ising}$ model. Since the bulk CFT of the p-wave-paired state is not the Gaussian model, this motivated Moore and Read to suggest that the p-wave state is a non-Abelian state. Indeed it was shown that the

quasiparticles in the p-wave state have very different structures from those in Abelian states [23, 71].

To have some basic understanding of the new structures of the quasiparticles in the non-Abelian state, let us start with the charge $e/2$ quasihole created by inserting a unit flux quantum:

$$\prod_i (\xi - z_i) \Phi_p(\{z_i\}). \quad (6.34)$$

Owing to the presence of the pairing wavefunction

$$\Phi_{\text{Pf}} = \mathcal{A} \prod_{i \text{ odd}} \frac{1}{z_i - z_{i+1} + 1}$$

we can split the quasihole in equation (6.34) into two excitations each carrying a charge of $e/4$:

$$\Phi(\{z_i\}, \xi_1, \xi_2) = \mathcal{A} \prod_{i \text{ odd}} \frac{(z_i - \xi_1)(z_{i+1} - \xi_2)}{z_i - z_{i+1}} \Phi_m(\{z_i\}). \quad (6.35)$$

(Note that $\Phi(\{z_i\}; \xi, \xi)$ is just the wavefunction in equation (6.34).) An interesting phenomenon happens when we create four $e/4$ quasiholes. One can write down three different wavefunctions for the same four quasiholes at the same positions:

$$\begin{aligned} \Phi_1(\{z_i\}; \xi_1, \xi_2, \xi_3, \xi_4) &= \mathcal{A} \prod_{i \text{ odd}} \frac{(z_i - \xi_1)(z_{i+1} - \xi_2)(z_i - \xi_3)(z_{i+1} - \xi_4)}{z_i - z_{i+1}} \Phi_m(\{z_i\}), \\ \Phi_2(\{z_i\}; \xi_1, \xi_2, \xi_3, \xi_4) &= \mathcal{A} \prod_{i \text{ odd}} \frac{(z_i - \xi_1)(z_{i+1} - \xi_3)(z_i - \xi_2)(z_{i+1} - \xi_4)}{z_i - z_{i+1}} \Phi_m(\{z_i\}), \\ \Phi_3(\{z_i\}; \xi_1, \xi_2, \xi_3, \xi_4) &= \mathcal{A} \prod_{i \text{ odd}} \frac{(z_i - \xi_1)(z_{i+1} - \xi_2)(z_i - \xi_4)(z_{i+1} - \xi_3)}{z_i - z_{i+1}} \Phi_m(\{z_i\}). \end{aligned} \quad (6.36)$$

However, only two of them are linearly independent [23]. It is a unique property of non-Abelian states that the four-quasihole states have a twofold degeneracy even when all the positions of the quasiholes are fixed. As we exchange different pairs of quasiholes, non-Abelian Berry phases may be generated.

However, although we have found the existence of the degeneracy, at the moment, we are still unable to calculate the non-Abelian Berry phase owing to the lack of the plasma analogue for the pairing wavefunction Φ_p , but the numerical results in [71] do provide evidence for the existence of the non-Abelian statistics in the p-wave state.

In the following we shall study the edge excitations of the p-wave state. We find that the edge structure of the p-wave state is also very different from that of Abelian states. Thus provides additional evidences for the existence of the new topological orders in the p-wave state.

From equation (6.30) we see that the holomorphic part of the ground-state wavefunction can be written as a correlation of ψ_e with the operator Φ_N inserted at infinity. Following the discussion in the last section, we can create edge excitations by inserting other operators. Here we can use both ψ and j insertions. Repeating the derivation in the last section, we find that edge excitations (i.e. the zero-energy states of the three body Hamiltonian (6.29)) are generated by the descendant fields of Ψ_N :

$$\Phi_p^{\text{edge}} = \lim_{z_\infty \rightarrow \infty} (z_\infty)^{2h_N + 2l} \left\langle \Psi_N^{(n_1, \dots, m_1, \dots)}(z_\infty) \prod_{i=1}^N \psi_e(z_i) \right\rangle, \quad (6.37)$$

where

$$l = \sum_i n_i + \sum_k m_k \quad (6.38)$$

is the level of the descendant field

$$\Psi_N^{(n_1, \dots, m_1, \dots)}(z_\infty) = (\psi_{-n_1} \dots j_{-m_1} \dots) \Psi_N(z_\infty) \quad (6.39)$$

and the ψ_n are operators in the expansion of $\psi(z)$:

$$\psi(z) = \sum_n \frac{\psi_n}{z^{n+1/2}}, \quad n = \text{integer} + \frac{1}{2}. \quad (6.40)$$

Note that, in order for the correlation to be non-zero, $\Phi_N^{(n_1, \dots, m_1, \dots)}$ must contain even or odd numbers of ψ_n operators if there are even or odd numbers of electrons (i.e. N even or odd).

Not all descendant fields $\Phi_N^{(n_1, \dots, m_1, \dots)}$ with different indices are linearly independent. The linearly independent descendant fields generated by j_n can be calculated from the K–M algebra (6.23) as we did in the last section. The linearly independent descendant fields generated by ψ_n can be calculated from the fermion algebra satisfied by ψ_n fields

$$\{\psi_n, \psi_m\} = 2\pi\delta_{n+m}. \quad (6.41)$$

The algebra (6.41) can be derived from the OPE:

$$\psi(z_1)\psi(z_2) = -\psi(z_2)\psi(z_1) = \frac{1}{z_1 - z_2}. \quad (6.42)$$

Thus the space of the descendant fields is isomorphic to the direct product of the space of the K–M algebra generated by j and the space of the Majorana fermion theory generated by ψ . This is because ψ and j commute with each other. Similar to the pure $U(1)$ case in the last section, the edge excitations generated by level- l descendant fields can be shown to have angular momentum $L = M_0 + l$, where $M_0 = h_N - NH_e = (m/2)N(N-1) - (N/2)$ is the angular momentum of the ground state Φ_p . Here we have used equation (6.21), and $h_e = (m+1)/2$, $h_N = (m/2)N^2$ [64].

The number of the linear independent descendant fields at each level can be easily calculated by counting the number of states in a free-boson theory (generated by j) and a free-fermion theory (generated by ψ). The results can be expressed through characters. The sectors with even and odd ψ_n should be considered separately. Let us first concentrate on the space generated by ψ_n . Note that the character for the two states generated by ψ_n is $1 + \xi^n$. The total character for both the even and the odd sectors is given by

$$\prod_{n=1/2}^{\infty} (1 + \xi^n) = \text{ch}_{\text{even}}(\xi) + \text{ch}_{\text{odd}}(\xi). \quad (6.43)$$

To obtain $\text{ch}_{\text{even, odd}}$ separately we need to keep track of the numbers of the fermion operators. This can be achieved by introducing a generalized character $1 + \xi^n \eta$ which leads to

$$\text{ch}(\xi, \eta) = \prod_{n=1/2}^{\infty} (1 + \xi^n \eta). \quad (6.44)$$

It is easy to see that the power of η counts the numbers of fermions. Thus the characters for the even and the odd sectors are

$$\begin{aligned} \text{ch}_{\text{even}} &= \frac{1}{2}(\text{ch}(\xi, 1) + \text{ch}(\xi, -1)) = \frac{1}{2} \left(\prod_{n=1/2}^{\infty} (1 + \xi^n) + \prod_{n=1/2}^{\infty} (1 - \xi^n) \right), \\ \text{ch}_{\text{odd}} &= \frac{1}{2}(\text{ch}(\xi, 1) - \text{ch}(\xi, -1)) = \frac{1}{2} \left(\prod_{n=1/2}^{\infty} (1 + \xi^n) - \prod_{n=1/2}^{\infty} (1 - \xi^n) \right). \end{aligned} \quad (6.45)$$

The characters for edge excitations of the p-wave paired state, after including the Gaussian (or Abelian) part generated by j , can be written as products of the $U(1)$ character (6.24) and the non-Abelian character (6.45) and are given by

$$\begin{aligned} \text{Ch}_{\text{even}} &= \left(1 / \left[2 \prod_{n>0} (1 - \xi^n)^2 \right] \right) \left(\prod_{n=1/2}^{\infty} (1 + \xi^n) + \prod_{n=1/2}^{\infty} (1 - \xi^n) \right) \xi^{M_0^{(e)}}, \\ \text{Ch}_{\text{odd}} &= \left(1 / \left[2 \xi^{1/2} \prod_{n>0} (1 - \xi^n)^2 \right] \right) \left(\prod_{n=1/2}^{\infty} (1 + \xi^n) - \prod_{n=1/2}^{\infty} (1 - \xi^n) \right) \xi^{M_0^{(o)}}, \end{aligned} \quad (6.46)$$

where $M_0^{(e)}$ and $M_0^{(o)}$ are the angular momenta for the ground state with an even number and an odd number respectively of electrons.

Having edge states generated in this way, two questions immediately come to our mind. First, are the edge states generated with different descendants linearly independent to each other? Second, do the insertions with all descendant fields generated by j and ψ exhaust all possible edge states? These are very hard and important questions. Equation (6.46) is actually a counting of linearly independent descendant fields. To apply equation (6.46) to count the physical excitations at the edge we need to address the above two questions. Let us examine them in turn.

For the first question, obviously the descendants of primary fields with different dimensions (which are related to the angular momentum quantum numbers) generate linearly independent edge states. The hard part of the question is whether different descendants at the same level will generate linearly independent state or not. Generally in CFT, linearly independent descendants, as operators, should have different (or linearly independent) sets of correlations which contain *arbitrary* number of electron operators $\psi_{\pm}$. However, when the number N of electrons is finite and fixed, one *cannot* claim that different descendant fields generate linearly independent correlations, in particular for descendant fields at arbitrarily large levels. So we suggest that the insertions with different operators generate different edge excitations in the thermodynamic limit or in the large- N limit, although at the moment we do not know how to prove it within CFT.

Now turn to the question of whether the descendant fields discussed above can generate all possible edge excitations in the system. We would like first to point out that the edge wavefunctions constructed above preserve not only the structure of zeros in the ground state as two or three electrons approach each other but the structure of higher-order zeros for four or more electrons approaching each other. If the wavefunctions generated by the descendant fields do not exhaust the zero-energy sector of a three-body Hamiltonian, in principle it is possible to construct a more restrictive Hamiltonian that contains additional two-body, three-body, four-body, etc., interactions, so that the new Hamiltonian makes the wavefunctions constructed above occur and to exhaust its zero-energy states. Thus the question of whether the space of edge states contains more states depends on the dynamics of

electron interactions and cannot be addressed by merely studying the wavefunctions. In the following, we shall assume that the Hamiltonian satisfies the above conditions.

Under the above two assumptions we see that there is a one-to-one correspondence between the descendant fields and the edge states (in the large- N limit). Thus one can use the known results (6.46) about the descendant fields derived from the CFT to obtain the spectrum of low-lying edge states. Numerical diagonalizations for small systems have been done to test the predictions [63]. It was found that indeed the numerical results shows the violation of the suggested correspondence for finite N at large level l . On the other hand, they verify the validity of the correspondence at the levels less than a certain number of order N .

From equation (6.46), one finds the numbers of states (NOSs) of the low-lying edge excitations at total angular momenta $M_0 + \Delta M$ (where M_0 is the angular momentum of the ground state) [63]:

$$\begin{aligned} \Delta M = & 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \quad \text{sector,} \\ \text{NOS} = & \begin{cases} 1 & 1 & 3 & 5 & 10 & 16 & 28 & 43 & 70 & N \text{ even,} \\ 1 & 2 & 4 & 7 & 13 & 21 & 35 & 55 & 86 & N \text{ odd.} \end{cases} \end{aligned} \quad (6.47)$$

The spectrum (6.47) is very different from the edge spectrums of the Abelian states. The spectrum of one branch of edge excitations in Abelian states is given by

$$\begin{aligned} \Delta M = & 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8, \\ \text{NOS} = & 1 \quad 1 \quad 2 \quad 3 \quad 5 \quad 7 \quad 11 \quad 15 \quad 22, \end{aligned} \quad (6.48)$$

while the spectrum of two branches of Abelian edge excitations is

$$\begin{aligned} \Delta M = & 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8, \\ \text{NOS} = & 1 \quad 2 \quad 5 \quad 10 \quad 20 \quad 36 \quad 65 \quad 110 \quad 180. \end{aligned} \quad (6.49)$$

It is clear that, in addition to its special dependence on N , the number of the edge excitations in the p-wave state is more than one branch but less than two branches of edge excitations in Abelian states. The specific heat (per unit length) of the p-wave edge state turns out to be $\frac{3}{2}$ times of $(\pi/6)(T/v)$, the specific heat of one branch of edge excitations of the Abelian states. In this sense, we may say that the p-wave state contains one and a half branches of edge excitations.

In the above, we merely discussed the spectrum of the edge excitations. We can go one step further. Both the Gaussian model and the Majorana fermion theory are equipped with a natural definition of inner product in their Hilbert space. We may assume the natural inner product in the CFT happens to be the physical inner product between the edge wavefunctions generated by the corresponding operators (in the large- N limit). Under this assumption we may use the known correlation functions of the CFT to calculate the correlation functions of physical operators near the edge. For example, the electron operator is given by ψ_e in equation (6.31). From the CFT results $\langle \psi_e(z) \psi_e(0) \rangle = 1/z^{m+1}$ we find the electron propagator on the edge to have the form

$$G_e(x, t) \propto \frac{1}{(x - vt)^{g_e}}, \quad g_e = m + 1. \quad (6.50)$$

This result can be confirmed through a calculation of the electron occupation number

n_l in the single-particle state $|l\rangle$. Here l is the angular momentum. For the p-wave state of ten electrons with $m = 2$, it was found [63] that

$$n_{l_0} : n_{l_0-1} : \dots = 1 : 3.06 : 5.86 : 8.63 : \dots$$

The theoretical predictions of n_l calculated from the Green function (6.50) for $g_e = m + 1 = 3$ are given by [33] $1 : 3 : 6 : 10 : \dots$. Here l_0 is the angular momentum of the last occupied orbit ($l_0 = 17$ for ten electrons). The agreement suggests the validity of our assumption about the correspondence between the inner products [63].

Next let us consider the quasiparticle operator. The CFT that describes a p-wave edge state, the $U(1) \times$ Ising model, contains the following local operators: $\exp(i\alpha\phi)$, ψ and σ . Here σ is the disorder operator in the Ising model which changes the boundary condition of the fermion operator ψ . Not all the above operators are physical, that is create an allowed excitation in the electron system. A physical quasiparticle operator must have a single-valued correlation function with the electron operator. This condition is closely related to the single-valued property of the electron wavefunction in the presence of the quasiparticle. The condition can be expressed through the operator product expansion between the electron and quasiparticle operators:

$$\psi_e(w_1)\psi_q(w_2) \propto (w_1 - w_2)^\gamma \hat{O}(w_2), \quad (6.51)$$

where we require γ to be an integer. From the OPE in the $U(1) \times$ Ising model,

$$\begin{aligned} \exp[i\alpha\phi(w)] \exp[i\beta\phi(0)] &\propto w^{l(\alpha+\beta)2-\alpha^2-\beta^2/4} \exp[i(\alpha+\beta)\phi(0)], \\ \psi(w)\sigma(0) &\propto w^{1/2}\mu(0), \end{aligned} \quad (6.52)$$

we find that the following operators can be identified as quasiparticle operators: $\psi_q(x) = \exp[i(1/2m^{1/2})\phi(x)] \sigma(x)$ and $\psi'_q(x) = \exp[i(1/m^{1/2})\phi(x)]$. ψ_q carries charge $e/2m$ and corresponds to the non-Abelian quasiparticle discussed above [23]. ψ'_q carries charge $1/m$ and has an Abelian statistics $\theta = \pi/m$. ψ'_q is created by inserting a unit flux. From the CFT result $\langle\sigma(w)\sigma(0)\rangle \propto w^{1/8}$, we find that $\langle\psi_q(x, t)\psi_q(0)\rangle \propto (x - vt)^{-g_q}$ with $g_q = (m + 2)/8m$. The exponents g_e and g_q can be measured in tunnelling experiments between the edges and provide an experimental test of the non-Abelian states.

Another important non-Abelian FQH state is the Haldane–Rezayi (HR) state for spin- $\frac{1}{2}$ electrons [27]. The HR state is important because it is a candidate for the $\nu = \frac{5}{2}$ FQH state observed in experiments [26]. The HR state is a d-wave-paired spin-singlet FQH state:

$$\begin{aligned} \Phi_{\text{HR}}(z_i, w_i) &= \Phi_m(z_i, w_i) \Phi_{\text{ds}}(z_i, w_i), \\ \Phi_{\text{ds}}(z_i, w_i) &= \mathcal{A}_{z, w} \left(\frac{1}{(z_1 - w_1)^2} \frac{1}{(z_2 - w_2)^2} \cdots \right), \end{aligned} \quad (6.53)$$

$$\Phi_m(z_i, w_i) = \left(\prod_{i < j} (z_i - z_j)^m \prod_{i < j} (w_i - w_j)^m \prod_{i, j} (z_i - w_j)^m \right) \exp \left(-\frac{1}{4} \sum_i (|z_i|^2 + |w_i|^2) \right),$$

which has a filling fraction $1/m$ with m an even integer. Here $z_i(w_i)$ are the coordinates of the spin-up (spin-down) electrons, and the operator $\mathcal{A}_{z, w}$ performs separate antisymmetrizations among z_i and among w_i .

The non-Abelian part of the bulk wavefunction of the HR state was shown to be

described by a CFT with central charge $c = -2$ [64]. Using this $c = -2$ CFT and the $U(1)$ Gaussian model for the Abelian part, one can calculate the spectrum of edge excitations. One finds that the edge excitations of an N -electron system with total spin s and a fixed S_z component σ are described by the following character:

$$\text{Ch}_{N,s}(\xi) = \left((1 - \xi^{2s+1}) / \prod_n (1 - \xi^n)^2 \right) \xi^{M_0^{(s)}}, \quad (6.54)$$

where

$$M_0^{(s)} = h_{N,s} - Nh_c = \frac{m}{2} N(N-1) + \frac{1}{8}[(4s+1)^2 - 1] - N \quad (6.55)$$

is the minimum angular momentum in the spin- s sector. Here $h_{N,s} = (m/2)N^2 + \frac{1}{8}[(4s+1)^2 - 1]$, and $h_c = 1 + (m/2)$. According to equation (6.54), the number of edge excitations in the spin- s sector at angular momentum $L = M_0^{(s)} + l$, is given by

$$\begin{array}{cccccccl} l = 0 & 1 & 2 & 3 & 4 & 5 & \text{spin,} \\ & 1 & 1 & 3 & 5 & 10 & s = 0, \\ & 1 & 2 & 4 & 8 & 15 & 26 & s = \frac{1}{2}, \\ & 1 & 2 & 5 & 9 & 18 & 31 & s = 1, \\ & 1 & 2 & 5 & 10 & 19 & 34 & s = \frac{3}{2}, \end{array} \quad (6.56)$$

which again has very different structures from the Abelian edge states.

In summary, non-Abelian FQH states are a fascinating class of topological fluids, which very probably exist in real experimental samples (such as the $\nu = \frac{5}{2}$ state). Our understanding of the non-Abelian states is quite limited. In particular, we do not understand the general mathematical structures behind the non-Abelian topological orders, and this limits our ability to calculate the physics properties of non-Abelian states.

7. Remarks

Many important aspects of topological fluids were not covered. I recommend two review articles, [72] (which is fun to read) and [73] which address some similar issues discussed in this paper with different emphases. They also cover some subjects which are not discussed here. Many early papers on topological properties of QH states can be found in [74]. In this paper I include only what I believe to be the most fundamental properties of topological orders in FQH liquids, and in particular those properties that might have experimental consequences. Other important and/or actively studied properties of FQH fluids include the following:

- (1) phase transitions and phase structures of FQH systems [75];
- (2) hierarchical structures in multilayer FQH systems [76];
- (3) superfluid mode and symmetry breaking in some double-layer FQH states [40];
- (4) structure of ground states of topological fluids on closed space [6, 7, 13]; for an Abelian FQH state characterized by the $\mathbf{K}$ matrix, one can show it has a $(\det K)^g$ -fold degeneracy on the genus g Riemann surface;
- (5) $\mathbf{W}_\infty$ algebra in FQH states [77];
- (6) spin dynamics and symmetries in FQH states [38].

FQH liquids demonstrate extremely rich internal structures. The more we probe, the more we are fascinated by the endless richness that nature reveals to us.

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